Introduction

Events and Probability

Probability theory provides a framework for modelling uncertainity.

Stochastic Processes provide a framework for coupling uncertainity with dynamics. (Space / Time etc)

Probabilistic Ideas

Examples of applications of probability theory.

- models for ambient / reciever noise in communication systems. (Cellular wireless communication).
- stock market and economic models.
- noise (defects) in computer memory chips.
- noise in ML models.
- birth death models in demographics.
- noise in solid state devices.
- noise in navigation systems.

Approaches

a) Relative Frequency Approach

Heuristic notion that probabilities are associated with relative frequencies of a repeatable event.

In general - not always applicable
- not unique.

b) Axiomatic Approach

A.N. Kolmogorov 1903 - 1987
1933 : foundations of Theory of Probability.

Based on measure theory

Relative frequencies are an interpretation of probabilities (rather than a definition)

1) Probability Space or Probability Triple $(\Omega, \mathcal{A}, P)$

- Starting point : Experiment E whose outcome is not known a priori.

  Ex: E - Toss a coin, outcome : H or T

- To quantify the randomness of the outcomes of an experiment E, we need,

  List of all outcomes of E

  Measure of the likelihood of each outcome. (probability assignment)

Definition : E is an experiment. Sample space Ω associated with E is the set of all possible outcomes of E.

- Ex : E is a roll of a die $\Rightarrow \Omega = \{1,2,3,4,5,6\}$

  E keep tossing a coin until H shows $\Rightarrow \Omega = \{H, TH, TTH, \dots\}$

  E throwing a dart at a circular dart board $\Rightarrow \Omega = \{(x,y) \in \mathbb{R}^2 : x^2+y^2 < 1\}$

- Important : For each experiment E, only one outcome is observed.

2) Types of Sample Spaces of Interest

i) Ω finite set

Ω is a finite set of cardinality $n \in \mathbb{N}$

$\exists$ one to one correspondence between Ω and $\{1,2,\dots,n\}$

Ex: $\Omega = \{H, T\}$ $n=2$

ii) Ω is countably infinite

$\exists$ one to one correspondance between Ω and $\mathbb{N}$.

Ex: $\{1, 2, 3, \dots\}$ is countably infinite.

iii) Ω is uncountable

Ex: $\Omega = [0, 1]$

We can afford to give probabilities to any subset of Ω. $\therefore$ We can pick $a = 2^{\Omega}$

Ω is Finite

Consider the probability assignment of sample space $\Omega = \{\omega_1, \dots, \omega_n\}$

p_i = probability that the observed outcome of E is ω_i $i = 1, \dots, n$

$\Rightarrow \forall\ i = 1, 2, \dots, n \quad 0 \leq p_i \leq 1$

$\Rightarrow \sum_{i=1}^{n} p_i = 1$

Definition - Event $E \subseteq \Omega$ is any subset of Ω. Likelihood of occurence of event E (ie outcome $\omega_j \in E$ is observed) if E is

$$P(E) = \sum_{\omega_j \in E} p_i = \text{Probability of event } E$$

Probabilities are assigned not to ω_i's $\in$ but to singleton sets $\{\omega_i\} \in a$

Remark -

I) P is defined by the mapping (set function)

$$P : \mathcal{P}(\Omega) \longmapsto [0, 1]$$
$$E \longmapsto P(E)$$

where $\mathcal{P}(\Omega)$ is the powerset of sample space and $P(E) = \sum_{\omega_i \in E} p_i$ such that following axioms hold.

(P_1) $P(\Omega) = 1$

(P_2) $P(E^c) = P(\Omega / E) = 1 - P(E)$

(P_3) If $E \cap F = \phi$ where $E, F \in \mathcal{P}(\Omega) \Rightarrow P(E \cup F) = P(E) + P(F)$

II) Note that for $i = 1, \dots, n$ $p_i = P(\{\omega_i\}) =$ Subset of Ω

$\Rightarrow P(E) = \sum_{\omega_i \in E} P(\{\omega_i\})$

$\Rightarrow p_1, p_2, \dots, p_n$ forms the building block for P.

Exercise :- $\left(\begin{array}{l}\text{Knowing } p_1, p_2, \ldots, p_n \text{ together with} \\ \text{the definition } P(E) \triangleq \sum_{\omega_i \in E} p_i \\ E \in \mathcal{P}(\Omega)\end{array}\right) \iff \left(\begin{array}{c}\text{Knowing } P(E) \quad E \in \mathcal{P}(\Omega) \\ \text{with the axioms} \\ (P1), (P2), (P3) \text{ holds}\end{array}\right)$

Ω Countably Infinite

Consider the probability assignment of sample space $\Omega = \{\omega_n, n \in \mathbb{N}\}$

p_i = probability that the observed outcome of E is ω_i $\quad i \in \mathbb{N}$

$\Rightarrow \forall\, i \in \mathbb{N} \quad 0 \leq p_i \leq 1$

$\Rightarrow \sum_{i=1}^{\infty} p_i = 1$

Ex:- $\Omega = \{0\} \cup \mathbb{N}$ is Countable. Pick $\mathcal{A} = 2^{\Omega}$

$P(\{k\}) = \frac{e^{-\lambda}\lambda^k}{k!} \quad \lambda > 0$

Poisson distribution.

Definition - Same as in case I.

Remark -

Same as in case I, except for the $(P3)$ condition, which strengthens to,

$(P3)$ If $\{E_i\}_{i \in I}$ I - countable and are pairwise disjoint events, ie $E_i \cap E_j = \emptyset \quad \forall\, i \neq j$

$$P\left(\bigcup_{i \in I} E_i\right) = \sum_{i \in I} P(E_i)$$

Ω Uncountable

Probability assignment can no longer be made per outcome unlike when Ω was countable.

Claim :- When Ω is uncountable, then not more than a countable number of outcomes $\omega \in \Omega$ can be assigned non zero probabilities.

Exercise: If $\Omega = [0,1]$, $P(\Omega) = P([0,1]) = 1$, then claim suggests $\{\omega \in [0,1] : P(\omega) > 0\}$ is at most countable.

Proof: Define $A_n \triangleq \{\omega \in [0,1] : P(\omega) \geq 1/n\}$ $n = 1, 2, \ldots$

$\Rightarrow \frac{1}{n+1} < \frac{1}{n} \Rightarrow A_{n+1} \supseteq A_n \Rightarrow \{A_n\}_{n=1}^{\infty} \uparrow$ is increasing in n

$\Rightarrow A_\infty = \bigcup_{n=1}^{\infty} A_n$

$\Rightarrow \{\omega \in [0,1] : P(\omega) > 0\} = \bigcup_{n=1}^{\infty} \{\omega \in [0,1] : P(\omega) \geq 1/n\}$

$\Rightarrow 1 = P([0,1]) \geq \sum_{\omega \in A_n} P(\omega) \geq \frac{1}{n} \#(A_n)$ *

$\Rightarrow n \geq \#(A_n)$

$\Rightarrow$ each A_n is a finite set. Hence A_∞ is countable since its a countable union of finite sets.

When Ω is uncountable, a probability assignment per outcome does not make sense.

Recall : When Ω is uncountable, a probability associated per outcome is not appropriate.

Axiomatic Approach : Overcomes the problem by assigning probabilities to events i.e subsets of Ω (rather than to individual outcomes).

Select a class or set of events $\mathcal{A}$ (subset of Ω) and the probability assigned is defined as a mapping

$$P: \mathcal{A} \longmapsto [0,1]$$

Satisfying certain properties. How to select $\mathcal{A}$?

3) Algebra ($\mathcal{F}$)

Definition - A field $\mathcal{F}$ on Ω

Ω is a given set (= sample space).
A field of $\mathcal{F}$ (or Boolean algebra, or simply algebra) is a set of subsets of Ω s.t., the following holds.

$(\mathcal{F}_1)$ $\mathcal{F}$ is nonempty

$(\mathcal{F}_2)$ $A \in \mathcal{F} \Rightarrow A^c = \Omega \setminus A \in \mathcal{F}$

$(\mathcal{F}_3)$ $A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$

Remark : $\mathcal{F}$ is a collection of sets that is closed under all finite Boolean set operations, namely complementations and finite unions.

Alternate Definition-

A σ-field $\mathcal{A}$ on Ω is a set of subsets of Ω $\exists$ $(\mathcal{F}_1)$, $(\mathcal{F}_2)$ holds together with the property that $\forall$ countable index sets I,

$$\{A_i,\ i \in I\},\ A_i \in \mathcal{A} \Rightarrow \bigcup_{i \in I} A_i \in \mathcal{A}$$

- If the sample space (Ω) is finite field, $\mathcal{F}$ (algebra) $\Leftrightarrow$ σ-field $\mathcal{A}$ (σ-algebra)

  If the sample space is infinite field, $\mathcal{F} \Leftarrow \not\Rightarrow$ σ-field $\mathcal{A}$ *

Show: (i) $A, B \in \mathcal{A} \Rightarrow A \setminus B = A - B = A \cap B^c \in \mathcal{A}$

(ii) $\{A_i\}_{i \in I,\ \text{countable}}$, $A_i \in \mathcal{A} \Rightarrow \bigcap_{i \in I} A_i \in \mathcal{A}$

- Note: For a given Ω, can have many σ-fields.
- If $\mathcal{A}$ is a σ-algebra $\Rightarrow$ $\mathcal{A}$ is an algebra
- When Ω is uncountable, we cannot pick all $2^{|\Omega|}$ subsets as interesting, when creating $\mathcal{A}$.

  ↓

  We need to pick $\mathcal{A} < 2^{|\Omega|}$

  ↓

  (We need to throw something out of our power set.)

Example : $\Omega = [0, \infty)$

(i) C_1 = set of all intervals of form $[0, a)$, $a > 0$

C_1 is not a field.

$(\mathcal{F}_1)$ is satisfied since non empty.

$(\mathcal{F}_2)$ is not satisfied. : $[0, a)^c = [a, \infty) \notin C_1$

$(\mathcal{F}_3)$ is satisfied : $[0, a) \cup [0, b) = [0, \max(a, b)) \in C_1$

(ii) C_2 = set of all unions of a finite number of intervals of the form $[a, b)$ $0 \leq a < b \leq \infty$

Field

An element of C_2 looks like $\bigcup_{i=1}^{n} [a_i, b_i)$

$(\mathcal{F}_1)$ is satisfied since nonempty.

$(\mathcal{F}_2)$ is satisfied $\left(\bigcup_{i=1}^{n} [a_i, b_i)\right)^c = \bigcap_{i=1}^{n} [a_i, b_i)^c$

verify

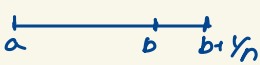
= a finite union of intervals choosen at left endpoint and open at right endpoint.

$(\mathcal{F}_3)$ ✓

Sigma Field

C_2 is not a sigma field

$\bigcap_{n=1}^{\infty} [a, b + 1/n) = [a, b] \notin C_2$

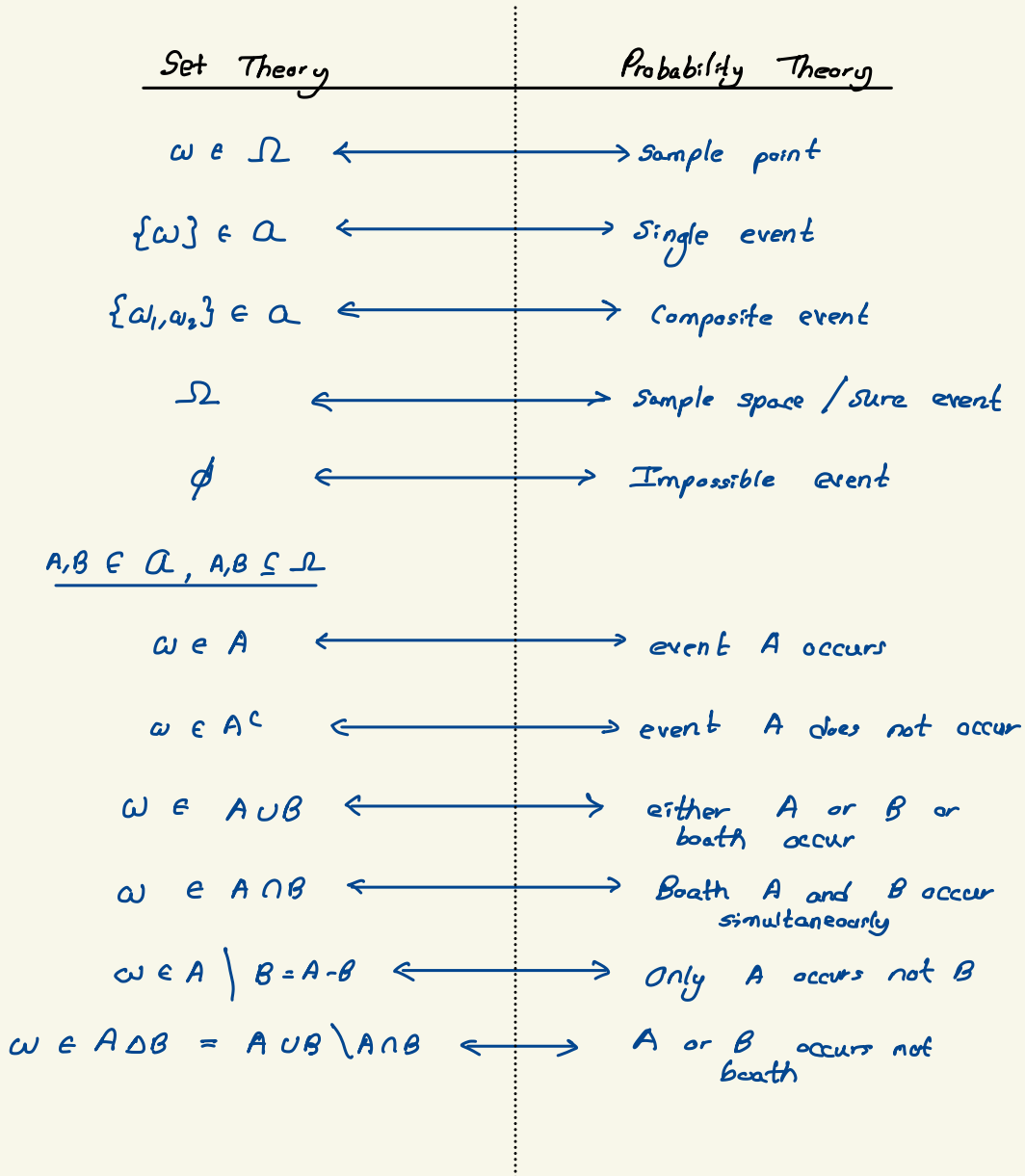


(iii) $C = \{\phi, [0, \infty)\}$ is boath a field and a σ-field.

Ω = set of outcomes = sample space.

Only one outcome is observed.

A Dictionary



4) Probability Measure (P)

Def : A probability measure P on a σ-field $\mathcal{A}$ on Ω is a mapping. (set function)

$$P : \mathcal{A} \longmapsto [0, 1]$$

s.t following axioms hold.

(P_1) $P(A) \geq 0 \quad A \in \mathcal{A}$

(P_2) P is **normed** ie $P(\Omega) = 1$

(P_3) P is **additive** ie $A, B \in \mathcal{A}$, $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$

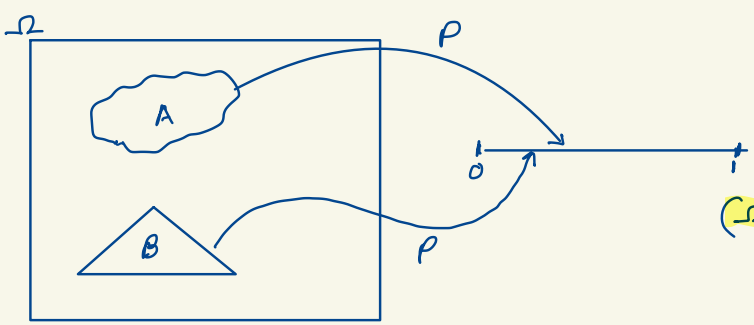
(P_3) $\Leftarrow \not\Rightarrow$ (P_4)

(P_4) P is **σ-additive** ie I is countable index set $\{A_i, i \in I\}$ **pairwise disjoint** sets in $\mathcal{A}$, ie. $\forall\ i \neq j \in I \quad A_i \cap A_j = \emptyset$ then

$$P\left(\bigcup_{i \in I} A_i\right) = \sum_{i \in I} P(A_i)$$

This summation does not oscilates since $P(A_i) > 0 \Rightarrow \sum_{i \in I} P(A_i)$ is monotonically increasing

$\therefore$ It is well defined



$(\Omega, \mathcal{A})$ = measurable space

To sum up

- $\mathcal{E}$ = expt
- $(\Omega, \mathcal{A}, P)$: **probability space** or probability triplet
- Elements of $\mathcal{A}$, (which are also called measurable sets) are events.

∴ Subsets of Ω in $\mathcal{A}$ are the only events. We had to do this without doing it to 2^Ω power set is things go wrong when Ω uncountable.

If Ω is countable $\mathcal{A}$ can be $2^{|\Omega|}$ power set. But $\mathcal{A}$ being uncountable is fine. ? why*

Properties of a probability measure

P on σ-field, $\mathcal{A}$ on Ω

(i) $P(\phi) = 0, \quad P(\Omega) = 1$

(ii) $P(A^c) = 1 - P(A) \quad A \in \mathcal{A}$

Special case of (iii)

Countable Additive $\Rightarrow$ Finite Additive

∴ Fill the n events by adding null events at the end

(iii) $\{A_i = i = 1, \ldots, n\}$ pairwise disjoint on $\mathcal{A}$ then

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i)$$

Montonicity

$B = A \cup B \setminus A$

$P(B) = P(A) + P(B \setminus A)$

$B \setminus A = B \cap A^c \in \mathcal{A}$

(iv) $A, B \in \mathcal{A}, \quad A \subseteq B$ then $P(A) \leq P(B)$.

(v) $A, B \in \mathcal{A}, \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B)$

(vi) $\{A_n\}_{n=1}^{\infty} \in \mathcal{A}, \quad P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{n} P(A_i)$

$P(A \cup B) \leq P(A) + P(B)$.

Union bound

Converse of union bound ie if $P(\cup A_i) = \sum P(A_i)$ does not mean A_i's are disjoint. It means intersections have '0' prob.

(vii) Continuity of Probability Measure - If $A_1, A_2, \ldots \in \mathcal{A}$ then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{m \to \infty} P\left(\bigcup_{i=1}^{m} A_i\right)$$

∴ Pairwise disjoint is not given here

Assume $P_M = P\left(\bigcup_{i=1}^{m} A_i\right)$

$A_M = \bigcup_{i=1}^{m} A_i$

A_m's are monotonically increasing ∴ This union exists.

(viii) If $A_1, A_2 \ldots \in \mathcal{A}$ are nested increasing events ie, $A_1 \subseteq A_2 \subseteq \ldots$ then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{m \to \infty} P(A_m)$$

(ix) If $A_1, A_2 \ldots \in \mathcal{A}$ are nested increasing events ie, $A_1 \supseteq A_2 \supseteq \ldots$ then

$$P\left(\bigcap_{i=1}^{\infty} A_i\right) = \lim_{m \to \infty} P(A_m)$$

Properties of $\mathcal{F}$ Show if $\mathcal{F}$ is a field on Ω, then

(i) $A, B \in \mathcal{F} \Rightarrow A \cap B \in \mathcal{F}$

(ii) $\Omega \in \mathcal{F}$ — Ω is a sure event

(iii) $\emptyset \in \mathcal{F}$ — $\emptyset$ is an impossible event

(iv) $A, B \in \mathcal{F} \Rightarrow A \setminus B \triangleq A - B \triangleq A \cap B^c \in \mathcal{F}$

(v) finite no. of operations, eg:-

$$\{A_i\}_{i=1}^{n}, \; A_i \in \mathcal{F} \Rightarrow \bigcup_{i=1}^{n} A_i \in \mathcal{F}$$

In order to consider sequences of events and their possible convergence, we need to allow countably infinitely many set operations on events to yield new events.

finite union is not enough

σ-field a can only be finite or uncountable

5) σ-Algebra / σ-Field (a)

Definition - A σ-field a on Ω (or Boolean σ-algebra, or simply σ-algebra) is a field on Ω ∋

$(\mathcal{F}_3)$ is replaced by

$$(\mathcal{F}_3)^* \quad \{A_i\}_{i=1}^{\infty}, \; A_i \in a \Rightarrow \bigcup_{i=1}^{\infty} A_i \in a$$

Remark: a is closed under all countable set operations, specifically, complements and complement unions.

a can have infinite unions, but it is countable union

Measure

- Countable Additivity Axiom

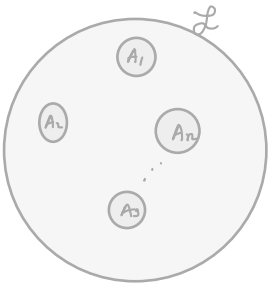
Assigning values to subsets of Ω, that are put into 'interesting' catagory in an Algebra.

A measure is a function $\mu : \mathcal{A} \longrightarrow [0, \infty]$ such that

(i) $\mu(\emptyset) = 0$

(ii) If $A_1, \ldots$ is a countable collection of disjoint $\mathcal{L}$-measurable sets, then

$$\mu\left(\underbrace{\bigcup_{i=1}^{\infty} A_i}_{\text{This is always in } \mathcal{A} \text{ from Closeness}}\right) = \sum_{i=1}^{\infty} \underbrace{\mu(A_i)}_{\substack{\text{Since } \mu(A_i) \text{ is positive} \\ \text{is monotonically increasing.} \\ \text{not oscilate.}}} \quad \therefore \text{does } \sum_{i=1}^{n} \mu(A_i)$$



- $(\Omega, \mathcal{A}, \mu)$ - Measurable Space

- Since $\emptyset \in \mathcal{A} \Rightarrow \emptyset^c = \Omega \in \mathcal{A} \Rightarrow \mu(\Omega)$ is well defined

$\mu(\Omega) < \infty$ - finite measure

$\mu(\Omega) = \infty$ - infinite measure

$\mu(\Omega) = 1$ - probability measure

Computation of Probabilities

Ex: Cointose experiment $\Omega = \{H, T\}$

$\mathcal{A} = \{\emptyset, \{H,T\}, \{H\}, \{T\}\}$

Suppose that $P(\{H\}) = p \quad 0 < p < 1$. Then

$P(\{H,T\}) = 1$ - Axiom (P_2)

$P(\{T\}) = 1-p$, $P(\emptyset) = 0$ } Axiom (P_3)

Important Idea - Can use axioms of probabilities to assign points to some events, and compute the events from axioms

"extension"

Proof of continuity of probability measure :-

▶ Since A_i's are not disjoint, let's create a sequence of B_i's but disjoint with the same union.

- **Property 5:-** If $\{A_i, i \geq 1\}$ are events, then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{m\to\infty} \mathbb{P}\left(\bigcup_{i=1}^{m} A_i\right). \tag{5.5}$$

This result is known as *continuity of probability measures.*

Proof:- Define a new family of sets $B_1 = A_1$, $B_2 = A_2 \setminus A_1$, ..., $B_n = A_n \setminus \bigcup_{i=1}^{n-1} A_i$, ...

Then, the following claims are placed:

Claim 1:- $B_i \cap B_j = \emptyset$, $\forall i \neq j$.

Claim 2:- $\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} B_i$.

Since $\{B_i, i \geq 1\}$ is a disjoint sequence of events, and using the above claims, we get

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} B_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(B_i).$$

Therefore,

We can't do this with Unions. But luckily we have a sumation

$$\begin{aligned}
\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) &= \sum_{i=1}^{\infty} \mathbb{P}(B_i) \\
&\overset{(a)}{=} \lim_{m\to\infty} \sum_{i=1}^{m} \mathbb{P}(B_i) \\
&\overset{(b)}{=} \lim_{m\to\infty} \mathbb{P}\left(\bigcup_{i=1}^{m} B_i\right) \\
&\overset{(c)}{=} \lim_{m\to\infty} \mathbb{P}\left(\bigcup_{i=1}^{m} A_i\right).
\end{aligned}$$

Here, (a) follows from the definition of an infinite series, (b) follows from *Claim 1* in conjunction with *Countable Additivity* axiom of probability measure and (c) follows from the intermediate result required to prove *Claim 2.*

Hence proved.

Proof of Union Bound Property

- **Property 8:-** Suppose $\{A_i, i \geq 1\}$ are events, then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i). \tag{5.8}$$

This result is known as the *Union Bound.* This is bound is trivial if $\sum_{i=1}^{\infty} \mathbb{P}(A_i) \geq 1$ since the LHS of (5.8) is a probability of some event. This is a very widely used bound, and has several applications. For instance, the union bound is used in the probability of error analysis in Digital Communications for complicated modulation schemes.

Proof:- Define a new family of sets $B_1 = A_1$, $B_2 = A_2 \setminus A_1$, ..., $B_n = A_n \setminus \bigcup_{i=1}^{n-1} A_i$, ...

Claim 1:- $B_i \cap B_j = \emptyset$, $\forall i \neq j$.

Claim 2:- $\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} B_i$.

Since $\{B_i, i \geq 1\}$ is a disjoint sequence of events, and using the above claims, we get

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} B_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(B_i).$$

Also, since $B_i \subseteq A_i$ $\forall i \geq 1$, $\mathbb{P}(B_i) \leq \mathbb{P}(A_i)$ $\forall i \geq 1$ (using Property 2). Therefore, the finite sum of probabilities follow

$$\sum_{i=1}^{n} \mathbb{P}(B_i) \leq \sum_{i=1}^{n} \mathbb{P}(A_i).$$

Eventually, in the limit, the following holds:

$$\sum_{i=1}^{\infty} \mathbb{P}(B_i) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

Finally we arrive at the result,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

Important Idea - Can we use the axioms of probability to assign proofs to some events and compute the event from axioms.

6) $\mathcal{B}(\mathcal{S})$ - Borel Sets

- Start with a collection of event $\mathcal{S}$ ($\subseteq \Omega$)
  $\mathcal{S}$ is closed under $\cap$. (ie. $A, B \in \mathcal{S} \Rightarrow A \cap B \in \mathcal{S}$). So $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ exist since $A \cap B \in \mathcal{S}$

  If we know $P(A)$ $\forall A \in \mathcal{S}$, then apply,
  - i) $P(A^c) = 1 - P(A)$
  - ii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

  repeatedly to determine the probability of every event that is obtained from events $\in \mathcal{S}$ by a finite series of unions, intersections and complementations. Denote the collection by $\mathcal{B}(\mathcal{S})$.

Example :- $\Omega = [0,1]$

$\mathcal{S}$ = collection of all semi open intervals of form $[a,b)$
$0 \leq a \leq b \leq 1$

Let $[a,a) = \phi$. Let P be defined by $P([a,b)) = b - a$

Verify,
- i) P is a valid probability measurement.
- ii) $\mathcal{S}$ is closed under $\cap$
- iii) $\mathcal{B}(\mathcal{S})$ = (collection of all sets of form $A = \hat{\bigcup}_{i=1}^{n} [a_i, b_i)$, $n \in \mathbb{N}$ and $[a_i, b_i) \cap [a_j, b_j) = \phi$ $i \neq j$)

Ans:- Then $P(A) = P\left(\bigcup_{i=1}^{n} [a_i, b_i)\right)$

$= \sum_{i=1}^{n} P([a_i, b_i))$ pairwise disjoint

$= \sum_{i=1}^{n} (b_i - a_i)$

Observe: (i) $\mathcal{S} \longrightarrow \mathcal{A}(\mathcal{S})$ needed only $P(A^c) = 1 - P(A)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

(ii) Can go further using Axioms P4 (σ-additivity)

$\Downarrow$

Need a concept of convergence of a sequence of sets.

7) Concept of Convergence of a Sequence

For a general sequence $\{A_k\}_{k=1}^{\infty}$, $A_k \subseteq \Omega$, set

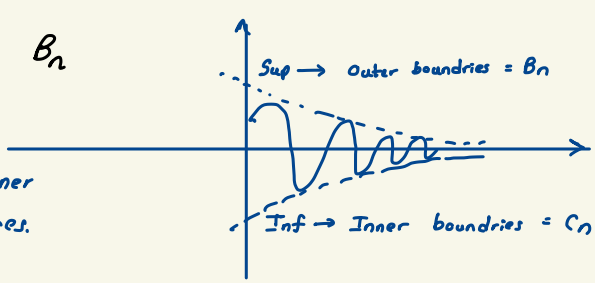
$B_n \triangleq \bigcup_{k \geq n} A_k \qquad C_n \triangleq \bigcap_{k \geq n} A_k$

clearly $\{B_n\} \downarrow$ decreasing ie $B_{n+1} \subseteq B_n \ \forall n$

$\{C_n\} \uparrow$ increasing ie $C_{n+1} \supseteq C_n$

Definition The sequence of sets $\{A_k\}_{k=1}^{\infty}$ $A_k \subseteq \Omega$, converges if

$$\lim_n C_n = \lim_n B_n$$



although set itself is fluctuating, if outer and inner boundries converg, set converges.

ie $\bigcup_{i=1}^{\infty} C_n = \bigcap_{n=1}^{\infty} B_n$

Mid 1 → Q1

ie $\bigcup_{n=1}^{\infty} \bigcap_{K \geq n} A_K = \bigcap_{n=1}^{\infty} \bigcup_{K \geq n} A_K$

ie $\liminf_{n \to \infty} A_n = \limsup_{n \to \infty} A_n$

$\underline{A}$ $\bar{A}$

If $\underline{A} = \bar{A}$ and we take the common limit to $\lim_{n \to \infty} A_n$.

Contained Sets

Consider a sequence of events (sets) $\{A_K\}_{K=1}^{\infty}$ $A_K \subseteq \Omega$

If $(A_K \in a)$. Suppose $\{A_K\}_{K=1}^{\infty}$ is an increasing on k. ie $A_K \subseteq A_{K+1}$. $\forall K$

Then $A_n = \bigcup_{K=1}^{n} A_K$ and so makes sense to say

$\lim_{n \to \infty} A_n = \bigcup_{K=1}^{\infty} A_K$

If $(A_K \in a)$ suppose $\{A_K\}_{K=1}^{\infty}$ is dicreasing on k. ie $A_{K+1} \subseteq A_K$ $\forall K$.

Then $\lim_{n \to \infty} A_n = \bigcap_{K=1}^{\infty} A_K$

Let $\{A_n\}_{n=1}^{\infty}$ be a sequance of events in a. sutch that $\{A_n\} \uparrow$ or $\{A_n\} \downarrow$. Then

$$P\left(\lim_{n \to \infty} A_n\right) = \lim_{n \to \infty} P(A_n)$$

Proof:

▸ First suppose $\{A_n\}$ is $\uparrow$ in Ω.

$$\lim_n A_n = \bigcup_{k=1}^{\infty} A_k \quad \cdots \quad \because A_n = \bigcup_{k=1}^{n} A_k$$

Note $\bigcup_{k=1}^{\infty} A_k = A_1 \cup (A_1^c \cap A_2) \cup (\underbrace{A_1^c \cap A_2^c}_{A_2^c} \cap A_3) \cup \ldots$

$$= A_1 \cup (A_2 - A_1) \cup (A_3 - A_2) \cup \ldots\ldots$$

Hence $P\left(\lim_{n\to\infty} A_n\right) = P\left(\bigcup_{k=1}^{\infty} A_k\right)$

$$= P\left(A_1 \cup (A_2 - A_1) \cup (A_3 - A_2) \cdots\cdots\right)$$

The sets here are pairwise disjoint. $\because$ from σ additivity in axiom 4

$$= P(A_1) + P(A_2 - A_1) + P(A_3 - A_2) + \cdots\cdots$$

Since $\{A_n\} \uparrow$, A_{n+1} is contoine in A_n. hence $P(A_{n+1} - A_n) = P(A_{n+1}) - P(A_n)$

$$= P(A_1) + P(A_2) - P(A_1) + P(A_3) - P(A_2)$$

$$= \lim_n P(A_n)$$

▸ Next suppose $\{A_n\}$ is $\downarrow$ in Ω.

Then $\{A_n^c\}$ is $\uparrow$. Hence

$$\lim_n A_n^c = \bigcup_{k=1}^{\infty} A_k^c \quad \cdots \quad \because A_n^c = \bigcup_{k=1}^{n} A_k^c$$

$$\Rightarrow P\left(\lim_{n\to\infty} A_n^c\right) = P\left(\bigcup_{k=1}^{\infty} A_k^c\right) = P\left(\bigcap_{k=1}^{\infty} A_k\right)^c$$

$$= \lim_{n\to\infty} P(A_n^c) \quad \text{from the previous part.}$$

$$\Rightarrow P\left(\bigcup_{k=1}^{\infty} A_k\right)^c = P\left(\bigcup_{k=1}^{\infty} A_k^c\right) = \lim_{n\to\infty} P(A_n^c)$$

$$\Rightarrow 1 - P\left(\bigcup_{k=1}^{\infty} A_k\right) = \lim_{n\to\infty} (1 - P(A_n))$$

$$\Rightarrow P\left(\bigcup_{k=1}^{\infty} A_k\right) = \lim_{n\to\infty} P(A_n)$$

$\|$ from previous part.

$$\Rightarrow P\left(\lim_{n\to\infty} A_n\right) = \lim_{n\to\infty} P(A_n)$$

Observation: ▸ By the previous Theorem under axioms $(P_1), (P_2), (P_4)$

σ additivity $\Longrightarrow$ Sequential Continuity.

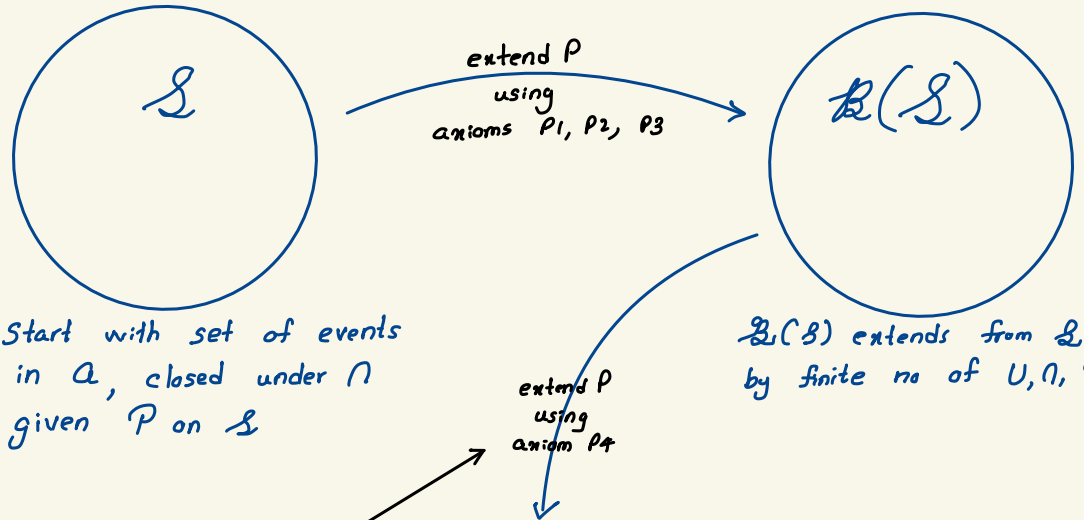
▸ Under axioms $(P_1), (P_2), (P_3)$

Sequential continuity $\Longrightarrow$ σ additivity.

Theorem :

σ additivity $\Longleftrightarrow$ Sequential Continuity.

Summary



Start with set of events in $\mathcal{A}$, closed under $\cap$ given $\mathcal{P}$ on $\mathcal{S}$

$\mathcal{B}(\mathcal{S})$ extends from $\mathcal{S}$ by finite no of $\cup, \cap, ^c$

Start with $\mathcal{B}(\mathcal{S})$, can compute $P(A)$ for any A that is the limit of sequence of events $\{A_n\}$ in $\mathcal{B}(\mathcal{S})$ by setting

$$P(A) = \lim_{n \to \infty} P(A_n)$$

$\mathcal{A}(\mathcal{S})$ obtained from $\mathcal{S}$ by using, $\cup, \cap, ^c$ and Sequential limits

$\cup$ - unions
$\cap$ - Intersections
^{c} - Complements

Sequential Continuity Theorem

Sequencial Continuity: Let $\{A_n\}_{n=1}^{\infty}$ be convergent sequence of events, not necessarily monotone. Then

$$P\left(\lim_{n\to\infty} A_n\right) = \lim_{n} P(A_n)$$

Proof: Set $B_n = \bigcup_{k\geq n} A_k \qquad C_n = \bigcap_{k\geq n} A_k$

Then $C_n \subseteq A_n \subseteq B_n$

And $\{B_n\}_{n=1}^{\infty}$ and $\{C_n\}_{n=1}^{\infty}$ are monotone. and B_n is $\downarrow$ in n and C_n is $\uparrow$ in n.

Hence,

$$P\left(\lim_{n\to\infty} B_n\right) = \lim_{n\to\infty} P(B_n) \geq \lim_{n\to\infty} P(A_n) \quad \cdots\cdots * \quad \cdots \text{by part I above.}$$

$$P\left(\lim_{n\to\infty} C_n\right) = \lim_{n\to\infty} P(C_n) \leq \lim_{n\to\infty} P(A_n) \quad \cdots\cdots **$$

By definition

$$\lim_{n\to\infty} A_n = \lim_{n\to\infty} B_n = \lim_{n\to\infty} C_n \quad \cdots\cdots ***$$

We get

$$P\left(\lim_{n\to\infty} A_n\right) \overset{***}{=} P\left(\lim_{n\to\infty} B_n\right) \overset{*}{\geq} \lim_{n\to\infty} P(A_n) \quad \cdots\cdots (1)$$

$$\lim_{n\to\infty} P(A_n) \overset{**}{\geq} P\left(\lim_{n\to\infty} C_n\right) \overset{***}{=} P\left(\lim_{n\to\infty} A_n\right) \quad \cdots (2)$$

From (1) and (2) and Sandwich logic,

$$P\left(\lim_{n\to\infty} A_n\right) = \lim_{n\to\infty} P(A_n)$$

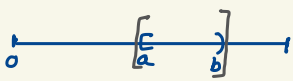
▸ What are the benefits of sequential continuity?

Ex: $\Omega = [0,1)$, $\mathcal{S} = \{[a,b);\ 0 \le a \le b \le 1\}$. verify that $\mathcal{S}$ is closed under $\cap$.

P is specified by: $P([a,b)) = b-a$

$P([a,a)) = 0$

Let's try to convert closed interval to open.



Note $[a,b] = \bigcap_{n:\, b+1/n \le 1} [a, b+1/n)$ $b<1$

$$\Rightarrow P[a,b] = P\left(\lim_{n\to\infty} [a, b+1/n)\right)$$

$$= \lim_{n\to\infty} P\left([a, b+\tfrac{1}{n})\right) \quad \text{by sequential continuity}$$

$$= \lim_{n\to\infty} \left(b+\tfrac{1}{n}\right) - a$$

$$= b-a$$

Hence $P([a,b]) = b-a$ given $P([b,a)) = b-a$

Since two sets only differ at closing point b, $[a,b] = [a,b) \cup \{b\}$

$$\Rightarrow P(\{b\}) = 0$$

By using the limit ideas above, every sub-intervals of $[0,1)$, closed or open at either end is in $\mathcal{A}(\mathcal{S})$; also

$P(\text{subinterval})$ = length of subinterval.

Ex: Experiment with an infinite no. of random bits at the output of a digital communication system.

Ω = set of all infinite sequences of 0's and 1's.

ω event has form: $\omega = (\omega_1, \omega_2, \ldots)$ $\omega_j = 0$ or 1

Suppose we wish: $\mathcal{A}$ = set of all subsets of Ω

Can start with $\mathcal{B}$ consisting of every set of ωs that are specified by fixing a finite no. of ω_js.

Ex:- All ωs such that $\omega_2 = 0$, $\omega_{37} = 1$, $\omega_{106} = 1$

One way of defining $\mathcal{B}$:

set $\Rightarrow A_i = \{\omega : \omega_i = 1\}$ $i = 1, 2, \ldots$

$\Rightarrow A_i^c = \{\omega : \omega_i = 0\}$

Thus every event in $\mathcal{B}$ is the $\cap$ of a finite # of A_i's and A_i^cs. For instance,

$\{\omega : \omega_2 = 0, \omega_{37} = 1, \omega_{106} = 1\} = A_2^c \cap A_{37} \cap A_{106}$

Conditional Probability and Independance

given $(\Omega, a P)$

Definition - Let $A \in \mathcal{A}$ with $P(A) > 0$. The conditional probability given A, $P(\cdot|A)$ defined on $\mathcal{A}$, and into $[0,1]$ ie

$$P(\cdot|A) : \mathcal{A} \longmapsto [0,1]$$

as follows

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \qquad B \in \mathcal{A}$$

$$= \text{conditional probability of } B \text{ given } A.$$

Remark - The set function $P(\cdot|A)$ is a probability measure on $\mathcal{A}$

Proof: (P_1) Non negativity $\longrightarrow$ $P(B|A) = \frac{P(A \cap B)}{P(A)} \geq 0 \quad \forall\, B \in \mathcal{A}$

(P_2) Normed $\longrightarrow$ $P(\Omega|A) = \frac{P(A \cap \Omega)}{P(A)} = \frac{P(A)}{P(A)} = 1$

$((P_4) \Rightarrow (P_3))$ σ additivity $\longrightarrow$ If $B_j \in \mathcal{A}$ $j = 1, 2, \ldots$ and $B_i \cap B_j = \emptyset$ $\forall\, i, j$ then

$$P\left(\bigcup_{j=1}^{\infty} B_j \,|\, A\right) = \frac{P\left(A \cap \left(\bigcup_{j=1}^{\infty} B_j\right)\right)}{P(A)}$$

$$= \frac{P\left(\bigcup_{j=1}^{\infty} (A \cap B_j)\right)}{P(A)}$$

$$= \frac{1}{P(A)} \cdot \sum_{j=1}^{\infty} P(A \cap B_j) \qquad \text{from } \sigma \text{ additivity}$$

$$= \sum_{j=1}^{\infty} \frac{P(A \cap B_j)}{P(A)}$$

$$= \sum_{j=1}^{\infty} P(B_j|A)$$

Thus all the axioms of P are satisfied by $P(\cdot|A)$.
$\hookrightarrow$ $P(\cdot|A)$ is a probability measure on $\mathcal{A}$.

■

Definition: Event A and B in $\mathcal{A}$ are (statistically / stochastically / in the probability sense) independant, denote $A \perp\!\!\!\perp B$, if

$$P(A \cap B) = P(A) \cdot P(B).$$

definition works even if one or boath of $P(A), P(B) = 0$

Observation: If $A, B \in \mathcal{A}$ with $P(A) > 0$ then

$$A \perp\!\!\!\perp B \implies P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A) \cdot P(B)}{P(A)} = P(B)$$

Similarly if $P(B) > 0$ then

$$A \perp\!\!\!\perp B \implies P(A|B) = P(A)$$

Definition: A finite collection of events A_j $j = 1, 2, \ldots n$ are mutually (or completely) independant if the following conditions hold for every $k = 2, 3, \ldots, n$ and $j_1, j_2, \ldots, j_k$ in $\{1, 2, \ldots, n\}$ with $1 \leq j_1 < j_2 \ldots < j_k \leq n$,

$$P(A_{j_1} \cap A_{j_2} \cap \ldots \cap A_{j_k}) = P(A_{j_1}) \cdot \ldots \cdot P(A_{j_k})$$

ie. for every finite subcollection $\{A_{j_1}, \ldots, A_{j_k}\}$ also the properly

$$P\left(\bigcap_{i=1}^{k} A_{j_i}\right) = \prod_{i=1}^{k} P(A_{j_i}) \longrightarrow 2^n - n - 1 \text{ conditions should be checked.}$$

Definition: Events A_j, $j = 1, 2, \ldots, n$ are pairwise independant if

$$P(A_i \cap A_j) = P(A_i) \cdot P(A_j) \quad \forall\, i \neq j \longrightarrow \binom{n}{2} = \frac{n(n-1)}{2}$$

conditions should be checked.

Note: Mutually $\perp\!\!\!\perp$ events $\implies$ Pairwise $\perp\!\!\!\perp$ events

$\not\Longleftarrow$

Ex: For a mutually independant events $n=3$ A_1, A_2, A_3 in $\mathcal{A}$

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) \cdot P(A_2) \cdot P(A_3)$$
$$P(A_1 \cap A_2) = P(A_1) \cdot P(A_2)$$
$$P(A_2 \cap A_3) = P(A_2) \cdot P(A_3)$$
$$P(A_3 \cap A_1) = P(A_3) \cdot P(A_1)$$

Definition An arbitary collection of events in $\mathcal{A}$, countable or uncountable, are mutually independant if every finite subcollection are mutually independant.

Ex: By S.M. Bernstein to show pairwise $\perp\!\!\!\perp$ $\not\Rightarrow$ mutual $\perp\!\!\!\perp$

Answer: $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$. Assume $P(\{\omega_i\}) = 0.25$. $i=1,2,3,4$
Assume $A = \{\omega_1, \omega_2\}$, $B = \{\omega_1, \omega_3\}$, $C = \{\omega_1, \omega_4\}$

$$\therefore P(A) = P(B) = P(C) = 0.5$$

$A \cap B = B \cap C = C \cap A = \{\omega_1\} \Rightarrow P(A \cap B) = P(B \cap C) = P(C \cap A) = 0.25 = (0.5)^2$

$\Rightarrow$ events A, B, C are pairwise $\perp\!\!\!\perp$

$P(A \cap B \cap C) = P(\{\omega_1\}) = 0.25 \neq (0.5)^3 \Rightarrow A, B, C$ are not mutually $\perp\!\!\!\perp$ ∎

Conditional Independance

Given $(\Omega, \mathcal{A}, P)$

Definition: A, B, C are events in $\mathcal{A}$ on Ω. $P(C) \neq 0$. A and B are conditionally independant given C, denote $A \perp\!\!\!\perp B \mid C$, if

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C)$$

Observations: Verify $A \perp\!\!\!\perp B \mid C$ ie $P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C)$

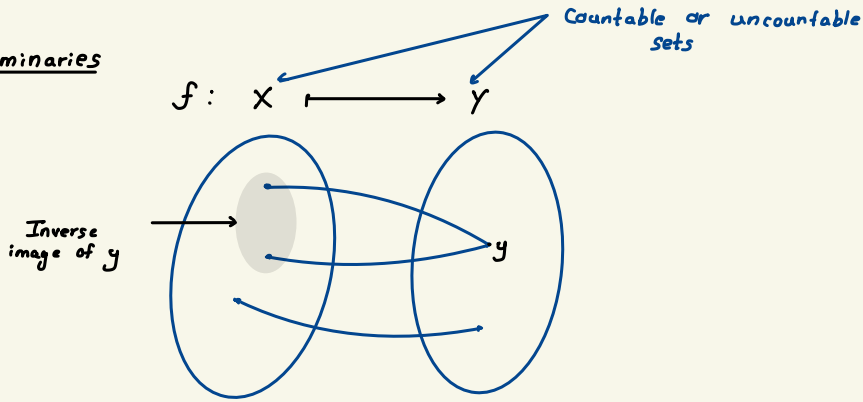
$$P(A \mid B \cap C) \triangleq P(A \mid B, C) = P(A \mid C) \quad \text{if } P(B \cap C) \neq 0$$
$$P(B \mid A \cap C) \triangleq P(B \mid A, C) = P(B \mid C) \quad \text{if } P(A \cap C) \neq 0$$

Remark - Can extend conditional independance as for independance in general, to mutual conditional independdat of arbitary no. of events, countable or uncountable given a conditioning event.

Note: $A \perp\!\!\!\perp B \mid C \not\Longrightarrow A \perp\!\!\!\perp B$ and $A \perp\!\!\!\perp B \not\Longleftarrow$

Random Variables

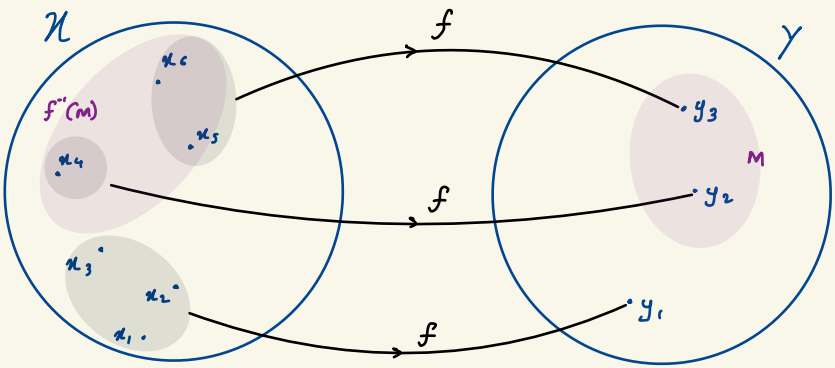
Preliminaries



- $\forall x \in X$ is mapped to some $y \in Y$
- $\forall x \in X$, $\exists!\ y \in Y$ such that $f(x) = y$
- multiple x's can be mapped to same y.
- $\forall y \in Y$, there corresponds a set x_s in X such that $f(x) = y$, called the inverse image of $y \in Y$
- If a $y \in Y$ is not in the range of f, the inverse image of y is ϕ.

Note: For $y \in Y$ $\quad f^{-1}(y) = \{ x \in X : f(x) = y \}$

$=$ inverse image of y in X.



In the example $f^{-1}(y_1) = \{x_1, x_2, x_3\}$

$f^{-1}(y_2) = \{x_4\}$

$f^{-1}(y_3) = \{x_5, x_6\}$

Definition: Similarly, to each y-set M $(\subseteq Y)$, there corresponds an x-set $f^{-1}(M)$ $(\subseteq X)$, termed the inverse image of M, consisting of all the x's in X that are mapped into $M \subseteq Y$ by f ie,

$$f^{-1}(M) = \{ x \in X : f(x) \in M \}$$

Ex: If $M = \{y_2, y_3\}$ then $f^{-1}(M) = \{x_4, x_5, x_6\}$

Returning to Conditional Probability, $\perp\!\!\!\perp$ etc

Exercise: $A \perp\!\!\!\perp B \iff A^c \perp\!\!\!\perp B \iff A^c \perp\!\!\!\perp B^c$ are equivalent

$A \perp\!\!\!\perp B^c$

Exercise: $A_1, A_2, \ldots, A_n$ are mutually $\perp\!\!\!\perp$ events in $\mathcal{A}$ on Ω. Suppose that $n_1 + n_2 + \ldots + n_k = n$ with $n_i \geq 1$ $\forall i$. Suppose: B_i (event in $\mathcal{A}$) is defined by Boolean operations (^{c}s, $\cup$s, $\cap$s) on the first n_i

[George Boole, 1815-1864]

events $A_1, A_2, \ldots, A_{n_1}$. B_2 is obtained similarly from $A_{n_1+1}, \ldots, A_{n_2}$ and so on. Then $B_1, B_2, \ldots, B_{n_K}$ are ||

$$\underbrace{A_1, A_2, \ldots, A_{n_1}}_{B_1}, \underbrace{A_{n_1+1}, \ldots, A_{n_2}}_{B_2}, \ldots\ldots, \underbrace{A_{n_{K-1}+1}, \ldots, A_{n_K}}_{B_{n_K}}$$

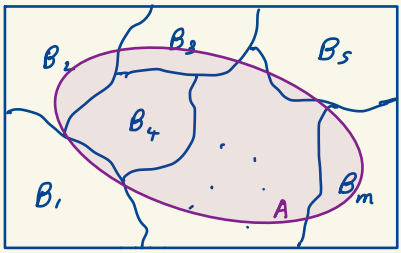
Important Ideas Emerging from Conditional Probability

Law of Total Probability

Theorem: $(\Omega, \mathcal{A}, P)$ Let A (effect) be an event in $\mathcal{A}$. Let the events $B_1, \ldots, B_m$ in $\mathcal{A}$ partition (or form a partition of) Ω ie.

$$\bigcup_{j=1}^{m} B_j = \Omega \quad \text{and} \quad B_i \cap B_j \neq \emptyset \ \forall\ i,j \ \text{(pairwise disjoint)}$$

$$P(A) = \sum_{j=1}^{m} P(A|B_j) \cdot P(B_j)$$



Proof: $\sum_{j=1}^{m} P(B_j) = 1$

Observe $P(A) = P[A \cap \Omega] = P\left[A \cap \left(\bigcup_{j=1}^{m} B_j\right)\right]$

$= P\left[\bigcup_{j=1}^{m} (A \cap B_j)\right]$ pairwise disjoint

$= \sum_{j=1}^{m} P(A \cap B_j)$

$= \sum_{j=1}^{m} P(A|B_j) \cdot P(B_j)$ if $P(B_j) \neq 0\ \forall\ j$

■

[Thomas Bayes, 1702 - 1761]

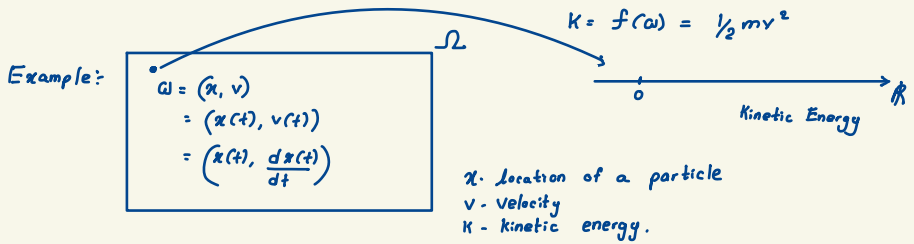
Baye's Rule

Using the definitions of Conditional Probability and the Law of Total Probability, leads to Baye's Rule.

Theorem : If $P(A) \neq 0$

$$P(B_j \mid A) = \frac{P(A \cap B_j)}{P(A)} = \frac{P(A \mid B_j) \cdot P(B_j)}{\sum_{j=1}^{m} P(A \mid B_j) \cdot P(B_j)}$$

Random Variables (RV)



Definition: Given $(\Omega, \mathcal{A}, P)$. A real valued function (mapping). $X: \Omega \to \mathbb{R}$ is a $\mathbb{R}$-real valued random variable (rv) if $\forall\, a \in \mathbb{R}$, it holds that $\{\omega \in \Omega : X(\omega) < a\}$ is an event in $\mathcal{A}$.

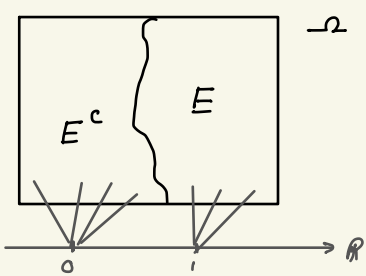
Shorthand: $X^{-1}(t<a) \in \mathcal{A}$.

Remark: A rv is a special case of a "measurable function"

Indicator Function

Definition: Let $E \in \mathcal{A}$ be an event (In general E can be any subset of Ω). The indicator function of E is

$$\mathbb{I}_E(\omega) = \begin{cases} 0, & \omega \notin E \\ 1, & \omega \in E \end{cases}$$



Claim: $\mathbb{I}_E$ is a RV

Clearly $\{\omega \in \Omega : \mathbb{I}_E(\omega) = 1\} = E$

$\{\omega \in \Omega : \mathbb{I}_E(\omega) = 0\} = E^c = \Omega \setminus E$

To see if $\mathbb{I}_E$ is a rv:

$$\{\omega : \mathbb{I}_E(\omega) < a\} = \begin{cases} \phi & a \leq 0 \\ E^c & 0 < a \leq 1 \\ \Omega & 1 < a \end{cases}$$

$\phi, E^c, \Omega \in \mathcal{A}$. (This can be obtained even without defining the sigma field since $E \in \mathcal{A}$.)

$\Downarrow$

Π_E is a rv

Remark: The requirement that $\{\omega : X(\omega) < a\}$ is an event in $\mathcal{A}$ $\forall\ a \in \mathbb{R}$ enables consideration of probabilities associated with X. Shall see later that all 'events' of interest in $\mathbb{R}$ can be generated countable operations on the collection of half lines $\{x \in \mathbb{R} : -\infty < x < a\}$, $a \in \mathbb{R}$. Thus for X to be a rv, it suffices to look at every set in Ω of form, $\{\omega : X(\omega) < a\}$ to be an event in $\mathcal{A}$.

Cumilative Distribution Function (CDF)

Definition: For a rv X, the probability of $\{\omega : X(\omega) < a\}$ defines a non negative valued $(\mathbb{R}^+ \cup \{0\})$ function

$$F_X(a), \quad -\infty < a < \infty \quad (\text{ie } a \in \mathbb{R})$$

is the probability distribution function (ditribution function / cumulitative distribution function / cdf) of X.

▸ Thus $F_X(a) = P(\{\omega : X(\omega) < a\})\ a \in \mathbb{R}$

$\leqslant$ - results in right continuity

$<$ - results in left continuity

Properties of $F_X(a)$

1) F_X is a non decreasing function with

$$F_X(-\infty) \triangleq \lim_{a \to -\infty} F_X(a) = 0$$

$$F_X(+\infty) \triangleq \lim_{a \to +\infty} F_X(a) = 1$$

2) F_X is left continuos.

$$\lim_{\varepsilon \to 0^-} F_X(a-\varepsilon) = F_X(a) \quad \forall\ a \in \mathbb{R}.$$

If $F_X(a) = P(\{\omega : X(\omega) \leqslant a\})$ $a \in \mathbb{R}$ then it will lead to right continuity.

Proofs :

(i) For $\varepsilon > 0$

$$F_X(a+\varepsilon) = P\left(\{\omega : X(\omega) < a+\varepsilon\}\right)$$

$$= P\left(\{\omega : X(\omega) < a\} \cup \{\omega : a \leqslant X(\omega) < a+\varepsilon\}\right)$$

$$= P(\{\omega : X(\omega) < a\}) + P\{\omega : a \leqslant X(\omega) < a+\varepsilon\}$$

$$\qquad\qquad = F_X(a) \qquad\qquad \geqslant 0$$

$$= F_X(a) + (\geqslant 0)$$

$$\geqslant F_X(a)$$

To Prove : $F_X(+\infty) = 1$

Since X takes values in $(-\infty, \infty)$, $F_X(+\infty)$ must be defined first.

Let $\{a_n\}_{n=1}^{\infty}$ be any non-decreasing sequence in $\mathbb{R}$ with $a_n \leqslant a_{n+1}$ $\forall n$ and $\lim_n a_n = +\infty$

$$\Rightarrow F_X(+\infty) \triangleq \lim_{n\to\infty} P\left(\{\omega : X(\omega) < a_n\}\right) \quad \cdots\cdots \left[\because \text{By assumption } (-\infty, a_n) \subseteq (-\infty, a_{n+1}) \ \forall n\right]$$

$$= P\left(\lim_{n\to\infty} \{\omega : X(\omega) < a_n\}\right) \quad \cdots\cdots \left[\because \text{Sequential continuity}\right]$$

$$= P\left(\{\omega : X(\omega) < \infty\}\right) \quad \cdots\cdots \lim_n a_n = \infty \ *$$

$$= P(\Omega)$$

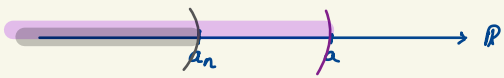
$$= 1$$

To Prove : $F_X(-\infty) = 0$

Exercise

(ii) Fix an arbitary $a \in \mathbb{R}$. Let $\{a_n\}_{n=1}^{\infty}$ be any non decreasing sequence in $\mathbb{R}$ with $a_n \leq a_{n+1}$ $\forall n$ and $\lim_{n\to\infty} a_n = a$

To prove: $\lim_n F_x(a_n) = F_x(a)$



By property ①

$$\Rightarrow \quad 0 \leq F_x(a) - F_x(a_n) = P\left(\{\omega : X(\omega) < a\} - P\{\omega : X(\omega) < a_n\}\right)$$

$$= P\left(\{\omega : a_n \leq X(\omega) \leq a\}\right) \quad \text{decreasing sequence of sets in } \mathcal{X}$$

$$\Rightarrow \lim_n \left(F_x(a) - F_x(a_n)\right) = \lim_n P\left(\{\omega : a_n \leq X(\omega) \leq a\}\right)$$

$$\Rightarrow \quad F_x(a) - \lim_n F_x(a_n) = P\left(\lim_n \{\omega : a_n \leq X(\omega) \leq a\}\right) \quad \text{sequential continuity}$$

$$= P\left(\bigcap_{n=1}^{\infty} \{\omega : \underset{a}{\lim_n a_n} \leq X(\omega) \leq a\}\right)$$

$$= P(\phi)$$

$$= 0$$

$$\Rightarrow \quad \lim_n F_x(a_n) = F_x(a) \in \mathbb{R}$$

■

Fact - All the probabilistic information casidering r.v X is contained in the cdf of F_x.

For us, F_x is usually of one of two forms

Probability Density Function (PDF)

I) X is descrete (with probability 1) ie.

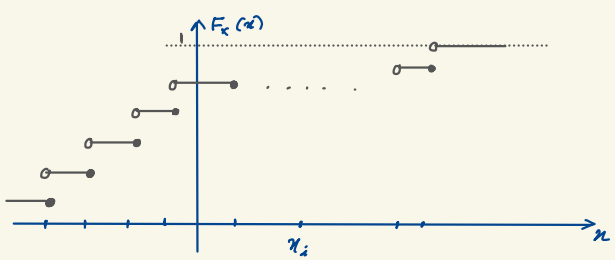
- $\exists$ countable subset S of $\mathbb{R}$ with $S = \{x_i, i \in I\}$. I - countable index set such that,

$$\sum_{x_j \in S} P\left(\{\omega \in \Omega : X(\omega) = x_i\}\right) = \sum_{x_j \in S} P_X(x_i) = 1$$

$(P_X(x_i), x_i \in S) \triangleq$ Probability mass function (pdf) of X

S - Set of all realizations of rv X.

- When X is descreate, F_X is continuos except for jumps at x_i, $i = 1, 2, \ldots$. $F_X(x)$ is termed Singular.



$$F_X(x_i) = P\left(\{\omega : X(\omega) < x_i\}\right) = \sum_{x_j < x_i} P_X(x_j)$$

The size of the jump at (any) x_i is $= P\left(\{\omega \in \Omega : X(\omega) = x_i\}\right)$

$$= F_X(x_i^+) - F_X(x_i).$$

Remark: S is countable. No. of jumps of F_X are at most Countably infinite.

II) X is Continuos

F_X is of form

$$F_X(a) = \int_{-\infty}^{a} f_X(x)\,dx$$

F_X is termed **absolute continuos** and $f_X : \mathbb{R} \to \mathbb{R}^+ \cup \{0\}$ is called the **probability density function** (pdf) of X.
If f_X is counted at a, then

$$f_X(a) = \frac{d}{da} F_X(a) = \frac{d}{dx} F_X(x)\Big|_{x=a}$$

The pdf f_X is non negative and $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$.

Interpretation

$f_X(x) \cong$ probability of event $\{\omega : x \leq X(\omega) < x+dx\}$

So that $f_X(x)$ is probability per unit interval around x, and hence "density". Precisely if $f_X(x)$ is continuos at $x \in \mathbb{R}$

$$f_X(x) = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \int_{x}^{x+\varepsilon} f_X(u)\,du$$

$$= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} P\left(\{\omega : x \leq X(\omega) < x+\varepsilon\}\right)$$

Example. Experiment E = toss a coin m times (independantly). Ω consist of $\omega = (\omega_1, \omega_2, \ldots, \omega_m)$ where $\omega_i = 0\ (T)$ $= 1\ (H)$

$P(\omega) = \frac{1}{2^m} \quad \forall \omega \in \Omega$

$X(\omega) \triangleq$ no. of 1's in ω. For $k = 0, 1, \ldots, m$

$$P\left(\{\omega : X(\omega) = k\}\right) = P_X(k) = \frac{1}{2^m}\binom{m}{k} = \frac{1}{2^m}\frac{m!}{k!(m-k)!}$$

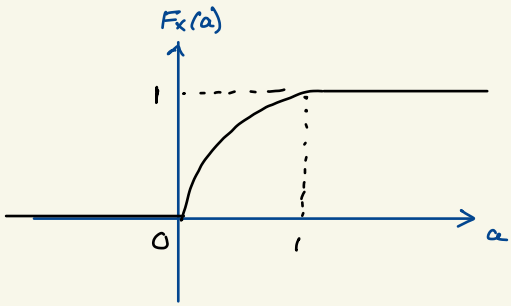
Example - $\Omega = [0,1)$. P(interval) = length of interval. $\mathcal{A}$ consistet of every sub interval of $[0,1)$, closed or open at either end.
Let $X(\omega) \triangleq \omega^2$, $0 \leq \omega < 1$. Verify X is a RV with values in $[0,1)$.

Similar to kinetic energy quetion

For $a \in [0,1)$.

$$F_X(a) = P\{\omega : X(\omega) = \omega^2 < a\} = P(\{\omega : \omega < \sqrt{a}\}) = \sqrt{a}$$

$$F_X(a) = \begin{cases} 0 & a < 0 \\ \sqrt{a}, & a \in [0,1) \\ 1, & a \in [1,\infty) \end{cases}$$



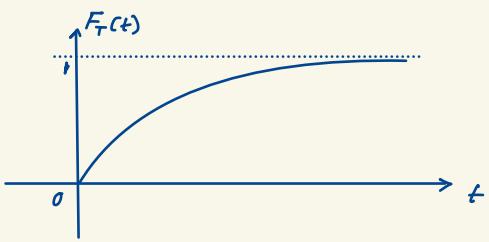
$\therefore F_X$ is not differentiable at $a = 0$

$$\therefore f_X(a) = \begin{cases} \frac{1}{2\sqrt{a}} & 0 < a < 1 \\ 0 & \text{otherwise} \end{cases}$$

Example Exponencial RV: T is a $\mathbb{R}^+ \cup \{0\}$ valued RV.

Exponetial pdf: $f_T(t) = \begin{cases} a \cdot e^{-at}, & t > 0 \\ 0, & \text{otherwise} \end{cases}$ $\quad a > 0$

cdf $F_T(t) = \int_{-\infty}^{t} f_T(u) \cdot du = \begin{cases} 0, & t \leq 0 \\ \int_0^t a \cdot e^{-au} du = 1 - e^{-at} & t > 0 \end{cases}$



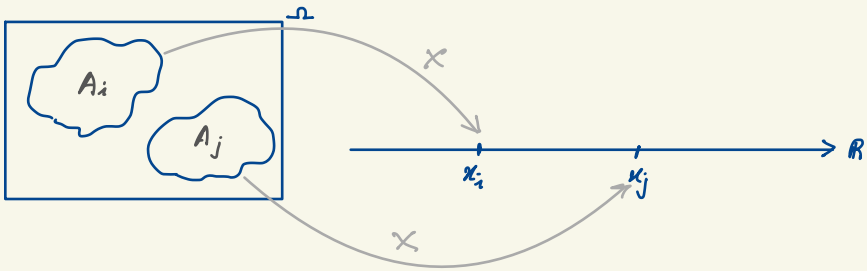
Expectation

Heuristic Interpretation : weighted average value of r.v with the weight equal (=) to probabilities.

Given $(\Omega, \mathcal{A}, P)$ Start with a simple r.v X, ie X takes only a finite number of values in $\mathbb{R}$, say, $x_1, x_2, \ldots, x_n$. Can write X as:

$$X = \sum_{i=1}^{n} x_i \mathbb{I}_{A_i} \longleftrightarrow \{\omega \in \Omega : X(\omega) = x_i\}$$

where $A_1, A_2, \ldots, A_n$ are pairwise disjoint events in Ω.



$$X(\omega) = \sum_{i=1}^{n} x_i \mathbb{I}_{A_i}(\omega)$$

Indicator Function

$$\mathbb{I}_E(\omega) = \begin{cases} 1 & \omega \in E \\ 0 & \omega \notin E \end{cases}$$

The expectation of (simple rv) X, denote $E[X]$ is

$$E[X] = \sum_{i=1}^{n} x_i \cdot P(X = x_i) \quad \cdots (*)$$

$$\begin{aligned} P(X = x_i) &= P_X(x_i) \\ &= P(\{\omega \in \Omega : X(\omega) = x_i\}) \\ &= P(\{\omega \in \Omega : \omega \in A_i\}) = P(A_i) \end{aligned}$$

$$E[X] = \sum_{i=1}^{n} x_i P(A_i)$$

A Few Properties of E in $*$

(I) E has a linearity property.

If X and Y are two simple rvs and a and b are arbitary constants in $\mathbb{R}$, then

$$E[aX+bY] = a\cdot E[X] + b\cdot E[Y]$$

Ex: show $aX+bY$ is a simple rv

Proof: $E[aX+bY] = \sum_{i=1}^{n}\sum_{j=1}^{m} (ax_i + by_j)\cdot P(X=x_i,\ Y=y_j)$

$= \{\omega\in\Omega : X(\omega)=x_i,\ Y(\omega)=y_j\}$

$= \{\omega\in\Omega : X(\omega)=x_i\} \cap \{\omega\in\Omega : Y(\omega)=y_j\}$

$$= a\sum_{i=1}^{n} x_i \left[\sum_{j=1}^{m} P(X=x_i,\ Y=y_j)\right] + b\cdot\sum_{j=1}^{m} y_j\left[\sum_{i=1}^{n} P(X=x_i,\ Y=y_i)\right]$$

$$= a\cdot\sum_{i=1}^{n} x_i\, P(X=x_i) + b\cdot\sum_{j=1}^{m} y_j\, P(Y=y_i)$$

$$= a\cdot E[X] + b\cdot E[Y].$$

■

(II) If $X\geq 0$ with probability 1 i.e

$$P(\{\omega\in\Omega : X(\omega)\geq 0\}) = 1 \implies E[X]\geq 0.$$

Example: If $X\geq Y$ with probability 1 i.e $P(\{\omega\in\Omega : X(\omega)\geq Y(\omega)\}) = 1$ then $E[X]\geq E[Y]$.

Proof: $X-Y$ is a rv. $X-Y\geq 0$ with probability 1.

$\Rightarrow$ $E[X-Y]\geq 0$ ..... [from II]

$\Rightarrow$ $E[X]-E[Y]\geq 0$ ..... [Linearity]

$\Rightarrow$ $E[X]\geq E[Y]$.

■

Expectation of rv with Infinitely Many Values in $\mathbb{R}$

This can be countable or uncountable.

I) Consider a nonnegative-valued rv. X (ie $\mathbb{R}^+ \cup \{\infty\}$-valued rv). $X(\omega) \geq 0$ $\forall\, \omega \in \Omega$. Define $E[X]$ "suitably"

II) For a general $\mathbb{R}$-valued rv X (ie. not necessarily non-negative), In this case we write, (decompose)

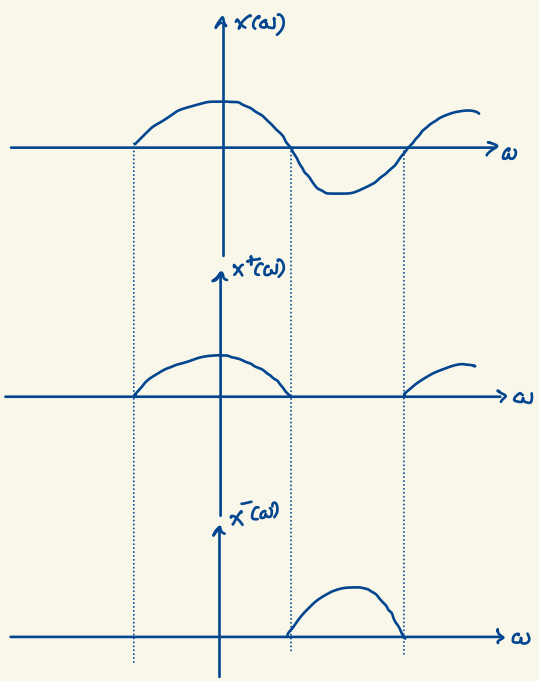
$$X(\omega) = X^+(\omega) - X^-(\omega)$$

where

$$X^+(\omega) = X(\omega) \cdot \mathbb{1}_{(X(\omega) \geq 0)} \qquad \geq 0 \ \forall \omega \in \Omega \longrightarrow \text{"Positive part of } X(\omega)\text{"}$$

$$X^-(\omega) = -X(\omega)\, \mathbb{1}_{(X(\omega) < 0)} \qquad \geq 0 \ \forall \omega \in \Omega \longrightarrow \text{"Negative part of } X(\omega)\text{"}$$

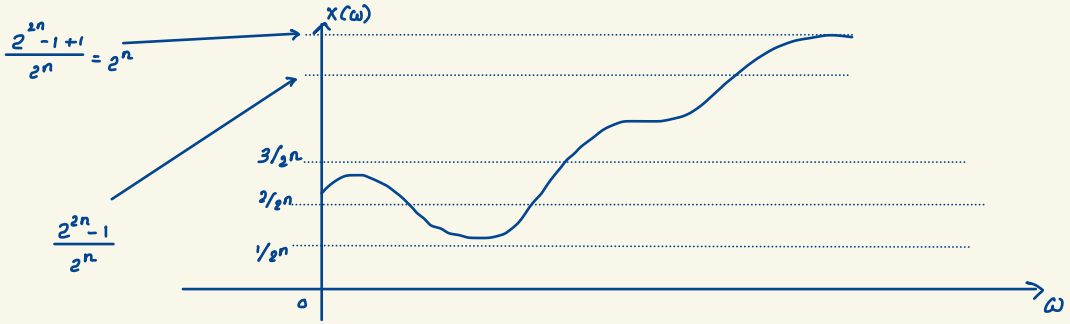
Finally examine $E[X] = E[X^+] - E[X^-]$ .... [$\because$ by linearity]



I) Non-negative Valued RVs

Suppose that $X(\omega) \geq 0 \quad \forall\, \omega \in \Omega$

Form a sequence $\{X_n(\omega),\ \omega \in \Omega\}_{n=1}^{\infty}$ of simple r.v.s such that $\{X_n(\omega)\} \uparrow n \ \forall\, \omega$ and $\lim_n X_n(\omega) = X(\omega) \ \forall\, \omega$.



Define $\{X_n\}_{n=1}^{\infty}$ as

$$X_n(\omega) = \begin{cases} k/2^n & \forall\, \omega \ \ni\ \frac{k}{2^n} \leq X(\omega) < \frac{k+1}{2^n} \quad k = 0,1,\ldots,2^{2n}-1 \\ 2^n & \forall\, \omega \ \ni\ X(\omega) \geq 2^n \end{cases}$$

Fix n : then $X_n(\omega)$ is a quantized version of $X(\omega)$

Observations

i) $X_n(\omega) \leq X(\omega) \quad \forall\, \omega \in \Omega$

ii) $\{X_n(\omega)\}$ is $\uparrow n \ \forall\, \omega$ (ie $X_{n+1}(\omega) \geq X_n(\omega) \ \forall \omega$)

iii) $\lim_{n\to\infty} X_n(\omega) = X(\omega) \ \forall \omega$

iv) In fact for each fixed ω, for $n \geq \lfloor \log_2 X(\omega) \rfloor$, show that,

$$\sup_{k \geq n} |X_k(\omega) - X(\omega)| \leq 2^{-n}$$

which $\to 0$ as $n \to \infty$

v) For each n, X_n takes at most $2^{2n}+1$ values.

vi) $E[X_n]$ is well defined in $*$

$$E[X_n] = \sum_{K=0}^{2^{2n}-1} \frac{K}{2^n} \underbrace{P\left(\frac{K}{2^n} \leqslant X(\omega) < \frac{K+1}{2^n}\right)}_{F_X\left(\frac{K+1}{2^n}\right) - F_X\left(\frac{K}{2^n}\right)} + 2^n \cdot \underbrace{P\left(X(\omega) \geqslant 2^n\right)}_{1 - F_X(2^n)}$$

- Note - $E[X_n]$ is determined by cdf of F_X for each n.

We define

$$E[X] \triangleq \lim_{n\to\infty} E[X_n]$$

$$E[X] = \lim_{n\to\infty} \sum_{K=0}^{2^{2n}-1} \frac{K}{2^n}\left[F_X\left(\frac{K+1}{2^n}\right) - F_X\left(\frac{K}{2^n}\right)\right] + 2^n\left[1 - F_X(2^n)\right] \quad (**)$$

- $\{E[X_n]\}_{n=1}^{\infty}$ is nondecreasing sequence in $\mathbb{R}$ since from $E[X]$ properties,

$$X_n(\omega) \leqslant X_{n+1}(\omega) \ \forall \omega, n \implies E[X_n] \leqslant E[X_{n+1}] \ \forall n$$

Hence the limit in $(**)$ always exist, but may go to $\pm\infty$.

- $E[X] = \lim_{n\to\infty} E[X_n]$ is determined by F_X.

Lebague Integral

Case: If X is nonnegative (ie. $P(\{\omega : X(\omega) \geqslant 0\}) = 1$) and has a pdf f_X, then $**$ takes the form,

$$E[X] = \lim_{n\to\infty} \sum_{K=0}^{2^{2n}-1} \left(\frac{K}{2^n}\right) \int_{K/2^n}^{K+1/2^n} f_X(x)\,dx + 2^n \int_{2^n}^{\infty} f_X(x)\cdot dx$$

$$E[X] = \int_0^{\infty} x\, f_X(x)\,dx \quad (***)$$

II) General Valued RV

Turning to general r.v X, write

$$X(\omega) = \underbrace{X^+(\omega)}_{\geq 0} - \underbrace{X^-(\omega)}_{\geq 0} \qquad \omega \in \Omega$$

$$\Rightarrow \quad E[X] = \underbrace{E[X^+]}_{\geq 0} - \underbrace{E[X^-]}_{\geq 0}$$

provided that at least one of $E[X^+]$ or $E[X^-]$ is finite. If both $E[X^+] = \infty$ and $E[X^-] = \infty$, then $E[X]$ is not defined.

X is Integrable

- If both $E[X^+]$ and $E[X^-]$ are finite, then

$$E[|X|] = E[X^+] + E[X^-] \quad \cdots\cdots \quad (\because |X| = X^+ + X^-)$$

is finite. This is known as X is integrable

- For a general r.v X, if $E[X]$ is well defined. (ie at least one of $E[X^+]$, $E[X^-]$ is finite) then

$$E[X^-] = \int_{-\infty}^{0} (-x) \cdot f_X(x) \cdot dx \quad \cdots\cdots \quad [\text{obtained similarly on } ***]$$

$$= -\int_{-\infty}^{0} x\, f_X(x)\, dx$$

and also we finally get,

$$E[X] = E[X^+] - E[X^-] = \int_{0}^{\infty} x \cdot f_X(x)\, dx - \int_{-\infty}^{0} x \cdot f_X(x)\, dx$$

$$= \int_{-\infty}^{\infty} x\, f_X(x)\, dx$$

$$E[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx \quad \cdots\cdots \quad (***)$$

If X has probability mass function p_x, then

$$E[X] = \sum_{x>0} x\, p_x(x) + \sum_{x<0} x\, p_x(x) \quad \cdots \text{(A)(B)(C)(D)(E)}$$

Example :- X has a Poison pmf ie. $P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$ $k = 0, 1, 2, \ldots$ $\lambda > 0$

$X \sim \text{Poison}(\lambda)$

$$E[X] = \sum_{k=0}^{\infty} k \frac{\lambda^k \cdot e^{-\lambda}}{k!} = \sum_{k=1}^{\infty} \frac{\lambda^k \cdot e^{-\lambda}}{(k-1)!} = \lambda \cdot \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} e^{\lambda} = 1$$

Example :- X has a Gaussian pdf ie $f_x(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ $X \sim N(0,1)$

$$E[X] = \int_{-\infty}^{\infty} x \cdot \frac{1}{\sqrt{2\pi}} \cdot e^{-x^2/2}\, dx$$

Example :- X has an Exponential pdf, ie $f_x(x) = \begin{cases} a e^{-ax} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$ $a > 0$

$$E[X] = \int_0^{\infty} x \cdot a e^{-ax}\, dx = \frac{1}{a}$$

$X \sim \exp(a)$

Probability Measures on Finite Dimension Spaces

[Emily Borel : 1871 - 1956]

Preliminaries - If $\mathcal{A}_1$ and $\mathcal{A}_2$ are σ-fields on Ω, then

$$\left[\mathcal{A}_1 \text{ is smaller than } \mathcal{A}_2 \ (\text{ie } \mathcal{A}_1 \subseteq \mathcal{A}_2)\right] \Rightarrow \left[A \in \mathcal{A}_1 \Rightarrow A \in \mathcal{A}_2\right]$$

Exercise :- If $\{\mathcal{A}_i, i \in I\}$, I is an index set which is countable or uncountable. is an index collection of σ-fields, then $\bigcap_{i \in I} \mathcal{A}_i$ is also a σ field on Ω.

does not work for U's

Exercise :- Given any collection $\mathcal{G}$ of events, $\exists$ a smallest σ-field $\mathcal{A}(\mathcal{G})$ which contains $\mathcal{G}$. (known as the σ-field generated by $\mathcal{G}$)

Proof: $\exists$ at least one σ-field containing $\mathcal{G}$, namely the powerset of Ω ($\mathcal{P}(\Omega)$ = set of all possible subsets of Ω)
Define, $\mathcal{A}(\mathcal{G})$ = $\cap$ {all σ-fields containing $\mathcal{G}$}

Borel Sets

$\Omega = \mathbb{R} = (-\infty, \infty)$

$\mathcal{G}$ = collection of subsets of $\mathbb{R}$ that can be expressed as finite unions of semi-open intervals

$$A \in \mathcal{G} \iff A = \bigcup_{i=1}^{n} [a_i, b_i)$$

open ?

Let

$\mathcal{A}(\mathcal{G})$ = Smallest σ-field containing $\mathcal{G}$

$\mathcal{A}(\mathcal{G})$: Borel σ-field on $\Omega = \mathbb{R}$

and the sets in $\mathcal{A}(\mathcal{G})$ are called **Borel Sets**

- Show $\mathcal{G}$ is a field but not a σ-field.
- The Borel σ-fields is the smallest σ-field that contains all the open subsets in $\mathbb{R}$.

Examples of Borel Sets

1) $[a,b] = \bigcap_{n=1}^{\infty} [a, b+1/n)$ ... closed interval is a Borel set

2) $(a,b) = \bigcup_{n: a+1/n<b} [a+1/n, b)$ .... open interval is a Borel set

3) $\{b\} = \bigcap_{n=1}^{\infty} [b, b+1/n)$

4) $(a,b] = [a,b] \setminus \{a\}$

- All semi-infinite intervals (open or close), $(-\infty, a)$, $(-\infty, a]$, (a, ∞), $[a, \infty)$ and $\mathbb{R}$ are Borel sets.
- Not every subset of $\mathbb{R}$ is a Borel set.

Notation

$\sigma(G)$ = Borel σ-field on $\mathbb{R}$ = $\mathcal{B}_{\mathbb{R}}$

$(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ is a measurable space. $\mathbb{R}$ — Sample space, $\mathcal{B}_{\mathbb{R}}$ — σ-field

$(\mathbb{R}, \mathcal{B}_{\mathbb{R}}) \equiv (\mathbb{R}, \mathcal{B}_{[0,1)})$

$\therefore$ Now we shrinked power set to Borel set. $\therefore$ This is our σ-field now.

Alternate Definition of a R.V

Definition: A $\mathbb{R}$-valued function $X: \Omega \longrightarrow \mathbb{R}$ is a $\mathbb{R}$-valued r.v if for every $A \in \mathcal{B}_{\mathbb{R}}$, $X^{-1}(A) \in \mathcal{A}$.

Exercise: Check that this definition of r.v is equivalent to previous definition.

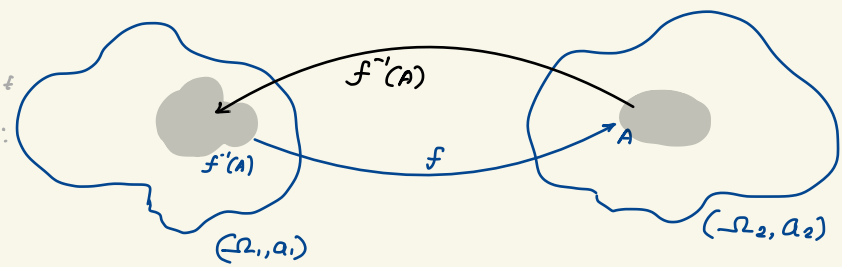
Measurable Mapping / Measurable Functions

Definition: Suppose that $(\Omega_1, \mathcal{A}_1)$ and $(\Omega_2, \mathcal{A}_2)$ are two measurable spaces and f is a mapping from $f: \Omega_1 \to \Omega_2$. Then f is a **measurable mapping** or a **measurable function** of $(\Omega, \mathcal{A}_1)$ into $(\Omega, \mathcal{A}_2)$ if

$$\forall A \in \mathcal{A}_2,$$
$$f^{-1}(A) = \{\omega \in \Omega_1 : f(\omega) \in A\} \in \mathcal{A}_1$$

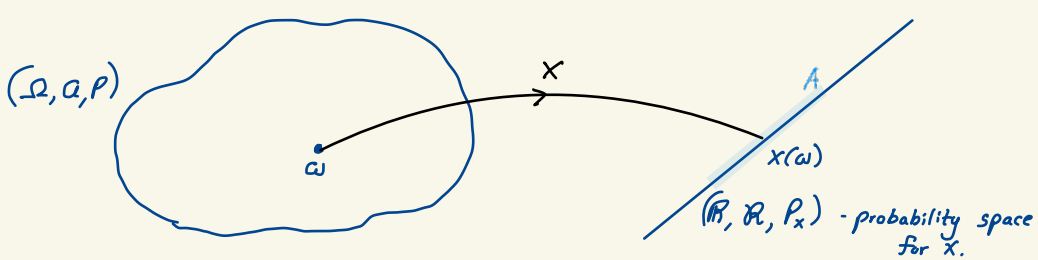
$f^{-1}(A)$ is the **inverse image** of A under f.

This is a simple mapping. We just want to find what values are mapped to A. ∴ No need of bijection.



Borel Probability Measure

- A rv is a measurable mapping from $(\Omega, \mathcal{A})$ into $(\mathbb{R}, \mathcal{R})$



For any $A \in \mathcal{R}$

$\because X^{-1}(A) = \{\omega : X(\omega) \in A\} \in \mathcal{A}$

we can compute

$$P(X \in A) = P\left(\{\omega \in \Omega : X(\omega) \in A\}\right)$$

$$P_X(A) \triangleq P(X \in A)$$

$\therefore P$ is actually on $\mathcal{A}$ in Ω.

Definition : $P_X(A)$ defines a probability measure on $(\mathbb{R}, \mathcal{R})$ and is called the Borel Probability Measure on $(\mathbb{R}, \mathcal{R})$.

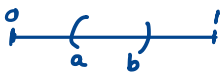
$$P_X(A) \triangleq P(X \in A) = P\left(\{\omega \in \Omega : X(\omega) \in A\}\right)$$

- This measure is defined by F_X and is uniquely defined.
- Often more convenient to work with $(\mathbb{R}, \mathcal{R}, P_X)$ where,

$$P_X(A) = P(X^{-1}(A))$$

$$= P\left(\{\omega \in \Omega : X(\omega) \in A\}\right) \qquad A \in \mathcal{R}$$

Intitution on why we need Borel sets



– We are obsessed with [0,1] interval since $\mu: \mathcal{A} \to [0,1]$ for a probability measure.

Consider the experiment of picking a real number at random from $\Omega = [0,1]$, such that every number is "equally likely" to be picked. It is quite apparent that a simple strategy of assigning probabilities to singleton subsets of the sample space gets into difficulties quite quickly. Indeed,

(i) If we assign some positive probability to each elementary outcome, then the probability of an event with infinitely many elements, such as $A = \{1, \frac{1}{2}, \frac{1}{3}, \cdots\}$, would become unbounded.

(ii) If we assign zero probability to each elementary outcome, this alone would not be sufficient to determine the probability of a uncountable subset of Ω, such as $\left[\frac{1}{2}, \frac{2}{3}\right]$. This is because probability measures are not additive over uncountable disjoint unions (of singletons in this case).

Thus, we need a different approach to assign probabilities when the sample space is uncountable, such as $\Omega = [0,1]$. In particular, we need to assign probabilities directly to specific subsets of Ω. Intuitively, we would like our 'uniform measure' μ on $[0,1]$ to possess the following two properties.

$P([a,b)) \sim b-a$ length.

(i) $\mu((a,b)) = \mu((a,b]) = \mu([a,b)) = \mu([a,b])$

(ii) Translational Invariance. That is, if $A \in [0,1]$, then for any $x \in \Omega$, $\mu(A \oplus x) = \mu(A)$ where, the set $A \oplus x$ is defined as

$$A \oplus x = \{a + x | a \in A, a + x \leq 1\} \cup \{a + x - 1 | a \in A, a + x > 1\}$$

However, the following impossibility result asserts that there is no way to consistently define a uniform measure on all subsets of $[0,1]$.

Theorem 7.1 (Impossibility Result) *There does not exist a definition of a measure $\mu(A)$ for all subsets of $[0,1]$ satisfying (i) and (ii).*

Power set of $2^{[0,1)}$ is too much for us to handle

Proof: Refer proposition 1.2.6 in [1].

Therefore, we must compromise, and consider a smaller σ-algebra that contains certain "nice" subsets of the sample space $[0,1]$. These "nice" subsets are the intervals, and the resulting σ-algebra is called the Borel σ-algebra. Before defining Borel sets, we introduce the concept of generating σ-algebras from a given collection of subsets.

▸ Borel sets have much lesser cardinality ($\sim \mathbb{R}$) than power set ($\sim 2^{\mathbb{R}}$).

Definition 7.3

It can contain complements etc ← All Borel sets does not have to be open intervals

(a) *Consider $\Omega = (0,1]$. Let $\mathcal{C}_0$ be the collection of all open intervals in $(0,1]$. Then $\sigma(\mathcal{C}_0)$, the σ - algebra generated by $\mathcal{C}_0$, is called the Borel σ - algebra. It is denoted by $\mathcal{B}((0,1])$.*

(b) *An element of $\mathcal{B}((0,1])$ is called a Borel-measurable set, or simply a Borel set.*

Thus, every open interval in $(0,1]$ is a Borel set. We next prove that every singleton set in $(0,1]$ is a Borel set.

Lemma 7.4 *Every singleton set* $\{b\},\ 0 < b \leq 1$, *is a Borel set, i.e.,* $\{b\} \in \mathcal{B}((0,1])$.

Proof: Consider the collection of sets set $\left\{\left(b-\frac{1}{n}, b+\frac{1}{n}\right), n \geq 1\right\}$. By the definition of Borel sets,

$$\left(b-\frac{1}{n}, b+\frac{1}{n}\right) \in \mathcal{B}((0,1]).$$

Using the properties of σ-algebra,

$$\begin{aligned}
&\left(b-\frac{1}{n}, b+\frac{1}{n}\right)^c \in \mathcal{B}((0,1]) \\
\Longrightarrow &\bigcup_{n=1}^{\infty}\left(b-\frac{1}{n}, b+\frac{1}{n}\right)^c \in \mathcal{B}((0,1]) \\
\Longrightarrow &\left(\bigcap_{n=1}^{\infty}\left(b-\frac{1}{n}, b+\frac{1}{n}\right)\right)^c \in \mathcal{B}((0,1]) \\
\Longrightarrow &\bigcap_{n=1}^{\infty}\left(b-\frac{1}{n}, b+\frac{1}{n}\right) \in \mathcal{B}((0,1]). \qquad (7.1)
\end{aligned}$$

Next, we claim that

$$\{b\} = \bigcap_{n=1}^{\infty}\left(b-\frac{1}{n}, b+\frac{1}{n}\right). \qquad (7.2)$$

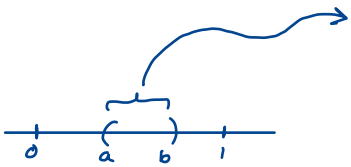
i.e., b is the only element in $\bigcap_{n=1}^{\infty}\left(b-\frac{1}{n}, b+\frac{1}{n}\right)$. We prove this by contradiction. Let h be an element in $\bigcap_{n=1}^{\infty}\left(b-\frac{1}{n}, b+\frac{1}{n}\right)$ other than b. For every such h, there exists a large enough n_0 such that $h \notin \left(b-\frac{1}{n_0}, b+\frac{1}{n_0}\right)$. This implies $h \notin \bigcap_{n=1}^{\infty}\left(b-\frac{1}{n}, b+\frac{1}{n}\right)$. Using (7.1) and (7.2), thus, proves that $\{b\} \in \mathcal{B}((0,1])$. ■

As an immediate consequence to this lemma, we see that every half open interval, $(a,b]$, is a Borel set. This follows from the fact that

$$(a,b] = (a,b) \cup \{b\},$$

and the fact that a countable union of Borel sets is a Borel set. For the same reason, every closed interval, $[a,b]$, is a Borel set.

Extending Probability Notion to Borel Sets



- Now we are settled on $\mathcal{B}((0,1])$ to be our interested σ-field.
- We have a notion that $P((0,b)) \sim b-a$
- But what this notion on assigning probabilities to (a,b) means for other sets in $\mathcal{B}((0,1])$ which are not just simple events.

Theorem 7.5 (Caratheodory's extension theorem) *Let $\mathcal{F}_0$ be an algebra of subsets of Ω, and let $\mathcal{F} = \sigma(\mathcal{F}_0)$ be the σ-algebra that it generates. Suppose that $\mathbb{P}_0$ is a mapping from $\mathcal{F}_0$ to $[0,1]$ that satisfies $\mathbb{P}_0(\Omega) = 1$, as well as countable additivity on $\mathcal{F}_0$.*
Then, $\mathbb{P}_0$ can be extended uniquely to a probability measure on $(\Omega, \mathcal{F})$. That is, there exists a unique probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ such that $\mathbb{P}(A) = \mathbb{P}_0(A)$ for all $A \in \mathcal{F}_0$.

Consider $\Omega = (0,1]$. Let $\mathcal{F}_0$ consist of the empty set and all sets that are finite unions of the intervals of the form $(a,b]$. A typical element of this set is of the form

$$F = (a_1, b_1] \cup (a_2, b_2] \cup \ldots \cup (a_n, b_n]$$

where, $0 \leq a_1 < b_1 \leq a_2 < b_2 \leq \ldots \leq a_n < b_n$ and $n \in \mathbb{N}$.

Lemma 7.6

a) $\mathcal{F}_0$ is an algebra

b) $\mathcal{F}_0$ is not a σ-algebra

c) $\sigma(\mathcal{F}_0) = \mathcal{B}$ - Sigma field of F.'s (unions of semi open intervals) and Borel sets are equivalent.

Proof:

a) By definition, $\Phi \in \mathcal{F}_0$. Also, $\Phi^C = (0,1] \in \mathcal{F}_0$. The complement of $(a_1, b_1] \cup (a_2, b_2]$ is $(0, a_1] \cup (b_1, a_2] \cup (b_2, 1]$, which also belongs to $\mathcal{F}_0$. Furthermore, the union of finitely many sets each of which are finite unions of the intervals of the form $(a,b]$, is also a set which is the union of finite number of intervals, and thus belongs to $\mathcal{F}_0$.

b) To see this, note that $\left(0, \frac{n}{n+1}\right] \in \mathcal{F}_0$ for every n, but $\bigcup_{n=1}^{\infty} \left(0, \frac{n}{n+1}\right] = (0,1) \notin \mathcal{F}_0$.

c) First, the null set is clearly a Borel set. Next, we have already seen that every interval of the form $(a,b]$ is a Borel set. Hence, every element of $\mathcal{F}_0$ (other than the null set), which is a finite union of such intervals, is also a Borel set. Therefore, $\mathcal{F}_0 \subseteq \mathcal{B}$. This implies $\sigma(\mathcal{F}_0) \subseteq \mathcal{B}$.

Next we show that $\mathcal{B} \subseteq \sigma(\mathcal{F}_0)$. For any interval of the form (a,b) in $\mathcal{C}_0$, we can write $(a,b) = \bigcup_{n=1}^{\infty} \left((a, b - \frac{1}{n}] \cap \Omega \right)$. Since every interval of the form $\left(a, b - \frac{1}{n}\right] \in \mathcal{F}_0$, a countable number of unions of such intervals belongs to $\sigma(\mathcal{F}_0)$. Therefore, $(a,b) \in \sigma(\mathcal{F}_0)$ and consequently, $\mathcal{C}_0 \subseteq \sigma(\mathcal{F}_0)$. This gives $\sigma(\mathcal{C}_0) \subseteq \sigma(\mathcal{F}_0)$. Using the fact that $\sigma(\mathcal{C}_0) = \mathcal{B}$ proves the required result.

■

Finite Collection of RVs / Jointly Distributed RVs

Finite Collection of RVs / Jointly Distributed RVs

$(\Omega, \mathcal{A}, P)$ $X_1, \ldots, X_n$ $n \geq 1$ are $\mathbb{R}$-valued rvs, each defined on the same sample space. ie. $X_i : \Omega \longrightarrow \mathbb{R}$ $i = 1, \ldots, n$.

Observations

1) For $i = 1, \ldots, n$ $\{\omega : X_i(\omega) < a\} \in \mathcal{A}$

2) For any $i \neq j$, $\{\omega : X_i(\omega) < a\} \cap \{\omega : X_j(\omega) < b\}$ is also $\in \mathcal{A}$.

3) Knowing probabilities of events of form $\{\omega : X_i(\omega) < a\}$ using cdf F_{X_i} $i = 1, \ldots, n$ does not enable the computation of probabilities of events such as
$$\{\omega : X_i(\omega) < a\} \cap \{\omega : X_j(\omega) < b\}$$

- If 3 is not sufficient what do we really need?

- Let $\mathcal{S}$ be a collection of events in $\mathcal{A}$ of the form
$$\bigcap_{i=1}^{n} \{\omega : X_i(\omega) < a_i\}$$
where $\{a, a_1, \ldots, a_n\}$ is any n-tuple of numbers in $\mathbb{R}$. We need the knowledge of
$$F_{X_1, \ldots, X_n}(a_1, \ldots, a_n) = F_{\underline{X}}(a_1, \ldots, a_n) \triangleq P\left(\bigcap_{i=1}^{n} \{\omega : X_i(\omega) < a_i\}\right)$$
$-\infty < a_i < \infty$ $i = 1, \ldots, n$

- $\underline{X} = X_1, \ldots, X_n$

- $F_{\underline{X}}$ is a function of n $\mathbb{R}$-valued variables and is termed the joint cdf or joint distribution of $\mathbb{R}$-valued rvs. $X_1, \ldots, X_n$

Simplest cases of joint cdf

(i) $X_1, \ldots, X_n$ are all $\mathbb{R}$-valued discrete rvs, ($F_{\underline{X}}$ is termed Singular.)

(ii) $F_{\underline{X}}$ can be termed as:

$$F_{\underline{X}}(a_1, \ldots, a_n) = \int_{-\infty}^{a_1} \cdots \int_{-\infty}^{a_n} f_{\underline{X}}(x_1, \ldots, x_n)\, dx_1 \ldots dx_n$$

($F_{\underline{X}}$ is absolutely continuous)

Properties

(a) $f_{\underline{X}}(a_1, \ldots, a_n) \geq 0 \quad \forall\, a_1, \ldots, a_n$

(b) $\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{\underline{X}}(x_1, \ldots, x_n)\, dx_1 \ldots dx_n = 1$

- Integrand $f_{\underline{X}}$ is the joint probability density function of $X_1, \ldots, X_n$ or joint pdf.
- If $f_{\underline{X}}$ is continuous at point $\underline{x} = (x_1, \ldots, x_n)$, then

$$f_{\underline{X}}(x_1, \ldots, x_n) = \frac{\partial^n}{\partial x_1 \ldots \partial x_n} F_{\underline{X}}(x_1, \ldots, x_n)$$

$$F_{X_1,\ldots,X_n}(a_1,\ldots,\underset{\substack{\uparrow\\ i^{th}\\ \text{position}}}{\infty},\ldots,a_n) \triangleq \lim_{a_i\to\infty} F_{X_1,X_2\ldots X_n}(a_1,\ldots,a_i,\ldots,a_n)$$

$$= F_{X_1,\ldots,X_{i-1},X_{i+1},\ldots,X_n}(a_1,\ldots,a_{i-1},a_{i+1},\ldots,a_n)$$

by sequential Continuity Theorem (i)

Special Cases of $F_{\underline{X}}$

1) $p_{\underline{X}}$ Joint PMF - $X_1,\ldots,X_n$ all $\mathbb{R}$-valued discrete rvs.
2) $F_{\underline{X}}$ Joint PDF - $F_{\underline{X}}$ is absolutely continuos

- Interpretation: $f_{\underline{X}}(x_1,\ldots,x_n)\,dx_1,\ldots dx_n$ ~ probability that $\underline{X}$ lies in an n-dimensional volume $dx_1,\ldots dx_n$ at the point $(x_1,\ldots,x_n)\in\mathbb{R}^n$.

- In general we can always write

Lebesgue Decomposition

$$F_{\underline{X}} = \alpha F_{\underline{X}}^{(1)} + (1-\alpha) F_{\underline{X}}^{(2)} \qquad 0\le\alpha\le 1$$

$F_{\underline{X}}^{(1)}$: absolutely continuos part; $F_{\underline{X}}^{(2)}$: Singular

Example. $\Omega = [0,1)$, $P(\text{interval}) = \text{length of the interval}$. $X_1(\omega) = \omega$, $X_2(\omega) = \omega^2$

(i) Check if X_1 and X_2 are actual valid pmfs
(ii) Calculate Cdf $X(\omega) = (X_1(\omega), X_2(\omega))$, where $X_2 = X_1^2$

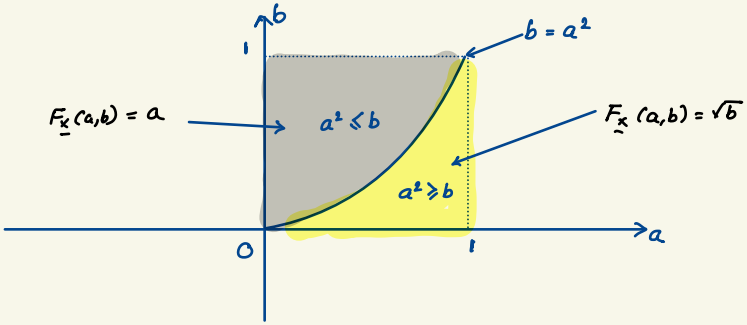
$\underline{X}$ takes values in $[0,1)^2 = [0,1)\times[0,1)$

For $0\le a,b\le 1$,

$$F_{\underline{X}}(a,b) = P\left(\{\omega : X_1(\omega) < a \text{ and } X_2(\omega) < b\}\right)$$

$$= P\left(\{\omega : \omega < a, \omega^2 < b\}\right)$$
$$= P\left(\{\omega : \omega < \min(a, \sqrt{b})\}\right)$$
$$= \min(a, \sqrt{b}), \quad 0 \leq a, b \leq 1$$

Do the same calculation for $-\infty < a, b < \infty$. Go from one interval to other



Although X_1, X_2 have individual pdfs, $\underline{X} = (X_1, X_2)$ does not have a joint pdf.

Events in $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ Determined by RVs

Single RV Case

- $\mathbb{R}$ valued RV X, given cdf F_X, $F_X(a) = P(\{\omega : X(\omega) < a\})$ $a \in \mathbb{R}$
- Steps $\mathcal{S} \longrightarrow \mathcal{B}(\mathcal{S}) \longrightarrow \mathcal{A}(\mathcal{S})$

extend P_X using

$P_X(A^c) = 1 - P_X(A)$ *

$P_X(A \cup B) = P_X(A) + P_X(B) - P_X(A \cap B)$

Using (*) we can compute the probability $P_X(A)$ where $A \in \mathcal{R}$. This can be done for some Borel sets $A \in \mathbb{R}$

$(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, P_X)$

$\mathcal{B}(\mathcal{S})$

$\mathcal{S} \subseteq \mathcal{R}$

$\mathcal{S}$ = {all events of form $\{\omega : X(\omega) < a\}$}

$\mathcal{S}$ is closed under $\cap$.

extend P_X using Axiom (P4) and σ-additivity

Starting with events in $\mathcal{B}(\mathcal{S})$, can compute probabilities in $\mathcal{A}(\mathcal{S})$ by using Sequential Continuity Theorem

$\mathcal{A}(\mathcal{S})$

Obtained from $\mathcal{S}$ by Us, $\cap$s, ^{c}s and sequential limits

Example: $\Omega \longrightarrow \mathcal{B}(S)$

$$P\left(\underbrace{\{\omega : X(\omega) \geq a\}}_{\text{event in } \mathcal{B}(S)}\right) = P\left(\{\omega : X(\omega) < a\}^c\right) = 1 - F_X(a)$$
$$= P_X(\underbrace{[a,\infty)}_{\text{Borel set in } \mathcal{R}})$$

Example: $\Omega \longrightarrow \mathcal{B}(S)$

$$P\left(\underbrace{\{\omega : X(\omega) \in [a,b]\}}_{\text{event in } \mathcal{B}(S)}\right) = P_X(\underbrace{[a,b]}_{\text{Borel set in } \mathcal{R}}) = F_X(b) - F_X(a)$$

Example: $\mathcal{B}(S) \longrightarrow \sigma(\Omega)$

- Since $\{\omega : a < X(\omega) < b + 1/n\}$ is a decreasing sequence in n

$$\underbrace{\{\omega : a \leq X(\omega) \leq b\}}_{\text{event in } \sigma(\Omega)} = \bigcap_{n=1}^{\infty} \underbrace{\{\omega : a \leq X(\omega) < b + 1/n\}}_{\text{event in } \mathcal{B}(S)}$$
$$= \lim_{n\to\infty} \{\omega : a \leq X(\omega) < b + 1/n\}$$

$$\Rightarrow P(\{\omega : a \leq X(\omega) \leq b\}) = P\left(\lim_{n\to\infty} \{\omega : a \leq X(\omega) < b + 1/n\}\right)$$
$$= \lim_{n\to\infty} P(\{\omega : a \leq X(\omega) < b + 1/n\}) \quad \cdots \text{(Sequential Continuity Theorem)}$$
$$= \lim_{n\to\infty} \left(F_X(b + 1/n) - F_X(a)\right)$$
$$= F_X(b^+) - F_X(a) \quad \cdots\cdots \text{Due to right continuousness}$$
$$= P_X(\underbrace{[a,b]}_{\text{Borel set in } \mathcal{R}})$$

Example: $\mathcal{B}(S) \longrightarrow \sigma(\Omega)$

$$P\left(\underbrace{\{\omega : a < X(\omega) \leq b\}}_{\text{event in } \sigma(\Omega)}\right) = P_X([a,b]) = F_X(b^+) - F_X(a^+)$$

In general height of the jump of F_x at b

Example :

$$P(\{\omega : X(\omega) = b\}) = P(\{\omega : b \leq X(\omega) \leq b\}) = F_x(b^+) - F_x(b)$$
$$= P_x(\underbrace{\{b\}}_{\text{Borel set in } \mathbb{R}})$$

- If F_x is continuos at b, $P_x(\{b\}) = P(\{\omega : X(\omega) = b\}) = 0$

General Case of a Finite no. of R.Vs

- $(\Omega, \mathcal{A}, P)$ $\underline{X} = (X_1, \ldots, X_n)$ where X is $\mathbb{R}$ valued random rvs. or $\mathbb{R}^n$ valued rvs.

$\mathcal{S}$ = Collection of events in $\mathcal{A}$ of form $\bigcap_{i=1}^{n} \{\omega : X_i(\omega) < a_i\}$, $a_i \in \mathbb{R}$ $i = 1, \ldots, n$

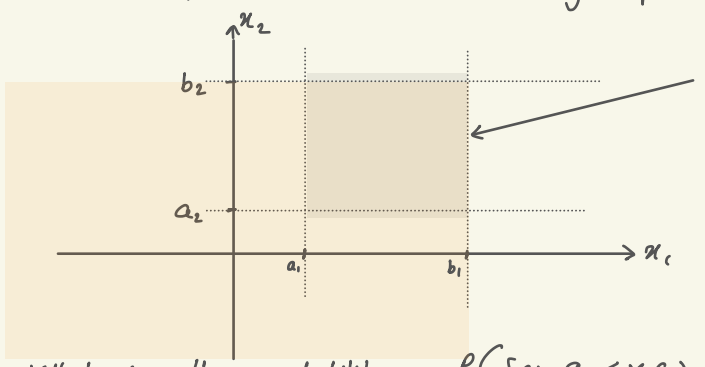
- Know F_X, Steps $\mathcal{S} \longrightarrow \mathcal{B}(\mathcal{S}) \longrightarrow \mathcal{A}(\mathcal{S})$
- $(\mathbb{R}^n, \mathcal{B}^n)$ measurable space where $\mathcal{B}^n$ = Borel σ-field on $\mathbb{R}^n$.
- Rectangle in $\mathbb{R}^n$ is of form : $\prod_{i=1}^{n} [a_i, b_i] \sim \bigtimes_{i=1}^{n} [a_i, b_i]$ (much more precise notation)

Definition : Hence Borel σ-field $\mathcal{B}^n$ is the smallest σ-field in $\mathbb{R}^n$ that contains all the rectangles in $\mathbb{R}^n$.

: $\mathcal{B}^n$ is the smallest σ-field in $\mathbb{R}^n$ that contains all the sets of form :

$$\prod_{i=1}^{n} (-\infty, a_i) = \{\underline{x} \in \mathbb{R}^n : -\infty < x_i < a_i \quad i = 1, \ldots, n\}$$

Example: $X(\omega) = (X_1(\omega), X_2(\omega))$ Given F_X - joint pdf.



What is the probability, $P(\{\omega : a_1 \leq X_1(\omega) < b_1, \; a_2 \leq X_2(\omega) < b_2\})$

- Write :

$$(-\infty, b_1) \times (-\infty, b_2) = (-\infty, a_1) \times (-\infty, a_2)$$

Disjoint union

$$+ [a_1, b_1) \times [a_2, b_2)$$
$$+ (-\infty, a_1) \times [a_2, b_2)$$
$$+ [a_1, b) \times [-\infty, a_2) \quad \cdots (1)$$

Does not come from joint pmf

- $(-\infty, a_1) \times [a_2, b_2] = (-\infty, a_1) \times (-\infty, b_2) - (-\infty, a_1) \times (-\infty, a_2) \quad \cdots (2)$

$[a_1, b) \times [-\infty, a_2) = (-\infty, b_1) \times (-\infty, a_2) - (-\infty, a_1) \times (-\infty, a_2) \quad \cdots (3)$

By (1), (2), (3)

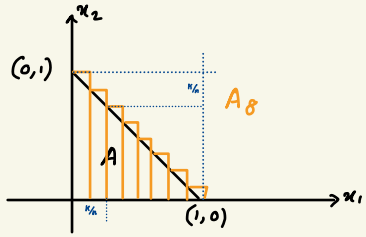
$$[a_1, b_1) \times [a_2, b_2) = (-\infty, b_1) \times (-\infty, b_2) - (-\infty, a_1) \times (-\infty, a_2)$$
$$- (-\infty, b_1) \times (-\infty, a_2) + (-\infty, a_1) \times (-\infty, b_2)$$

Consequently

$$P(\{\omega : X(\omega) \in [a_1, b_1) \times [a_2, b_2)\}) = P_X([a_1, b_1) \times [a_2 \times b_2))$$

$$= F_X(b_1, b_2) - F_X(a_1, b_2) - F_X(b_1, a_2)$$
$$+ F_X(a_1, a_2)$$

Ex: $X(\omega) = (X_1(\omega), X_2(\omega))$. Consider the event $\{\omega : X(\omega) \in A\}$. Suppose



$$f_{\underline{X}}(x_1, x_2) = \begin{cases} 1 & 0 < x_1, x_2 < 1 \\ 0 & \text{o-w} \end{cases}$$

- $\{\omega : X(\omega) \in A\} = \{\omega : X_1(\omega) \geq 0,\ X_2(\omega) \geq 0,\ X_1(\omega) + X_2(\omega) \leq 1\}$
- Give $F_{\underline{X}}$ we seek $P_X(A) = P\{\omega : X(\omega) \in A\}$

  Approximate A by the union of the pairwise disjoint rectangles.

$$A_n = \bigcup_{k=0}^{n-1} \left\{ \left[\frac{k}{n}, \frac{k+1}{n}\right), \left[0, 1-\frac{k}{n}\right) \right\}$$

Exercise: verify : $\{A_n\} \downarrow n$ ie $A_n \supseteq A_{n+1}$ $\forall n$

$\{A_n\} \downarrow A$ (how? precisely ?)

Then

$$P(\{\omega : \underline{X}(\omega) \in A\}) = P\left(\lim_{n\to\infty} \{\omega : \underline{X}(\omega) \in A_n\}\right)$$

$$= \lim_{n\to\infty} P(\{\omega : \underline{X}(\omega) \in A_n\}) \quad \cdots \text{Sequential continuity theory.}$$

$$= \lim_{n\to\infty} \sum_{k=0}^{n-1} \left\{ F_{\underline{X}}\left(\frac{k+1}{n}, 1-\frac{k}{n}\right) - F_{\underline{X}}\left(\frac{k}{n}, 1-\frac{k}{n}\right) - F_{\underline{X}}\left(\frac{k+1}{n}, 0\right) + F_{\underline{X}}\left(\frac{k}{n}, 0\right) \right\}$$

Remark

If X has a joint pmf $p_{\underline{X}}$ or joint pdf $f_{\underline{X}}$, then this step can be simplified

I) $\underline{X} = (x_1, x_2, \ldots, x_n)$ is descrete with joint pmf $p_{\underline{X}}$

then

$$LHS = \sum_{\underline{x} \in A} p_{\underline{X}}(\underline{x})$$

II) $\underline{X}$ is continuos with joint pdf of $f_{\underline{X}}$. Then by definition

$$f_{\underline{X}}(x_1,\dots,x_n)\,dx_1\dots dx_n = P\left(\{\omega : X(\omega) \in (x_1, x_1+dx) \times \dots \times (x_n, x_n+dx)\}\right)$$

Covering any set $A \subseteq R^n$ by n-dimensional rectangles,

$$LHS = \int_A f_{\underline{X}}(x_1,\dots,x_n)\,dx_1\dots dx_n$$

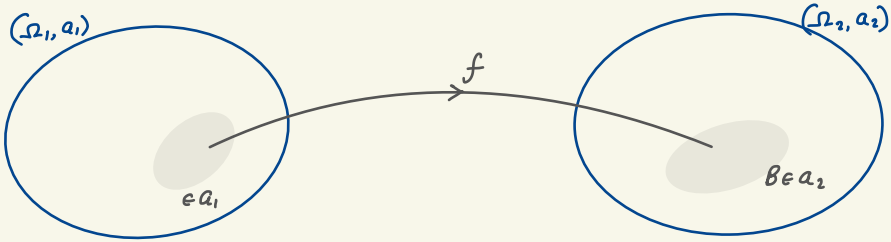
$$P\left(\{\omega : X_1(\omega) \geq 0,\ X_2(\omega) \geq 0,\ X_1(\omega)+X_2(\omega) \leq 1\}\right)$$

$$= \int_A f_{\underline{X}}(x_1,x_2)\,dx_1\,dx_2$$

$$= \int_0^1 dx_1 \int_0^{1-x_1} dx_2 \quad \dots \quad = 1/2$$

Functions of RVs

Functions of RVs



f is measurable in Ω_1 into Ω_2 if $\forall B \in a_2$, the set $f^{-1}(B) = \{\omega \in \Omega_1, f(\omega) \in B\}$ is in a_1.

Borel Functions

Definition- If f is a measurable function on

$$f: (\mathbb{R}^n, \mathcal{R}^n) \longrightarrow (\mathbb{R}^m, \mathcal{R}^m) \qquad n, m \in \mathbb{Z}^+$$

then f is a Borel function.

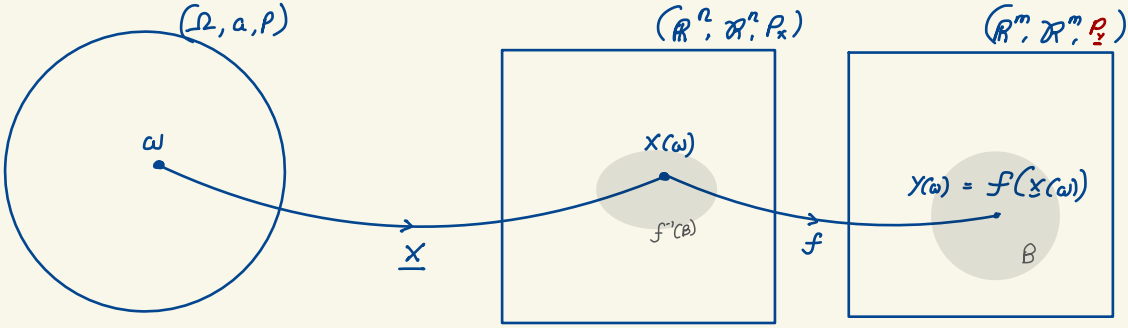
- Examples for Borel Functions
  - All continuos functions are Borel functions.
  - The pointwise limits of a sequence of Borel functions is also a Borel function.

- The class of Borel functions is very large.

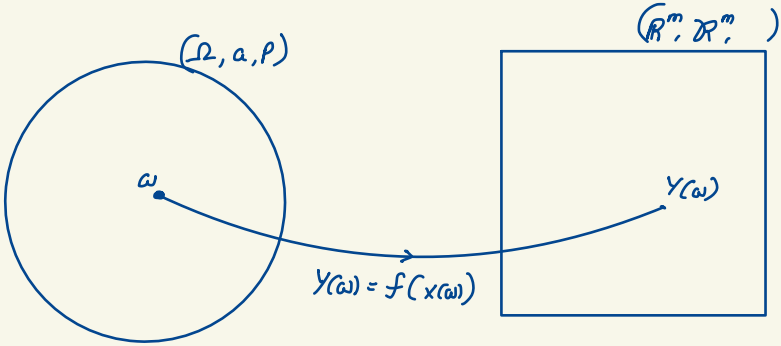
When is a Function of a RV also a RV

$\underline{X}(\omega) = (X_1(\omega), \ldots, X_n(\omega))$ $\mathbb{R}^n$ valued rv. defined on (Ω, a, P)

how to define $P_{\underline{Y}}$?



r.v $\underline{X}: \Omega \to \mathbb{R}^n$ and $f: \mathbb{R}^n \to \mathbb{R}^m \implies Y: \Omega \to \mathbb{R}^m$ given by $\underline{Y}(\omega) = f(\underline{X}(\omega))$ ie $\underline{Y} = f(\underline{X})$



- We want Y to be a rv. Under what condition is $\underline{Y}$ a $\mathbb{R}^m$ valued R.V ?

$\forall\, B \in \mathcal{R}^m$ we need

$$\underline{Y}^{-1}(B) \in a \implies \{\omega : \underline{Y}(\omega) \in B\} \in a \subseteq \Omega$$

$$\implies \{\omega : f(\underline{X}(\omega)) \in B\} \in a$$

$$\implies \{\omega : X(\omega) \in f^{-1}(B)\} \in a$$

inverse image of B under $f = \{\underline{x} \in \mathbb{R}^n : f(\underline{x}) \in B\}$.

This happen if $\implies f^{-1}(B) \in \mathcal{R}^n \;\forall\, B \in \mathcal{R}^m$.

$\implies$ if f is a Borel function

Derived Probability Measure $P_{\underline{Y}}$ on $(\mathbb{R}^m, \mathcal{R}^m)$

cdf $F_{\underline{Y}}$ and $P_{\underline{Y}}$ on $(\mathbb{R}^m, \mathcal{R}^m)$

Depends on $P_{\underline{X}}$ or $F_{\underline{X}}$ and f, $y \in \mathbb{R}^m$

$$F_{\underline{Y}}(\underline{y}) = P(\underline{Y} < \underline{y}) \quad \cdots \text{componentwise}$$
$$= P(f(\underline{X}) < \underline{y})$$
$$= P(\underline{X} \in f^{-1}(-\infty, \underline{y}))$$

Inverse image of $\mathcal{R}^n$ of $(-\infty, y) \in \mathcal{R}^m$

$$= P_{\underline{X}}(f^{-1}(-\infty, \underline{y})) \quad y \in \mathcal{R}^m \quad \#$$

$$= \text{Borel probability measure on } \mathcal{R}^n$$

For any $B \in \mathcal{R}^m$ what is $P_{\underline{Y}}(B)$? ie [$P_{\underline{Y}}$ is the Borel probability measure on $(\mathbb{R}^m, \mathcal{R}^m)$]

$$P_{\underline{Y}}(B) = P(\{\omega : \underline{Y}(\omega) \in B\})$$
$$= P(\{\omega : f(\underline{X}(\omega)) \in B\})$$
$$= P(\{\omega : \underline{X}(\omega) \in f^{-1}(B)\})$$

Inverse image in $\mathcal{R}^n$ of $B \in \mathcal{R}^m$

$$= P_{\underline{X}}(\{\underline{x} \in \mathbb{R}^n : \underline{x} \in f^{-1}(B)\})$$
$$= P_{\underline{X}}(f^{-1}(B)), \quad B \in \mathcal{R}^m \quad \#\#$$

$\#$ and $\#\#$ are equivalent

Special Cases

1) $F_{\underline{X}}$ is singular, $\underline{Y}$ is descrete, $F_{\underline{Y}}$ is singular and has a pmf.

$$p_{\underline{Y}}(y) = P_{\underline{Y}}\{\{y\}\}$$

$$= \sum_{\underline{x} \in f^{-1}(y)} p_{\underline{X}}(\underline{x})$$

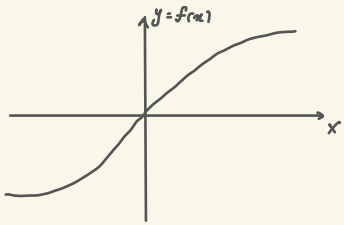
Sufficiency Approach

2) $F_{\underline{X}}$ is absolutely continuous.: $\underline{X}$ is a $\mathbb{R}^n$ valued rv with $f_{\underline{X}}$ being pdf.

Claim: Let $f: \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be one to one and invertible. (hence, onto its range too) such that booth f and f^{-1} have continuos partial derivatives w.r.t co-ordinates Let $g \triangleq f^{-1}$, $g: \mathbb{R}^n \longrightarrow \mathbb{R}^n$ Then

$\underline{Y} = f(\underline{X})$ has cdf $F_{\underline{Y}}$ that is absolutely continuos with pdf $f_{\underline{Y}}$ given by

$$f_{\underline{Y}}(y) = f_{\underline{X}}(\underbrace{g(y)}_{f^{-1}(y)}) \left| \det J_g(y) \right| \qquad y \in \mathbb{R}^n$$



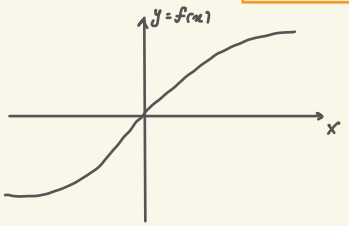
where $J_g = J_g(y)$ is the $n \times n$ Jacobian matrix defined by

$$[J_g(y)]_{ij} = \frac{\partial g_i}{\partial y_j}(y) = \frac{\partial f_i^{-1}}{\partial y_j}(y)$$

Claim: Let $f: \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be one to one and invertible. (hence, onto its range too) such that both f and f^{-1} have continuous partial derivatives w.r.t co-ordinates Let $g \triangleq f^{-1}$, $g: \mathbb{R}^n \longrightarrow \mathbb{R}^n$ Then

$\underline{Y} = f(\underline{X})$ has cdf $F_{\underline{Y}}$ that is absolutely continuous with pdf $f_{\underline{Y}}$ given by

$$f_{\underline{Y}}(y) = f_{\underline{X}}\underbrace{(g(y))}_{f^{-1}(y)} \left| \det J_g(y) \right| \qquad y \in \mathbb{R}^n$$



Where $J_g = J_g(y)$ is the $n \times n$ Jacobian matrix defined by

$$[J_g(y)]_{ij} = \frac{\partial g_i}{\partial y_j}(y) = \frac{\partial f_i^{-1}}{\partial y_j}(y)$$

Proof for $n = 1$

$\therefore f: \mathbb{R} \to \mathbb{R}$ is one-one, invertible (hence onto) function such that both f and f^{-1} have continuous derivatives. To prove is

$$f_{\underline{Y}}(y) = f_{\underline{X}}(g(y)) \cdot \left| \det J_g(y) \right| = f_{\underline{X}}(g(y)) \cdot \left| \frac{d}{dy} g(y) \right|$$

A continuous function is invertible $\iff$ If it is strictly monotone

$\therefore$ f is differentiable $\implies$ f continuous $\iff$ f is strictly montone

Assume $f: R_f \longrightarrow \mathbb{R}$ is strictly Increasing

$\therefore f^{-1} = g: \mathbb{R} \longrightarrow R_f$ is also strictly $\uparrow$. Hence

$$\begin{aligned} F_Y(y) = P(Y < y) &= P(f(x) < y) \\ &= P(X < f^{-1}(y)) \quad \cdots \text{ Strictly } \uparrow \text{ property} \\ &= P(X < g(y)) \end{aligned}$$

$$= \int_{-\infty}^{g(y)} f_x(x)\, dx$$

$\therefore$ From Leibniz rule,

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} \int_{-\infty}^{g(y)} f_x(x)\, dx$$

$$= f_x(g(y)) \frac{d}{dy} g(y) - f_x(-\infty) \cdot 0 + \int_{-\infty}^{g(y)} 0 \cdot dx$$

$$= f_x(g(y)) \cdot \left(\frac{d}{dy} g(y)\right) \quad (>0) \quad \cdots ①$$

Assume $f: R_f \longrightarrow \mathbb{R}$ is strictly Decreasing

$\therefore f^{-1} = g : \mathbb{R} \longrightarrow R_f$ is also strictly $\downarrow$. Hence

$$F_Y(y) = P(Y < y) = P(f(x) < y)$$

$$= P(X > f^{-1}(y)) \quad \cdots \text{Strictly } \downarrow \text{ property}$$

$$= P(X > g(y))$$

$$= 1 - P(X < g(y))$$

$$= 1 - \int_{-\infty}^{g(y)} f_x(x)\, dx$$

$$\therefore f_Y(y) = \frac{d}{dy} F_Y(y) = -\frac{d}{dy} \int_{-\infty}^{g(y)} f_x(x)\, dx$$

$$= -f_x(g(y)) \cdot \left(\frac{d}{dy} g(y)\right) \quad (<0) \quad \cdots ②$$

from ①, ② $\quad f_Y(y) = f_x(g(y)) \left| \frac{d}{dy} g(y) \right|$

■

Leibniz Integration Rule

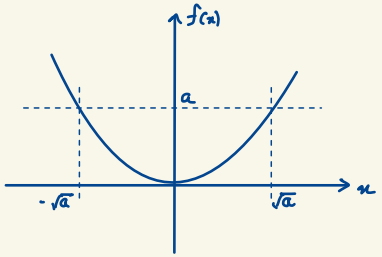
This uses Leibniz chain rule for integration,

$$\therefore \frac{d}{du} \int_{a(u)}^{b(u)} h(u,v)\, dv = h(u, b(u)) \cdot \frac{d}{du} b(u) - h(u, a(u)) \cdot \frac{d}{du} a(u) + \int_{a(u)}^{b(u)} \frac{\partial}{\partial u} h(u,v)\, dv$$

Example - Show f_Y where $Y = f(X)$ can be computed from f_X and f when f^{-1} does not exist.

Assume $X \sim N(0,1) \Rightarrow f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad x \in \mathbb{R}$

And $f: \mathbb{R} \to \mathbb{R}^+ \cup \{0\}$ given by $f(x) = x^2 \quad x \in \mathbb{R}$



$Y = f(x) = x^2$ does not have an inverse

$$F_Y(a) = P(Y < a) = \begin{cases} 0 & a \leq 0 \\ P(-\sqrt{a} < x < \sqrt{a}) & a > 0 \end{cases}$$

$\therefore a > 0$

$$\Rightarrow F_Y(a) = P(-\sqrt{a} < x < \sqrt{a}) = \int_{-\sqrt{a}}^{\sqrt{a}} f_X(x)\, dx$$

$$= \int_{-\sqrt{a}}^{\sqrt{a}} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\sqrt{a}} e^{-x^2/2}\, dx$$

$$\Rightarrow f_Y(a) = \frac{d}{dy} F_Y(a) = \frac{d}{dy}\left(\sqrt{\frac{2}{\pi}} \int_0^{\sqrt{a}} e^{-x^2/2}\, dx\right)$$

$$= \frac{1}{\sqrt{2\pi a}} e^{-a/2} \quad \ldots\ldots \text{ from Leibniz Integration}$$

$$\therefore f_Y(a) = \begin{cases} 0 & a \leq 0 \\ \frac{1}{\sqrt{2\pi a}} e^{-a/2} & a > 0 \end{cases}$$

To Show: $|\det J_g(y)| = \frac{1}{|\det \underline{A}|}$ $\forall\, y \in \mathbb{R}^2$ where A^{-1} exists and $y = \underline{A} \cdot g(y)$

Assume $A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \Rightarrow \det A = ad - bc \neq 0$

$\therefore A^{-1} = \frac{\begin{vmatrix} d & -b \\ -c & a \end{vmatrix}}{ad - bc}$ exists

$y = \underline{A} \cdot g(y) \Rightarrow g(y) = \underline{A}^{-1} y = \frac{\begin{vmatrix} d & -b \\ -c & a \end{vmatrix}}{ad - bc} \begin{vmatrix} y_1 \\ y_2 \end{vmatrix} = \begin{vmatrix} \frac{dy_1 - by_2}{ad - bc} \\ \frac{-cy_1 + ay_2}{ad - bc} \end{vmatrix}$ ($g_1(y)$, $g_2(y)$)

Get the Jacobian of $g(y)$ by using partial derivatives of g_1 and g_2, we get,

$$J_g(y) = \frac{1}{\det \underline{A}} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\Rightarrow \det J_g(y) = \det \begin{bmatrix} d/\det \underline{A} & -b/\det \underline{A} \\ -c/\det \underline{A} & a/\det \underline{A} \end{bmatrix}$$

$$= \frac{da - bc}{(\det \underline{A})^2}$$

$$= \frac{1}{\det A}$$

$$\Rightarrow |\det J_g(y)| = \frac{1}{|\det A|} \quad \forall\, y \in \mathbb{R}^2$$

Example "Jacobian Approach"

Let $\underline{X} = (X_1, X_2)$ is $\mathbb{R}^2$- valued rv with joint pdf of

$$f_{\underline{X}}(x_1, x_2) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x_1^2 + x_2^2)} \qquad (x_1, x_2) \in \mathbb{R}^2$$

If, $Y = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \underline{A} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ where $\underline{A}$ is a constant non-singular matrix with $\underline{A}^{-1}$ exists, Find pdf of $f_{\underline{Y}}$

Denote $\underline{x}^T = [x_1, x_2] \Rightarrow x_1^2 + x_2^2 = \underline{x}^T \underline{x}$ hence $f_{\underline{X}}(x_1, x_2) = \frac{e^{-\frac{1}{2}\underline{x}^T\underline{x}}}{\sqrt{2\pi}}$

$$f_{\underline{Y}}(y) = f_{\underline{X}}(g(y)) \cdot |\det J_g(y)|$$

$g = \underline{A}^{-1}$

$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

$$f_{\underline{X}}(g(y)) = f_{\underline{X}}(\underline{A}^{-1} y)$$

$$= \frac{1}{2\pi} e^{-\frac{1}{2}(\underline{A}^{-1}y)^T(\underline{A}^{-1}y)}$$

$$= \frac{1}{2\pi} e^{-\frac{1}{2} \cdot y^T(\underline{A}\underline{A}^T)^{-1}y}$$

$\frac{1}{|\det A|}$ from previous example

$Y = AX$

$A^{-1}Y = X$

$$\therefore f_{\underline{Y}}(y) = \frac{1}{2\pi |\det \underline{A}|} \cdot e^{-\frac{1}{2} y^T(\underline{A}\underline{A}^T)^{-1}y} \qquad y \in \mathbb{R}$$

$Y = f(\underline{X})$, How to Find $\mathbb{E}[\underline{Y}]$

$(\Omega, \mathcal{A}, P)$ is a probability measure with $\underline{X}: \Omega \longrightarrow \mathbb{R}^n$

$\underline{Y}: \mathbb{R}^n \longrightarrow \mathbb{R}$ $\quad Y = f(\underline{X})$

How to compute $\mathbb{E}[\underline{Y}]$ when it exists.

Approach 1 If $F_{\underline{X}}$ (known), f is $\longrightarrow$ Compute F_Y $\longrightarrow$ Compute $\mathbb{E}[Y]$

Approach 2 Direct approach by using $F_{\underline{X}}$ and f, without finding F_Y.

- Approximate rvs $\underline{X}$ by simple $\Longrightarrow$ $\underline{X} = (\underline{X}_1, \ldots, \underline{X}_n)$

  $\underline{X}^{(k)} = \left(\underline{X}_1^{(k)}, \ldots, \underline{X}_n^{(k)}\right)$

- Compute $\mathbb{E}[f(\underline{X}^{(k)})]$, which depends on $F_{\underline{X}}$ and f

- Take $\lim_{k \to \infty}$ to find $\mathbb{E}[f(\underline{X})]$ from above. In particular if both $\underline{X}$ and $\underline{Y}$ have pdfs, then we must have

$$\mathbb{E}[\underline{Y}] = \int_{-\infty}^{\infty} y \, f_{\underline{Y}}(y) \, dy$$

$$= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, \ldots, x_n) \cdot f_{\underline{X}}(x_1, \ldots, x_n) \, dx_1 \ldots dx_n$$

Proof: Approximate f by piecewise constant functions

"Law of Unconscious Statistician" (LOTUS)

- For a general Borel $f: \mathbb{R}^n \longrightarrow \mathbb{R}^m$ $\quad \mathbb{E}[\underline{Y}] = \begin{bmatrix} \mathbb{E}[Y_1] \\ \vdots \\ \mathbb{E}[Y_m] \end{bmatrix}$ $\quad Y_i = f_i(\underline{X})$

Generating Functions and Inequalities

1) k^{th} Moment

Definition:- $(\Omega, \mathcal{A}, P)$ is a probability measure. X is a $\mathbb{R}$-valued rv. The k^{th} Moment of X ($\mathbb{E}[X^k]$) for $k = 1, 2, 3, \ldots$ is,

$$\mathbb{E}[X^k] = \sum_j x_j^k \, P(X = x_j) = \sum_j x_j^k \, p_X(x_j)$$

$$= \int_{-\infty}^{\infty} x_j^k \cdot f_X(x) \, dx$$

- Provided that the integration exist $\longrightarrow$ k^{th} absolute moment of X = $\mathbb{E}[|X|^k]$

$\mathbb{E}[|X|^k] = \mathbb{E}[|X^k|]$ always exists, but may go to ∞.

Properties of k^{th} Moment

1) If k is odd then $\mathbb{E}[X^k]$ need not exist

2) If k is even then $\mathbb{E}[X^k]$ always exists, but may $= +\infty$

3) $\mathbb{E}[X^k]$ is finite $\iff$ $\mathbb{E}[|X|^k]$ is finite.

hint : $Y(\omega) = \underset{\geq 0}{Y^+(\omega)} - \underset{\geq 0}{Y^-(\omega)}$ $\quad \omega \in \Omega$

$\Rightarrow \mathbb{E}[Y] = \mathbb{E}[Y^+] - \mathbb{E}[Y^-]$

$\Rightarrow \mathbb{E}[|Y|] = \mathbb{E}[Y^+] - \mathbb{E}[Y^-]$ .... Since $Y^+, Y^- \geq 0$

4) If $\mathbb{E}[X^k)$ exist and is finite for some k $\Longrightarrow$ $\mathbb{E}[X^j]$ exists and is finite $\forall\ 1 \leq j \leq k$

Proof: Assume $\mathbb{E}[X^k)$ exists and $< \infty$. $\iff$ $\mathbb{E}[|X|^k] = \mathbb{E}[|X^k|] < \infty$ .... property (iii)

Claim: $\mathbb{E}[|X|^j] < \infty$ $\quad j = 1, 2, \ldots, k$

$\mathbb{E}[|X|^j] = \mathbb{E}[|X|^j \cdot 1]$

$$= E\left[|x|^j \cdot \mathbb{1}_{(|x|\le 1)} + |x|^j \cdot \mathbb{1}_{(|x|>1)}\right]$$ By using the partition $1 = \mathbb{1}_{(|x|\le 1)} + \mathbb{1}_{(|x|>1)}$

$$= \underbrace{E\left[|x|^j \mathbb{1}_{(|x|\le 1)}\right]}_{\le 1} + E\left[\underbrace{|x|^j}_{\le |x|^k} \cdot \mathbb{1}_{(|x|>1)}\right]$$

$$\le 1 + E\left[|x|^k \cdot \mathbb{1}_{(|x|>1)}\right] \le 1 + E[|x|^k] < \infty$$

$\therefore E[|x|^j] < \infty \iff E[x^j] < \infty$ and exist ..... from property (iii)

■

Remarks:-

(i) First moment of X :

$$E[x] = \mu = \mu_x = \text{mean of } x$$

(ii) Second moment of $X - \mu_x$:

$$E[(x-\mu)^2] \triangleq \sigma^2 = \sigma_x^2 = \text{variance of } x = \text{Second centralized moment of } x$$

$$\sigma^2 = E[(x-\mu_x)^2] = E[x^2] - (E[x])^2 = E[x^2] - \mu_x^2 \ge 0$$

(iv) Standard deviation of X:

$$\sigma(x) \triangleq \sigma = +\sqrt{\text{Variance } (x)}$$

Note:

- If $E[x^k] < \infty \quad \forall \; k = 1, 2, \ldots$, these moments uniquely determine F_x.
- Finite moments give useful info about pmf or pdf.

2) Effects of Linear Transformations

$$Y = \alpha X + \beta \qquad \alpha, \beta \in \mathbb{R} \text{ constants}$$

$$\mathbb{E}[Y] = \mu_Y = \alpha \cdot \mu_X + \beta$$

$$\text{Var}[Y] = \sigma_Y^2 = \alpha^2 \sigma_X^2$$

Normalization

Definition: If X has mean μ_X and variance $\sigma_X^2 < \infty$, then

$$Y \triangleq \frac{X - \mu_X}{\sigma_X} \quad \text{has} \quad \mu_Y = 0 \quad \sigma_Y^2 = 1$$

This is an invertible transformation.

Co-variance and Correlation

Definition: If X_1, X_2 are jointly distributed $\mathbb{R}$-valued rvs with F_{X_1, X_2} known, with finite means μ_1 and μ_2

$$\begin{aligned} \text{Covariance of } X_1, X_2 &= \text{Cov}[X_1, X_2] = \sigma_{12} = \sigma_{21} \\ &= \mathbb{E}[(X_1 - \mu_1)(X_2 - \mu_2)] \\ &= \mathbb{E}[X_1 X_2] - \mathbb{E}[X_1] \cdot \mathbb{E}[X_2] \\ &= \mathbb{E}[X_1 X_2] - \mu_1 \mu_2 \end{aligned}$$

$$\text{Correlation of } X_1, X_2 = \mathbb{E}[X_1 X_2]$$

Show: $\sigma_{X_1}^2 + \sigma_{X_2}^2 < \infty \implies \text{Cov}[X_1, X_2]$ exist.

$(\sigma_{X_1}^2 < \infty, \sigma_{X_2}^2 < \infty)$

Uncorrelated

If $\mathrm{Cov}[X_1, X_2] = 0 \implies X_1$ and X_2 are uncorrelated rvs $\iff$ Notation $X_1 \perp X_2$

Fact: $\mathrm{Var}[X_1 + X_2] = \mathbb{E}\left[\{(X_1 + X_2) - \mathbb{E}[X_1 + X_2]\}^2\right]$

$= \mathbb{E}\left[(X_1 - \mathbb{E}[X_1])^2\right] + \mathbb{E}\left[(X_2 - \mathbb{E}[X_2])^2\right] + 2\,\mathbb{E}\left[(X_1 - \mathbb{E}[X_1])(X_2 - \mathbb{E}[X_2])\right]$

$= \mathrm{Var}[X_1] + \mathrm{Var}[X_2] + 2 \cdot \mathrm{Cov}[X_1, X_2]$

X_1 and X_2 are uncorrelated $\iff \mathbb{E}[X_1 X_2] = \mathbb{E}[X_1] \cdot \mathbb{E}[X_2]$

$X_1 \perp X_2$

$\mathrm{Var}[X_1 + X_2] = \mathrm{Var}[X_1] + \mathrm{Var}[X_2] \iff$ Correlation Coefficient

$\sigma^2_{X_1+X_2} = \sigma^2_{X_1} + \sigma^2_{X_2}$ $\qquad \rho_{X_1 X_2} \triangleq \dfrac{\mathrm{Cov}[X_1, X_2]}{\sigma_{X_1}\sigma_{X_2}} = 0$

Example: $(\Omega, \mathcal{A}, P)$ $\qquad \mathbb{1}_A(\omega) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}$ $\qquad A \in \mathcal{A}$

$\mathbb{E}[\mathbb{1}_A] = P(A)$

$\mathrm{Var}[\mathbb{1}_A] = \mathbb{E}[\mathbb{1}_A^2] - (\mathbb{E}[\mathbb{1}_A])^2 = P(A) - P(A)^2 = P(A)[1 - P(A)]$

$= P(A) \cdot P(A^c)$

Andrey Markov
1856 - 1922

3) Markov Inequality

Given $(\Omega, \mathcal{A}, P)$. Y is a $\mathbb{R}$-valued rv. Assume $\mathbb{E}[|Y|^k] < \infty$ for a given integer $k \geq 1$

$$P(|Y| > \varepsilon) \leq \frac{\mathbb{E}[|Y|^k]}{\varepsilon^k}$$

Proof: For any fixed $\varepsilon > 0$

$$\Rightarrow |Y| = |Y| \cdot \mathbb{1}_{(|Y| \leq \varepsilon)} + |Y| \cdot \mathbb{1}_{(|Y| > \varepsilon)} \geq |Y| \cdot \mathbb{1}_{(|Y| > \varepsilon)}$$

$$\Rightarrow |Y|^k \geq |Y|^k \cdot \mathbb{1}_{(|Y| > \varepsilon)} \geq \varepsilon^k \mathbb{1}_{(|Y| > \varepsilon)}$$

$$\Rightarrow \mathbb{E}[|Y|^k] \geq \mathbb{E}[\varepsilon^k \mathbb{1}_{(|Y| > \varepsilon)}] = \varepsilon^k \cdot \mathbb{E}[\mathbb{1}_{(|Y| > \varepsilon)}]$$

$$\Rightarrow \mathbb{E}[|Y|^k] \geq \varepsilon^k P(|Y| > \varepsilon)$$

$$\Rightarrow P(|Y| > \varepsilon) \leq \frac{\mathbb{E}[|Y|^k]}{\varepsilon^k}$$

Tail probability

■

- The bound is useful only when RHS < 1

$$Y \longrightarrow Y - \mu_Y$$
$$\therefore P(|Y - \mu_Y| > \varepsilon) \leq \frac{\mathbb{E}[|Y - \mu_Y|^k]}{\varepsilon^k}$$

Simply Markov Inequality $(k=1 \,/\, Y \to X)$

Assume $\mathbb{E}[|X|] < \infty$, $\forall\, \varepsilon > 0$

$$P(|X| > \varepsilon) \leq \frac{\mathbb{E}[|X|]}{\varepsilon}$$

$$P(|X - \mu_X| > \varepsilon) \leq \frac{\mathbb{E}[|X - \mu_X|]}{\varepsilon}$$

Pafnuty Chebyshev
1821 - 1894

4) Chebyshev Inequality

Assume $\mathbb{E}[X^2] < \infty$. Then $\forall\ \varepsilon > 0$

$$P(|X| > \varepsilon) \leq \frac{\mathbb{E}[X^2]}{\varepsilon^2}$$

$$P(|X - \mu_X| > \varepsilon) \leq \frac{\mathbb{E}[|X-\mu_X|^2]}{\varepsilon^2} = \frac{\text{Var}[X]}{\varepsilon^2}$$

This is the Markov Inequality where, $n = 2$ $Y \to X$.

Ex: Application in estimating probabilities.
Study shows that the expected no. of simultaneous phone calls at a base station $\simeq C_{avg} = 100$. The actual no. of calls is random and station is designed to handle up to $C_{max} = 150$ calls. Find the Probability that station recieves more than 150 calls.

X - rv showing the actual no. of phone calls at once
$E[X] = 100$
By Markov Inequality,

$$P(X \geq 151) \leq \frac{E[X]}{151} = \frac{100}{151} = 0.66$$

Assume that the additional information of variance = 50 is given. Find a better bound.

$$P(X \geq 151) \leq \frac{E[X^2]}{151^2} = \frac{\text{Var}[X] + E[X]^2}{151^2}$$

$$\leq \frac{50 + 100^2}{151^2} = 0.44$$

5) Generalized Markov / Chebyshev Inequality

X is a $\mathbb{R}$-valued random variable. Let $g: \mathbb{R} \longrightarrow \mathbb{R}^+ \cup \{0\}$ g is a non-negative valued and non-decreasing Borel mapping, s.t $\mathbb{E}[g(x)]$ exists and is $< \infty$. $\forall \varepsilon > 0$

$$P(X \geq \varepsilon) \leq \frac{\mathbb{E}[g(x)]}{g(\varepsilon)}$$

Proof: $g(x) = \underbrace{g(x) \cdot \mathbb{1}_{(x<\varepsilon)}}_{\geq 0} + \underbrace{g(x) \cdot \mathbb{1}_{(x \geq \varepsilon)}}_{\geq 0}$

$$\mathbb{E}[g(x)] = \mathbb{E}[g(x) \cdot \mathbb{1}_{(x<\varepsilon)}] + \mathbb{E}[g(x) \cdot \mathbb{1}_{(x \geq \varepsilon)}]$$

$$\geq \mathbb{E}[g(x) \cdot \mathbb{1}_{(x \geq \varepsilon)}]$$

$$\geq \mathbb{E}[g(\varepsilon) \cdot \mathbb{1}_{(x \geq \varepsilon)}] \quad \cdots\cdots \text{ } g \text{ is non decreasing.}$$

$$\geq g(\varepsilon) \cdot \mathbb{E}[\mathbb{1}_{(x \geq \varepsilon)}]$$

$$\geq g(\varepsilon) \cdot P(x > \varepsilon)$$

$$\Rightarrow P(x > \varepsilon) \leq \frac{\mathbb{E}[g(x)]}{g(\varepsilon)}$$

■

Exercise: Obtain the "Simple Markov" inequality and Chebyshev inequality as special cases.

6) Cauchy - Schwarz Inequality

- Recall from Linear Algebra:

$\alpha, \beta \in V$ = inner product space

$$|\langle \alpha, \beta \rangle| \leq \|\alpha\| \, \|\beta\|$$

With the equality if $\alpha \neq 0$ and $\longrightarrow$ $\beta = \dfrac{\langle \beta, \alpha \rangle}{\|\alpha\|^2} \cdot \alpha$

$\langle \beta, \alpha \rangle = \overline{\langle \alpha, \beta \rangle}$ Complex Conjugate

Equality $\iff$ α, β are linearly independant

- Cauchy - Schwarz Inequality in our case for $(\Omega, \mathcal{A}, P)$

Let X_1, X_2 be jointly distributed $\mathbb{R}$-valued rvs (with given CDF F_{X_1,X_2})
If $\sigma_{X_1}^2 < \infty$ and $\sigma_{X_2}^2 < \infty$, then

$$\left(\mathbb{E}[X_1 X_2]\right)^2 \leq \mathbb{E}[X_1^2] \cdot \mathbb{E}[X_2^2] \quad *$$

- Useful bound for the correlation of X_1, X_2

Equality holds when

a) $P(X_1 = 0)$ or $P(X_2 = 0) = 1$

and

b) iff X_1 and X_2 are linearly related

ie. $P(X_1 = a X_2) = 1$ for some $a \neq 0$ in $\mathbb{R}$

ie. $\left(X_2 = \frac{1}{a} X_1\right)$

- Furthermore

Cauchy Schwarz Inequality $\Longrightarrow$ $-1 \leq \rho_{X_1,X_2} \triangleq \dfrac{\mathrm{Cov}[X_1, X_2]}{\sigma_{X_1} \sigma_{X_2}} \leq 1$ **

with equality $= 1 \iff X_2 = a X_1 \quad a > 0$

$= -1 \iff X_2 = -a X_1$

Proof: ▸ Observe that ⊛ is trivial for the cases

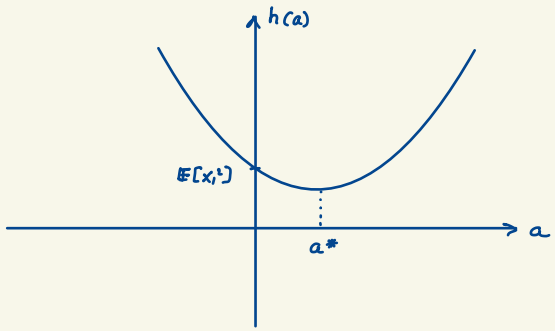
$P(X_2 = 0) = 1$ or $P(X_1 = 0) = 1$

or

$P(X_1 = a X_2) = 1$ for any $a \neq 0 \in \mathbb{R}$

▸ Set $h(a) \triangleq \mathbb{E}[(X_1 - aX_2)^2]$ $-\infty < a < \infty$. Clearly $h(a) \geq 0$ $\forall a \in \mathbb{R}$

$\therefore h(a) = \mathbb{E}[X_1^2] + a^2 \mathbb{E}[X_2^2] - 2a\,\mathbb{E}[X_1 X_2]$



Assume minimum is attained at $a = a^*$

$$\frac{d}{da} h(a) = 2a\,\mathbb{E}[X_2^2] - 2\,\mathbb{E}[X_1 X_2]$$

$$\Rightarrow \left.\frac{d}{da} h(a)\right|_{a=a^*} = 2a^*\,\mathbb{E}[X_2^2] - 2\,\mathbb{E}[X_1 X_2] = 0$$

$$\Rightarrow a^* = \frac{\mathbb{E}[X_1 X_2]}{\mathbb{E}[X_2^2]}$$

$\therefore$ It is sufficient to prove that $h(a^*) \geq 0$. Then we get CS. $\therefore$ If

$\Rightarrow 0 \leq h(a^*) = \mathbb{E}[X_1^2] + a^{*2}\,\mathbb{E}[X_2^2] - 2a^*\,\mathbb{E}[X_1 X_2]$ ..... substitute a^* from above

$$= \mathbb{E}[X_1^2] - \frac{(\mathbb{E}[X_1 X_2])^2}{\mathbb{E}[X_2^2]}$$

$\Leftrightarrow (\mathbb{E}[X_1 X_2])^2 \leq \mathbb{E}[X_1^2] \cdot \mathbb{E}[X_2^2]$

■

Equality Case

$\Leftrightarrow \quad h(a^*) = 0$

$\Leftrightarrow \quad \mathbb{E}[(X_1 - a^* X_2)^2] = 0 \quad \left(\text{ie } P(X_1 = a^* X_2) = 1 \right.$

$\left. \text{ie } P\left(X_1 = \frac{\mathbb{E}[X_1 X_2]}{\mathbb{E}[X_2^2]} \cdot X_2\right) = 1 \right)$

$\Leftrightarrow \quad P(|X_1 - a^* X_2| > \varepsilon) \leq \frac{\mathbb{E}[|X_1 - a^* X_2|^2]}{\varepsilon^2} \leq 0 \quad \forall \varepsilon > 0$

from Chebyshevs Inequality

$\Leftrightarrow \quad P(|X_1 - a^* X_2| = 0) = 1$

$\Leftrightarrow \quad P(X_1 = a^* X_2) = 1$

Probability Generating Function

- Probability Generating function for $\mathbb{N}$-valued rvs

  $\mathbb{N} = \{0, 1, 2, \dots\}$

  $G_X(z) \triangleq \mathbb{E}[z^X] = \sum_{n=0}^{\infty} z^n p_X(n)$ z-Transform

- Probability Generatig function for $\mathbb{R}$-valued rvs

  $\psi_X(s) \triangleq \mathbb{E}[e^{sX}] = \int_0^{\infty} e^{sx} \cdot f_X(x)\, dx$ Laplace Transform

Examples

- $P(X = c) = 1 \Rightarrow G_X(z) = \mathbb{E}[z^X] = z^c$
- $P(X = 1) = p \Rightarrow G_X(z) = \mathbb{E}[z^X] = p \cdot z^1 + (1-p) z^0 = 1 - p + p \cdot z$

Theorem : $\mathbb{E}[X] = G'(1)$

Theorem : X, Y are Independant $\Rightarrow G_{X+Y}(z) = G_X(z) \cdot G_Y(z)$

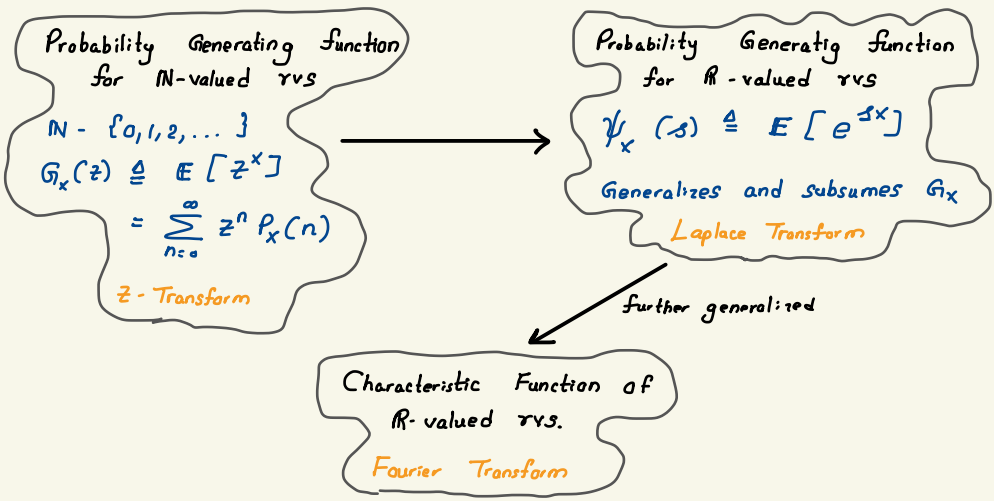
Definition : Joint Probability Generating Function $= G_{X_1, X_2}(z_1, z_2) = \mathbb{E}[z_1^{X_1} z_2^{X_2}]$

Independance $\Longleftrightarrow G_{X_1 X_2}(z_1, z_2) = G_{X_1}(z_1) \cdot G_{X_2}(z_2)$

(Generalized case of a Probability Generating Function)

7) Characteristic Function of a Random Variable

- Characteristic function of a RV $\sim$ Triansformation Method
- From this characteristic function, all the moments of a RV can be extracted.



Definition: The characteristic function of a $\mathbb{R}$- valued rv $\underline{X} = (X_1, \ldots, X_n)$ is the mapping,

$$\phi_{\underline{X}} : \mathbb{R}^n \longrightarrow \mathbb{C}$$

where

$$\phi_{\underline{X}}(\underline{u}) = E\left[e^{i \underline{u}^T \underline{X}}\right] \qquad i = \sqrt{-1} \quad \underline{u} \in \mathbb{R}^n$$

$$= E[\cos \underline{u}^T \underline{X}] + i\, E[\sin \underline{u}^T \underline{X}]$$

- Here $\underline{u} = (u_1, \ldots, u_n) \in \mathbb{R}^n$

  $\underline{u}^T \underline{X} = u_1 X_1 + \ldots + u_n X_n$

- If $\underline{X}$ has a joint pdf $f_{\underline{X}}$, then

$$\phi_{\underline{X}}(\underline{u}) = \int_{-\infty}^{\infty} \ldots \int_{-\infty}^{\infty} e^{i \underline{u}^T \underline{x}} f_{\underline{X}}(\underline{x}) \cdot dx_1 \ldots dx_n$$

Properties of Characteristic Function

1) $\phi_{\underline{x}}(\underline{u})$ always exists for $\forall\ \underline{u} \in \mathbb{R}^n$

since $|\cos \underline{u}^T\underline{x}| \leq 1$ and $|\sin \underline{u}^T\underline{x}| \leq 1$

2) $\phi_{\underline{x}}(\underline{0}) = 1$ and $|\phi_{\underline{x}}(\underline{u})| \leq 1$

3) The mapping $\phi_{\underline{x}} : \mathbb{R}^n \longrightarrow \mathbb{C}$ is uniformly continuous on $\mathbb{R}^n$.

$\forall \varepsilon > 0 \quad \exists\ \delta(\varepsilon) > 0$ such that

$$|\underline{u}_1 - \underline{u}_2| \leq \delta(\varepsilon) \Longrightarrow |\phi_{\underline{x}}(\underline{u}_1) - \phi_{\underline{x}}(\underline{u}_2)| < \varepsilon$$

4) Knowing $\phi_{\underline{x}} \iff$ Knowing $F_{\underline{x}}$

a) If $F_{\underline{x}}$ is absolutely continuous with pdf $f_{\underline{x}}$, then

$$\phi_{\underline{x}}(\underline{u}) = \int_{\mathbb{R}^n} e^{i\underline{u}^T\underline{x}} \cdot f_{\underline{x}}(\underline{x})\, d\underline{x} \qquad \underline{u} \in \mathbb{R}^n$$

$$= \mathcal{F}(f_{\underline{x}}(\underline{u}))$$

$$=$$ Fourier Transform of $f_{\underline{x}}$

$\therefore$ By using the inversion formula for Fourier Transform,

$$f_{\underline{x}}(\underline{x}) = (\mathcal{F}^{-1}\phi_{\underline{x}})(\underline{x})$$

$$= \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \phi_{\underline{x}}(\underline{u}) \cdot e^{-i\underline{u}^T\underline{x}}\, d\underline{u}$$

b) If $F_{\underline{x}}$ is singular. ie the pdf is of discrete type

$$\phi_{\underline{x}}(\underline{u}) = \mathbb{E}[e^{i\underline{u}^T\underline{x}}] = \sum_{\underline{x} \in \mathcal{S}_{\underline{x}}} e^{i\underline{u}^T\underline{x}} \cdot P(\underline{x} = \underline{x}) \qquad \underline{u} \in \mathbb{R}^n$$

$$= \sum_{\underline{x} \in \mathcal{S}_{\underline{x}}} e^{i\underline{u}^T\underline{x}} \cdot P_{\underline{x}}(\underline{x})$$

$\therefore$ By using the inversion formula for Fourier Transform,

$$\lim_{U\to\infty} \frac{1}{(2U)^n} \int_{-U}^{U} \cdots \int_{-U}^{U} \phi_{\underline{x}}(\underline{u}) \cdot e^{-i\underline{u}^T\underline{x}} \, d\underline{u} = \begin{cases} \rho_{\underline{x}}(\underline{x}) & \underline{x} \in \mathcal{S}_{\underline{x}} \\ 0 & \text{otherwise} \end{cases}$$

5) Generation of Moments

If X is a $\mathbb{R}$ valued rv with a finite k^{th} order moments. (ie $\mathbb{E}[X^k] < \infty \implies \mathbb{E}[X^l] < \infty \quad 1 \le l \le k$). Then $\phi_{\underline{x}}$ has a k^{th} derivative at $u = 0$ and

$$\mathbb{E}[X^l] = \frac{1}{i^l} \frac{d^l}{du^l} \phi_x(u) \Big|_{u=0} \qquad 1 \le l \le k$$

Proof: A rough and plausible proof

By Taylor series expansion,

$$\Rightarrow e^{iux} = 1 + iux + \frac{(iux)^2}{2!} + \cdots + \frac{(iux)^k}{k!} + O((ux)^k)$$

$$\Rightarrow \phi_x(u) = \mathbb{E}[e^{iux}]$$

$$= 1 + iu \cdot \mathbb{E}[x] + \frac{(iu)^2}{2!}\mathbb{E}[x^2] + \cdots + \frac{(iu)^k}{k!}\mathbb{E}[x^k] + O(u^k)$$

$\therefore$ For $1 \le l \le k$

$$\frac{d^l}{du^l} \phi_x(u) \Big|_{u=0} = i^l \mathbb{E}[X^l]$$

(6) Effects of Linear Transformations

X - $\mathbb{R}^n$ valued RV and A - $m \times n$ matrix with $\mathbb{R}$ valued entries $\underline{b} \in \mathbb{R}^n$. If

$$\underline{Y} \triangleq \underline{A}\underline{X} + \underline{b} \Rightarrow \phi_{\underline{Y}}(\underline{v}) = e^{i\underline{b}^T\underline{v}} \cdot \phi_{\underline{X}}(\underline{A}^T v) \qquad v \in \mathbb{R}^m$$

Applications

a) X, Y - $\mathbb{R}$ valued rvs and if $Y \triangleq \frac{X - \mu_X}{\sigma_X}$ normalization,

$$\phi_Y(v) = e^{i\frac{\mu_X v}{\sigma_X}} \cdot \phi_X\left(\frac{v}{\sigma_X}\right) \qquad v \in \mathbb{R}$$

b) $\underline{X} = [X_1, X_2]$, $\underline{A} = [1, 1]$, $b = 0$ and if $Y = X_1 + X_2$

$$\phi_Y(v) = \phi_{\underline{X}}(v, v) \qquad v \in \mathbb{R}$$

Ex:- X_1, X_2 are $\mathbb{R}$-valued rvs with joint pdf $\underline{X} = (X_1, X_2)$ given by, $-\infty < x_1, x_2 < \infty$

$$f_{\underline{X}}(x_1, x_2) = \frac{1}{2\pi} e^{-\frac{1}{2}(x_1^2 + x_2^2)}$$

This is "Joint Gaussian" distribution. then $f_Y(y) = N(0, 2)$

Proof: $\phi_{\underline{X}}(\mu_1, \mu_2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{i(u_1 x_1 + \mu_2 x_2)} \cdot f_{\underline{X}}(x_1, x_2)\, dx_1\, dx_2$

$$= e^{-\frac{1}{2}(u_1^2 + u_1^2)}$$

Let $Y \triangleq X_1 + X_2$. Then by property (6).

$$\phi_Y(v) = \phi_{\underline{X}}(v, v) = e^{-v^2} \qquad v \in \mathbb{R}$$

$$\Rightarrow f_Y(y) = \left(\mathcal{F}^{-1}\phi_Y\right)(y) \qquad -\infty < y < \infty$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-v^2} \cdot e^{-ivy}\, dv$$

$$\vdots$$

$$= \frac{1}{\sqrt{4\pi}} e^{-y^2/4} \sim N(0, 2) \qquad -\infty < y < \infty$$

Moment Generating Function of a Random Variable

Definition: The moment generating function of a $\mathbb{R}^n$ valued rv of $\underline{X} = (X_1, X_2, \ldots, X_n)$ is a mapping $\psi_{\underline{X}} : \mathbb{R}^n \to [0, \infty)$ is defined by

$$\psi_{\underline{X}}(\underline{s}) = \mathbb{E}\left[e^{\underline{s}^T \underline{X}} \right] \qquad \underline{s} = (s_1, \ldots s_n) \in \mathbb{R}^n$$

Properties of Moment Generating Function

(i) $0 \leq \psi_{\underline{X}}(\underline{s}) \leq +\infty$

(ii) $\psi_{\underline{X}}$ can be used to generate the moment of $\underline{X}$, if they exist. If all the moments of $\underline{X}$ are finite, so is $\psi_{\underline{X}}$.

Take $n = 1 \Rightarrow e^{\underline{s}^T \underline{X}} = e^{sX} = 1 + sX + \ldots + \frac{(sX)^k}{k!} + O(s^k X^k)$

$$\Rightarrow \psi_X(s) = \mathbb{E}\left[e^{\underline{s}^T \underline{X}} \right] = 1 + s\mathbb{E}[X] + \ldots + \frac{s^k}{k!}\mathbb{E}[X^k] + O(s^k)$$

$$\Rightarrow \frac{d^l}{ds^l} \psi_X(s) \Big|_{s=0} = \mathbb{E}[X^l] \qquad 1 \leq l \leq k$$

(iii) $\psi_{\underline{X}}$ shares many properties with $\phi_{\underline{X}} : s \longleftrightarrow iu$ (Laplace Transform: s; Fourier Transform: iu)

Cross-moments / Mixed-moments of $\mathbb{R}^n$-valued Random Variables

Definition: $\underline{X} : \Omega \to \mathbb{R}^n$ is an $\mathbb{R}^n$ valued rv. For any n-tuple of integers $(k_1, \ldots, k_n)$

$$\mathbb{E}\left[X_1^{k_1} X_2^{k_2} \ldots X_n^{k_n} \right] = \text{mixed moment / Cross moment of order } (k_1, \ldots, k_n) \text{ of rv } \underline{X} \text{ provided it exists}$$

Independant Random Variables

Independant Random Variables

Definition : (Ω, a, P) is the probability measure. Let $\underline{X}_i : \Omega \longrightarrow \mathbb{R}^{n_i}$ be a $\mathbb{R}^{n_i}$ valued random variable, $1 \leq i \leq k$. Here $n_1, \ldots, n_k \in \mathbb{Z}^+$. The rvs $\{\underline{X}_i \mid 1 \leq i \leq k\}$ are mutually independent if for every choice of Borel Sets $B_i \in \mathbb{R}^{n_i}$

$$P\left[\bigcap_{i=1}^{k} \underline{X}_i^{-1}(B_i)\right] = \prod_{i=1}^{k} P\left[\underline{X}_i^{-1}(B_i)\right]$$

these probabilities are in (Ω, a, P)

equivalent statement

$$P_{\underline{X}_1, \ldots, \underline{X}_k}(B_1, \ldots, B_k) = \prod_{i=1}^{k} P_{\underline{X}_i}(B_i)$$

these probabilities are in $\left(\mathbb{R}^{n_1+\ldots+n_k}, \mathcal{B}^{n_1+\ldots+n_k}, P_{\underline{X}_1 \ldots \underline{X}_k}\right)$

equivalent statement

$$F_{\underline{X}_1 \ldots \underline{X}_k}(\underline{x}_1, \ldots, \underline{x}_k) = \prod_{i=1}^{k} F_{\underline{X}_i}(\underline{x}_i) \qquad \underline{x}_i \in \mathbb{R}^{n_i} \quad 1 \leq i \leq k$$

Remark

(i) For any subset $\{l_1, \ldots, l_m\} \subseteq \{1, \ldots, k\}$ $m \leq k$, we get

$$F_{\underline{X}_{l_1}, \ldots \underline{X}_{l_m}} = \prod_{i=1}^{m} F_{\underline{X}_{l_i}}(\underline{x}_{l_i})$$

(ii) Mutually $\perp\!\!\!\perp$ $\Longrightarrow$ Pairwise $\perp\!\!\!\perp$

$\not\Longleftarrow$

Precisely,

Let $\{\underline{X}_i\}_{i=1}^{k}$ be mutually $\perp\!\!\!\perp$ rvs. $\Longrightarrow$ any pairs $\{\underline{X}_i, \underline{X}_j\}$ $i \neq j$ are pairwise $\perp\!\!\!\perp$

$$\Longrightarrow F_{\underline{X}_i, \underline{X}_j}(\underline{x}_i, \underline{x}_j) = F_{\underline{X}_i}(\underline{x}_i) \cdot F_{\underline{X}_j}(\underline{x}_j) \quad \forall\, \underline{x}_i, \underline{x}_j$$

Note :- If ⊛⊛⊛ hold $\forall$ pairs (i,j), $1 \leq i,j \leq k$ it does not imply that $\{\underline{X}_i\}_{i=1}^k$ are mutually $\perp\!\!\!\perp$.

Ex :- X_1, X_2 $\{0,1\}$ - valued rvs with $\underline{X}_1 \perp\!\!\!\perp \underline{X}_2$ and $P_{X_1}(1) = P_{X_2}(1) = \frac{1}{2}$

Define $X_3 = X_1 \oplus X_2$ wher $\oplus$ is modulo 2 addition.

Show that

$X_1 \perp\!\!\!\perp X_2$, $X_1 \perp\!\!\!\perp X_3$, $X_2 \perp\!\!\!\perp X_3$. $\iff F_{X_1X_2X_3}(x_1,x_2,x_3) \neq F_{X_1}(x_1) \cdot F_{X_2}(x_1) \cdot F_{X_3}(x_3)$

However X_1, X_2, X_3 are not mutually $\perp\!\!\!\perp$.

Special Cases of Mutually $\perp\!\!\!\perp$ rvs

1) For $1 \leq i \leq k$, $F_{\underline{X}_i}$ is absolutely continuos. Then $\{\underline{X}_i\}_{i=1}^k$ are mutually $\perp\!\!\!\perp$ iff the joint CDF of $\{\underline{X}_i\}_{i=1}^k$ is abslutely continuos and

$$f_{X_1 \ldots X_k}(\underline{x}_1 \ldots \underline{x}_k) = \prod_{i=1}^k f_{\underline{X}_i}(\underline{x}_i)$$

2) For $1 \leq i \leq k$, $F_{\underline{X}_i}$ is of discrete type. ie Singular. Then $\{\underline{X}_i\}_{i=1}^k$ are mutually $\perp\!\!\!\perp$ iff the joint CDF of $\{\underline{X}_i\}_{i=1}^k$ is of discrete type with

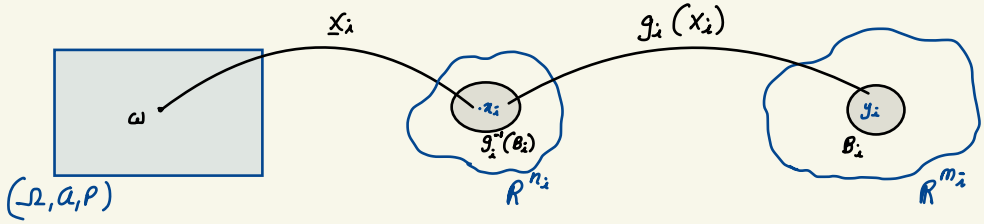
$$S_{\underline{X}} = \prod_{i=1}^k S_{X_i} \qquad \left(\text{ie } \underset{\underline{X}}{\text{Support of}} = \prod \underset{\underline{X}_i}{\text{Support of}}\right)$$

and

$$P(\underline{X}_i = \underline{x}_i \quad 1 \leq i \leq k) = \prod_{i=1}^k P(\underline{X} = \underline{x}_i) \qquad \underline{x}_i \in S_{\underline{X}_i} \quad 1 \leq i \leq k$$

Theorem :- For $1 \leq i \leq k$, $g_i : \mathbb{R}^{n_i} \longrightarrow \mathbb{R}^{m_i}$, Borel and set $\underline{Y}_i = g_i(\underline{X}_i)$. If $\{\underline{X}_i\}_{i=1}^{k}$ are mutually $\perp\!\!\!\perp$, then so are the rvs $\{\underline{Y}_i\}_{i=1}^{k}$

Proof :-



For $B_i \in \mathcal{B}^{m_i}$, $g_i^{-1}(B_i) = \mathcal{B}^{n_i}$

$$\Rightarrow \underline{Y}_i^{-1}(B_i) = \underline{X}_i^{-1}[g^{-1}(B_i)] \in \mathcal{A}$$

$\{\underline{X}_i\}_{i=1}^{k}$ are mutually independant $\Rightarrow$ $\underline{X}_i^{-1}[g^{-1}(B_i)]$ are mutually independant in Ω

$\Rightarrow$ $\underline{Y}_i^{-1}(B_i)$ are mutually independant

$\Rightarrow$ $\{\underline{Y}_i\}_{i=1}^{k}$ are mutually independant

■

Theorem: If the $\mathbb{R}$-valued rvs $\{X_i\}_{i=1}^{K}$ are mutually $\perp\!\!\!\perp$, and (either $X_i \geq 0$ with probability 1 / or $\mathbb{E}[|X_i|] < \infty$ $1 \leq i \leq k$) then, $X \triangleq \prod_{i=1}^{K} X_i$ has a well defined expectation given by

$$\mathbb{E}[X] = \mathbb{E}\left[\prod_{i=1}^{K} X_i\right] = \prod_{i=1}^{K} \mathbb{E}[X_i]$$

Proof: Assume that $\{X_i\}_{i=1}^{K}$ have cdfs that are absolutely continuos Then,

$$\mathbb{E}[X] = \mathbb{E}[X_1 \dots X_K]$$

$$= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} (x_1 * x_2 * \dots * x_K) \cdot \underbrace{f_{X_1 \dots X_K}(x_1, \dots x_K)}_{f_{X_1}(x_1) \overset{\text{independance}}{\dots} f_{X_K}(x_K) \text{ from}} \, dx_1 \dots dx_K$$

$$= \prod_{i=1}^{K} \int_{-\infty}^{\infty} x_i \cdot f_{X_i}(x_i) \, dx_i \quad \cdots \quad \text{from Fubini theorem}$$

$$= \prod_{i=1}^{K} \mathbb{E}[X_i]$$

∎

Corollary: Extension to $\mathbb{C}$-valued rvs

Corollary: If the $\mathbb{R}^n$ valued rvs $\underline{X}_i : \Omega \to \mathbb{R}^{n_i}$ $i = 1, \dots, k$ are mutually idependant, then $\forall\, g_i : \mathbb{R}^{n_i} \to \mathbb{R}$ Borel, it holds

$$\mathbb{E}[g_1(\underline{X}_1) \dots g_K(\underline{X}_K)] = \prod_{i=1}^{K} \mathbb{E}[g_i(\underline{X}_i)]$$

provided all expectations exist.

Proof: By 2 previous theorems.

Theorem: A collection of $\mathbb{R}^{n_\ell}$ valued rvs, $\underline{X}_\ell : \Omega \longrightarrow \mathbb{R}^{n_\ell}$ $1 \le \ell \le k$ are mutually independant iff

$$\phi_{\underline{X}_1 \dots \underline{X}_k}(\underline{u}_1, \dots, \underline{u}_k) = \prod_{\ell=1}^{k} \phi_{\underline{X}_\ell}(\underline{u}_\ell)$$

$\forall (\underline{u}_1, \dots, \underline{u}_k) \in \mathbb{R}^n \quad n = n_1 + \dots + n_k$

Proof: If X_ℓ are mutually independant

Immediate consequence of previous two theorems with a light extension to complex-valued functions

From the definition of characteristic function,

$$\phi_{\underline{X}_1 \dots \underline{X}_k}(\underline{u}_1, \dots, \underline{u}_k) = \mathbb{E}\left[e^{i \sum_{\ell=1}^{k} \underline{u}_\ell^T \underline{X}_\ell} \right]$$

$$= \mathbb{E}\left[\prod_{\ell=1}^{k} e^{i \underline{u}_\ell^T \underline{X}_\ell} \right]$$

$$= \prod_{\ell=1}^{k} \mathbb{E}\left[e^{i \underline{u}_\ell^T \underline{X}_\ell} \right] = \prod_{\ell=1}^{k} \phi_{\underline{X}_\ell}(\underline{u}_\ell) \quad (*)$$

Assume $\phi_{\underline{X}_1, \dots, \underline{X}_k}$ has the product form

By the Fourier Inversion formula applied to $*$ we can show that

$$\Rightarrow F_{\underline{X}_1 \dots \underline{X}_k}(\underline{x}_1, \dots \underline{x}_k) = \prod_{\ell=1}^{k} F_{\underline{X}_\ell}(\underline{x}_\ell)$$

$$\Rightarrow \{\underline{X}_\ell\}_{\ell=1}^{k} \text{ are mutually } \perp\!\!\!\perp$$

QED

Theorem:- Let X and Y be $\mathbb{R}$-valued rvs with $\sigma_X^2, \sigma_Y^2 < \infty$. Then

$$X \perp\!\!\!\perp Y \implies \text{Cov}[X,Y] \triangleq \mathbb{E}[(X-\mu_X)(Y-\mu_Y)] = 0$$

or

$$X \perp\!\!\!\perp Y \implies X \perp Y$$

ie, two independant rvs are uncorrelated.

Proof: Set $g_1(x) \triangleq x - \mu_X$, $g_2(y) \triangleq y - \mu_Y$ $g_1(x), g_1(y)$ are $\mathbb{R}$ valued rvs.

$$\implies \mathbb{E}[g_1(x)] = \mathbb{E}[g_2(y)] = 0 \quad \cdots\cdots (1)$$

$\therefore$ Assume X, Y are independant $\implies g_1(x) \perp\!\!\!\perp g_2(y)$

$$\implies \mathbb{E}[g_1(x)\cdot g_2(y)] = \mathbb{E}[g_1(x)\cdot g_2(y)] = 0 \quad \cdots \text{from } (1)$$

$$\implies \text{Cov}[x,y] = 0$$

$$\implies X, Y \text{ are uncorrelated.}$$

Remark: $X \perp\!\!\!\perp Y \implies X \perp Y$, $\not\Leftarrow$

Example: For $U \perp V$ but $U \not\perp\!\!\!\perp V$

Let $X \sim$ Uniform $[-\pi/2, \pi/2]$

and let

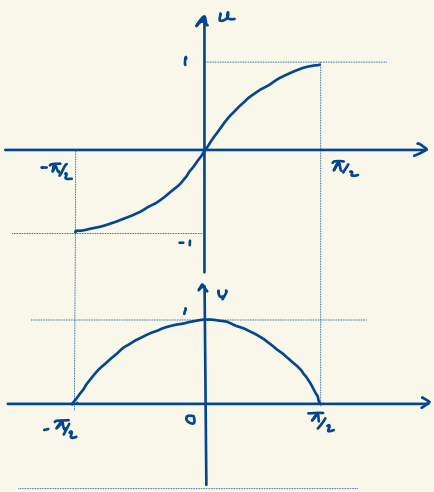
$U \triangleq \sin X$

$V \triangleq \cos X$

Always $U^2 + V^2 = 1$

Consider the events $\{U \in [0, 1/2]\}$ and $\{V \in [0, 1/2]\}$

$$P_U([0,1/2]) = P(0 \le U \le 1/2) > 0$$
$$P_V([0,1/2]) = P(0 \le V \le 1/2) > 0$$



But since $U^2 + V^2 = 1$ both cannot be less than $1/2$ $\implies P(0 \le U \le 1/2,\ 0 \le V \le 1/2) = 0$

$\therefore U \not\perp\!\!\!\perp V$

$$\mathbb{E}[U] = \int_{-\pi/2}^{\pi/2} \frac{1}{\pi} \sin x \, dx = 0$$

$$\mathbb{E}[UV] = \int_{-\pi/2}^{\pi/2} \frac{1}{\pi} \cdot \sin x \cdot \cos x \, dx = \int_{-\pi/2}^{\pi/2} \frac{\sin 2x}{2\pi} \, dx = 0$$

But

$$\mathrm{Cov}[U,V] = \mathbb{E}[UV] - \mathbb{E}[U]\,\mathbb{E}[V] = 0 \implies U \perp V$$

■

Definition: Let I be a countable Index set (can be infinite). Let $\{\underline{X}_i \; i \in I\}$ be a collection of rvs, $\underline{X}_i : \Omega \longrightarrow \mathbb{R}^{n_i}$ The collection $\{\underline{X}_i \; i \in I\}$ is mutually $\perp\!\!\!\perp$ if the collection $\{\underline{X}_j \; j \in J\}$ are mutually $\perp\!\!\!\perp$ for every finite set of indices $J \subseteq I$.

Conditional Probabilities and Distributions

Conditional Probabilities and Distributions

1) Conditional Probabilities

- For $(\Omega, \mathcal{A}, P)$ probability measure $A, B \in \mathcal{A}$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \qquad P(B) \neq 0$$

- $P(\cdot|B)$ is a probability measure on $\mathcal{A}$

- For $P(B) > 0$ $P(\cdot|B)$ is a unique probability measure

$$P(\cdot|B) : \mathcal{A} \longrightarrow [0,1]$$

(i) $P(A|B) \cdot P(B) = P(A \cap B)$

(ii) Law of Total Probability

$\{B_i \in \mathcal{A}, i \in I\}$ for countable partition of Ω where $B_i \cap B_j = \emptyset$ $\forall i \neq j$ and $\bigcup_{j \in I} B_j = \Omega$ If $P(B_i) > 0$ $\forall i \in I$

$$P(A) = \sum_{i \in I} P(A \cap B_i) = \sum_{i \in I} P(A|B_i) \cdot P(B_i)$$

New Definition:

Natural Definition of $P(\cdot|B)$

For any arbitary $B \in \mathcal{A}$ is that of a probability measure $\mathcal{A} \longrightarrow [0,1]$ such that

$$P(A|B) \cdot P(B) = P(A \cap B) \quad \forall A \in \mathcal{A}$$

- Above is the new definition for conditional probabilities.

- Dont' need the fact that $P(B_j) > 0$ $\forall j \in I$.

- $\therefore$ Law of Total Probabilities: becomes

$$\forall A \in \mathcal{A} \qquad P(A) = \sum_{j \in I} P(A|B_j) \cdot P(B_j) = \sum_{j \in I} P(A \cap B_j)$$

2) Conditional Distributions

$(\Omega, \mathcal{A}, P)$ is the probability measure. $\underline{X}$ is $\mathbb{R}^n$ valued and $\underline{Y}$ is $\mathbb{R}^m$ valued. Given joint CDF $F_{\underline{X}\underline{Y}}$.

Likelihood of $\underline{X} \in B_{\underline{X}} \in \mathcal{B}^n$ given $\underline{Y} \in B_{\underline{Y}} \in \mathcal{B}^m$

$$= P\left(\underline{X} \in B_{\underline{X}} \mid \underline{Y} \in B_{\underline{Y}}\right) \quad \text{(Formally)}$$

$$= P\left(\{\omega \in \Omega : \underline{X}(\omega) \in B_{\underline{X}}\} \,\middle|\, \{\omega \in \Omega : \underline{Y}(\omega) \in B_{\underline{Y}}\}\right)$$

$$\triangleq \frac{P\left(\underline{X} \in B_{\underline{X}} \text{ and, } \underline{Y} \in B_{\underline{Y}}\right)}{P\left(\underline{Y} \in B_{\underline{Y}}\right)}$$

$$= \frac{P_{\underline{X}\underline{Y}}\left(B_{\underline{X}} \times B_{\underline{Y}}\right)}{P_{\underline{X}\underline{Y}}\left(\mathbb{R}^n \times B_{\underline{Y}}\right)}$$

$$= \frac{P_{\underline{X}\underline{Y}}\left(B_{\underline{X}}, B_{\underline{Y}}\right)}{P_{\underline{X}\underline{Y}}\left(\mathbb{R}^n, B_{\underline{Y}}\right)}$$

from conditional probability law

Conditional Probability Law of $\underline{X}$ given that $\underline{Y} \in B_{\underline{Y}}$

$$= \underbrace{P_{\underline{X}|\underline{Y}}\left(B_{\underline{X}} \mid B_{\underline{Y}}\right)}_{\text{Notation used in the lecture}} = P_{\underline{X}}\left(B_{\underline{X}} \mid \underline{Y} \in B_{\underline{Y}}\right)$$

Remark

- Thus $P_{\underline{X}|\underline{Y}}\left(\cdot \mid B_{\underline{Y}}\right)$ is determined solely by $P_{\underline{X}\underline{Y}}$ or $F_{\underline{X}\underline{Y}}$, at least when $P_{\underline{Y}}\left(B_{\underline{Y}}\right) > 0$

- If we set $B_{\underline{X}} = (-\infty, \underline{x})$ ... Infinite rectangles in $\mathbb{R}^n$ we get the conditional probability distribution of $\underline{X}$ given $\underline{Y} \in B_{\underline{Y}}$ denoted by,

$$F_{\underline{X}|\underline{Y}}\left(x \mid B_{\underline{Y}}\right) \triangleq P_{\underline{X}}\left((-\infty, x) \,\middle|\, \underline{Y} \in B_{\underline{Y}}\right)$$

$$= P_{\underline{X}|\underline{Y}}\left((-\infty, \underline{x}) \mid B_{\underline{Y}}\right)$$

- If $\underline{X} \perp\!\!\!\perp \underline{Y}$

(i) $P(\underline{X} \in B_{\underline{X}}, \underline{Y} \in B_{\underline{Y}}) = P(\underline{X} \in B_{\underline{X}}) \cdot P(\underline{Y} \in B_{\underline{Y}})$

$\forall\ B_{\underline{X}} \in \mathcal{B}^n \quad B_{\underline{Y}} \in \mathcal{B}^m$

(ii) $P_{\underline{X}|\underline{Y}}(B_{\underline{X}} | B_{\underline{Y}}) = P_{\underline{X}}(B_{\underline{X}})$

$P_{\underline{Y}|\underline{X}}(B_{\underline{Y}} | B_{\underline{X}}) = P_{\underline{Y}}(B_{\underline{Y}})$

I) $\underline{Y}$ is of Discrete Type :

Definition :- $S_{\underline{Y}} = \{\underline{y}_i \in \mathbb{R}^n,\ i \in I \text{ (Countable Index Set)}\}$

Assume $P(\underline{Y} = \underline{y}_i) > 0 \quad \forall\ \underline{y}_i \in S_{\underline{Y}}$

Let $\underline{y}_i \in \mathbb{R}^m$, take the Borel sets as $B_{\underline{Y}} = \{\underline{y}_i\}$

$\therefore (\underline{Y} \in B_{\underline{Y}}) = (\underline{Y} = \underline{y}_i)$

For any $B_{\underline{X}} \in \mathcal{B}^n$

$$P(\underline{X} \in B_{\underline{X}} \mid \underline{Y} \in B_{\underline{Y}}) = \frac{P(\underline{X} \in B_{\underline{X}}, \underline{Y} = \underline{y}_i)}{P(\underline{Y} = \underline{y}_i)}$$

$$F_{\underline{X}|\underline{Y}}(\underline{x} \mid \underline{y}_i) = \frac{P(\underline{X} \leq \underline{x}, \underline{Y} = \underline{y}_i)}{P(\underline{Y} = \underline{y}_i)}$$

Remark :-

- If $\underline{X}$ too is of discrete type, that means $S_{\underline{X}} = \{\underline{x}_k \in \mathbb{R}^n,\ k \in K\}$ where K is countable index set. Then $F_{\underline{X}|\underline{Y}}(\underline{x} | \underline{y}_i)$ is also of the discrete type

$$P(\underline{X} \in B_{\underline{X}} \mid \underline{Y} \in B_{\underline{Y}}) = P(\underline{X} = \underline{x}_k \mid \underline{Y} = \underline{y}_i) \triangleq P_{\underline{X}|\underline{Y}}(\underline{x}_k | \underline{y}_i)$$

$$= \frac{P(\underline{X} = \underline{x}_k, \underline{Y} = \underline{y}_i)}{P(\underline{Y} = \underline{y}_i)}$$

$$= \frac{P(\underline{X} = \underline{x}_k, \underline{Y} = \underline{y}_i)}{\sum_{\underline{x}_{k'} \in S_{\underline{X}}} P(\underline{X} = \underline{x}_{k'}, \underline{Y} = \underline{y}_i)}$$

- Check $\sum_{\underline{x}_{k'} \in S_{\underline{X}}} P(\underline{X} = \underline{x}_k \mid \underline{Y} = \underline{y}_i) = 1 \quad \forall\ \underline{y}_i \in S_{\underline{Y}}$

II) $\underline{Y}$ is a rv of the Absolute Continuos Type

Definition :- For $y \in \mathbb{R}^m$ take $B_{\underline{y}} = \{y\} \in \mathcal{B}^m$ (Singleton events which are Borel) Note :- $P(\underline{Y} = y) = 0$

Define the conditional probability of $(\underline{X} \in B_{\underline{X}})$ given that $(\underline{Y} = y)$ by probability measure

$$P(\cdot \mid \underline{Y} = y) : \mathcal{A} \longrightarrow [0,1]$$

Such that $\forall A \in \mathcal{A}$ ie $A = \{\omega \in \Omega : \underline{X}(\omega) \in B_{\underline{X}}\}$

$$P(A \mid \underline{Y} = y) \cdot P(\underline{Y} = y) = P\left(A \cap (\underline{Y} = y)\right) \quad \cdots \textcircled{*}$$

- Due to $P(\underline{Y} = y) = 0$ in continuos cases, we need additional conditions to make $\textcircled{*}$ into a valid definition of conditional probability

$$\Downarrow$$

Law of Total Probability ?

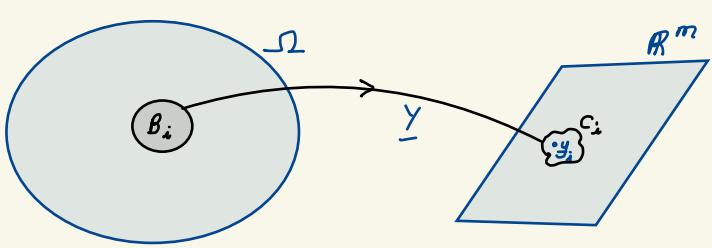
- How to connect $\textcircled{*}$ to $\textcircled{***}$ using Law of Total Probability.

$P(\cdot \mid \underline{Y} = y)$ $y \in \mathbb{R}^m$ must be selected such that

$$P(A) = \int_{\mathbb{R}^m} P(A \mid \underline{Y} = y) \cdot f_{\underline{Y}}(y) \cdot dy \qquad A \in \mathcal{A} \quad \cdots \textcircled{**}$$

Randon - Nikodyn Derivative

Proof :-



Idea - Since $P(Y = y) = 0$ in continuos case, let's consider a small m dimensional 'cell' in $\mathbb{R}^m$ around $y \in \mathbb{R}^m$. Then argue that whole $\mathbb{R}^m$ space of interest can be partiotioned in to such 'cells' and use 'Law of Total Probability' on them.

Consider a small volume 'C' around y If it's very small,

$$P(A \mid Y = y) \cong P(A \mid Y \in C) \qquad A \in \mathcal{A}$$

Let $\{C_j\}_{j=-\infty}^{\infty}$ be a countable partition (ex: Borel Cubes) of $\mathbb{R}^m$ and assume B_j are their inverse mappings in $\mathcal{A}$ under $\underline{Y}$.

$$\therefore \quad B_j = \underline{Y}^{-1}(C_j) \in \mathcal{A}$$

$\{C_j\}_{j=-\infty}^{\infty}$ are a partitioning of $\mathbb{R}^m$ (Does such a partitioning always exist)

$\Longrightarrow \{B_j\}_{j=-\infty}^{\infty}$ are a partitioning of Ω (why?)

$\Longrightarrow P(A) = \sum_j P(A \mid B_j) \cdot P(B_j)$ ---- ($A \in \mathcal{A}$, From The Law of Total Probability)

$\Longrightarrow \quad = \sum_j P(A \mid \underline{Y} \in C_j) \cdot P_{\underline{Y}}(C_j)$

$\Longrightarrow \quad = \sum_j P(A \mid \underline{Y} \in C_j) \, f_{\underline{Y}}(y_j) \cdot \mathrm{vol}(C_j)$

$\Longrightarrow \quad = \int_{\mathbb{R}^m} P(A \mid \underline{Y} = y) \cdot f_{\underline{Y}}(y) \cdot dy$ ---- (Assuming $\sup_j \mathrm{vol}(C_j)$ goes to '0'.)

$\therefore$ Thus $P(\cdot \mid Y = y) \quad y \in \mathbb{R}^m$ must be selected such that (*) (*) is true.

■

QED

Provided $f_{\underline{Y}}(\underline{y}) > 0$

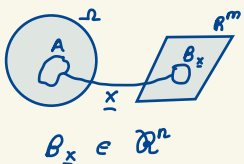
$$F_{\underline{X}|\underline{Y}}(\underline{x}/\underline{y}) = \frac{\int_{(-\infty,\underline{x})} f_{\underline{X}\underline{Y}}(\underline{u},\underline{y})\cdot d\underline{u}}{\int_{\mathbb{R}^n} f_{\underline{X}\underline{Y}}(\underline{u},\underline{y})\,d\underline{u}} = \frac{\int_{(-\infty,\underline{x})} f_{\underline{X}\underline{Y}}(\underline{u},\underline{y})\cdot d\underline{u}}{f_{\underline{Y}}(\underline{y})}$$

Proof: $P(A \mid \underline{Y}=\underline{y}) = \lim_{\substack{Vol(C)\to 0 \\ C\in \underline{y}}} P(A \mid \underline{Y}\in C)$

$$= \lim_{\substack{Vol(C)\to 0 \\ C\in \underline{y}}} \frac{P(A\cap(\underline{Y}\in C))}{P(\underline{Y}\in C)} \quad \cdots\cdots ①$$

Consider a set A specioulizing to $A = \{\omega\in\Omega : \underline{X}(\omega)\in B_{\underline{X}}\} \triangleq (\underline{X}\in B_{\underline{X}})$

Then above equation becomes



$B_{\underline{X}} \in \mathcal{B}^n$

Conditional Probability Law of $\underline{X}$ given $\underline{Y}=\underline{y}$ $\triangleq$ $P_{\underline{X}|\underline{Y}}(B_{\underline{X}}\mid \underline{y}) = P(\underline{X}\in B_{\underline{X}} \mid \underline{Y}=\underline{y})$

$$= \lim_{\substack{Vol(C)\to 0 \\ C\in \underline{y}}} \frac{P(\underline{X}\in B_{\underline{X}},\ \underline{Y}\in C)}{P(\underline{Y}\in C)} \quad \cdots \text{ from } ①$$

$$= \lim_{\substack{Vol(C)\to 0 \\ C\in \underline{y}}} \frac{P(\underline{X}\in B_{\underline{X}},\ \underline{Y}\in C)/Vol(C)}{P(\underline{Y}\in C)/Vol(C)} \quad \cdots ②$$

Instead of picking just some Borel set $B_{\underline{X}}$, pick the left sided $\mathbb{R}^m$ dimensional rectangles $B_{\underline{x}} = (-\infty, \underline{x})$ $\underline{x}\in\mathbb{R}^n$

The $P(\cdot)$ becomes $F(\cdot)$

$\Longrightarrow$ $F_{\underline{X}}(\underline{x}\mid \underline{Y}=\underline{y})$

$$F_{\underline{X}|\underline{Y}}(\underline{x}\mid\underline{y}) = P(\underline{X}\leqslant\underline{x} \mid \underline{Y}=\underline{y})$$

$$= P(\underline{X}\in(-\infty,\underline{x}) \mid \underline{Y}=\underline{y})$$

$$= P_{\underline{X}|\underline{Y}}(B_{\underline{X}}\mid\underline{y})$$

$$= \lim_{\substack{Vol(C)\to 0 \\ C\in \underline{y}}} \frac{P(\underline{X}\leqslant\underline{x},\ \underline{Y}\in C)/Vol(C)}{P(\underline{Y}\in C)/Vol(C)} \quad \cdots \text{ from } ②$$

If $F_{\underline{X}\underline{Y}}$ is of absolutely continuous type probabilities can be found from integrating f

$$= \lim_{\substack{Vol(C)\to 0 \\ C\in \underline{y}}} \frac{\left(\int_{(-\infty,\underline{x})\times C} f_{\underline{X}\underline{Y}}(\underline{u},\underline{v})\cdot d\underline{u}\,d\underline{v}\right)/Vol(C)}{\left(\int_{\mathbb{R}^n\times C} f_{\underline{X}\underline{Y}}(\underline{u},\underline{v})\,d\underline{u}\,d\underline{v}\right)/Vol(C)}$$

$$= \lim_{\substack{\mathrm{Vol}(C)\to 0 \\ C\ni y}} \frac{\left(\int_{(-\infty,x)\times C} f_{\underline{X}\underline{Y}}(u,v)\cdot du\,dv\right)\cdot \mathrm{Vol}(C)\,/\,\mathrm{Vol}(C)}{\left(\int_{\mathbb{R}^n\times C} f_{\underline{X}\underline{Y}}(u,v)\,du\,dv\right)\cdot \mathrm{Vol}(C)\,/\,\mathrm{Vol}(C)}$$

$$= \frac{\int_{(-\infty,x)\times C} f_{\underline{X}\underline{Y}}(u,v)\cdot du\,dv}{\int_{\mathbb{R}^n\times C} f_{\underline{X}\underline{Y}}(u,v)\,du\,dv} \qquad f_{\underline{Y}}(y)$$

Q.E.D.

Provided $f_{\underline{Y}}(y) > 0$

$$f_{\underline{X}|\underline{Y}}(x\,|\,y) = \frac{f_{\underline{X}\underline{Y}}(x,y)}{f_{\underline{Y}}(y)} \qquad \cdots\cdots (*)(*)(*)$$

Observe from the above equation since $F_{\underline{X}|\underline{Y}}(\cdot\,|\,y)$ is of absolute continuous type

Conditional pdf of $\underline{X}$ given $\underline{Y}=y$ $= \dfrac{f_{\underline{X}|\underline{Y}}(x\,|\,y)}{\int_{\mathbb{R}^n} f_{\underline{X}\underline{Y}}(u,y)\,du = f_{\underline{Y}}(y)}$ provided $f_{\underline{Y}}(y) > 0$

- We can integrate (*)(*)(*) to get cdf.

3) Two Consequences of Importance

1) Law of Total Probability

PDF $$f_{\underline{X}}(\underline{x}) = \int_{\mathbb{R}^m} f_{\underline{X}|\underline{Y}}(\underline{x}|\underline{y}) \cdot f_{\underline{Y}}(\underline{y}) \cdot d\underline{y}$$

Proof:

$$f_{\underline{X},\underline{Y}}(\underline{x},\underline{y}) = f_{\underline{X}|\underline{Y}}(\underline{x}|\underline{y})\, f_{\underline{Y}}(\underline{y})$$

By integration over $\underline{y}$ $\Rightarrow$ $$\int_{\mathbb{R}^m} f_{\underline{X},\underline{Y}}(\underline{x},\underline{y}) \cdot d\underline{y} = \int_{\mathbb{R}^m} f_{\underline{X}|\underline{Y}}(\underline{x}|\underline{y}) \cdot f_{\underline{Y}}(\underline{y}) \cdot d\underline{y}$$

marginal probabilities $\Rightarrow$ $$f_{\underline{X}}(\underline{x}) = \int_{\mathbb{R}^m} f_{\underline{X}|\underline{Y}}(\underline{x}|\underline{y}) \cdot f_{\underline{Y}}(\underline{y}) \cdot d\underline{y}$$ ∎

CDF $$F_{\underline{X}}(\underline{x}) = \int_{\mathbb{R}^m} F_{\underline{X}|\underline{Y}}(\underline{x}|\underline{y}) \cdot f_{\underline{Y}}(\underline{y}) \cdot d\underline{y}$$

Proof:

From ⊛⊛⊛ $\Rightarrow$ $$\int_{(-\infty,\underline{x})} f_{\underline{X},\underline{Y}}(\underline{u},\underline{y})\, d\underline{u} = F_{\underline{X}|\underline{Y}}(\underline{x}|\underline{y}) \cdot f_{\underline{Y}}(\underline{y})$$

By integration over $\underline{y}$ space $\Rightarrow$ $$\int_{\mathbb{R}^m} \left(\int_{(-\infty,\underline{x})} f_{\underline{X},\underline{Y}}(\underline{u},\underline{y}) \cdot d\underline{u} \right) d\underline{y} = \int_{\mathbb{R}^m} F_{\underline{X}|\underline{Y}}(\underline{x}|\underline{y}) \cdot f_{\underline{Y}}(\underline{y}) \cdot d\underline{y}$$

From Fubiniz principle, let's change integration $\Rightarrow$ $$\int_{(-\infty,\underline{x})} \Big(\underbrace{\int_{\mathbb{R}^m} f_{\underline{X},\underline{Y}}(\underline{u},\underline{y}) \cdot d\underline{y}}_{f_{\underline{X}}(\underline{u}) \text{ from marginal probabilities}} \Big) \cdot d\underline{u} = \int_{\mathbb{R}^m} F_{\underline{X}|\underline{Y}}(\underline{x}|\underline{y}) \cdot f_{\underline{Y}}(\underline{y}) \cdot d\underline{y}$$

$$\Rightarrow \underbrace{\int_{(-\infty,\underline{x})} f_{\underline{X}}(\underline{u})\, d\underline{u}}_{F_{\underline{X}}(\underline{x})} = \int_{\mathbb{R}^m} F_{\underline{X}|\underline{Y}}(\underline{x}|\underline{y}) \cdot f_{\underline{Y}}(\underline{y}) \cdot d\underline{y}$$

$$\Rightarrow F_{\underline{X}}(\underline{x}) = \int_{\mathbb{R}^m} F_{\underline{X}|\underline{Y}}(\underline{x}|\underline{y}) \cdot f_{\underline{Y}}(\underline{y}) \cdot d\underline{y}$$ ∎

2) Bayes Formula

$$f_{\underline{X}|\underline{Y}}(x|y) = \frac{f_{\underline{Y}|\underline{X}}(y|x) \cdot f_{\underline{X}}(x)}{f_{\underline{Y}}(y)}$$

Proof: From Law of Total Probability of PDFs,

$$f_{\underline{X},\underline{Y}}(x,y) = f_{\underline{X}|\underline{Y}}(x|y) \cdot f_{\underline{Y}}(y) = f_{\underline{Y}|\underline{X}}(y|x) \cdot f_{\underline{X}}(x)$$

$$\Rightarrow f_{\underline{X}|\underline{Y}}(x|y) = \frac{f_{\underline{Y}|\underline{X}}(y|x) \cdot f_{\underline{X}}(x)}{f_{\underline{Y}}(y)}$$

■

Ex: $\underline{X}$ - message signal (Cause)

$\underline{Y}$ - recieved signal (result) = $\underline{X} + \underline{N}$ (Channel Noise)

Ex: Application

X, Y are $\mathbb{R}$ valued random variables with joint cdf F_{XY} that is of the absolute continuos type. Assume $X \perp\!\!\!\perp Y$ hence $F_{XY} = F_X \cdot F_Y$ $Z = X + Y$ where X - Signal and Y - noise. Hence Z is a $\mathbb{R}$ valued rv. Find $f_Z(z)$ $z \in \mathbb{R}$.

$$F_Z(z) = P(Z < z) = P(X + Y < z)$$

$$= \int_{-\infty}^{\infty} P(X+Y<z \mid Y=y) \cdot f_Y(y)\, dy$$ from law of total probability

This use the law of Total probability to use underlying probabilistic features given in the question

$$= \int_{-\infty}^{\infty} P(X+y<z \mid Y=y) \cdot f_Y(y)\, dy$$

$$= \int_{-\infty}^{\infty} P(X<z-y \mid Y=y) \cdot f_Y(y) \cdot dy$$

$$= \int_{-\infty}^{\infty} P(X<z-y) \cdot f_Y(y) \cdot dy$$ Since X and Y are independant we can drop the condition

$$= \int_{-\infty}^{\infty} F_X(z-y) \cdot f_Y(y) \cdot dy$$

$\therefore$ By differentiation

$$f_Z(z) = \frac{d}{dz} F_Z(z) = \frac{d}{dz}\left(\int_{-\infty}^{\infty} F_X(z-y) \cdot f_Y(y) \cdot dy \right)$$

Function of z

$$= \int_{-\infty}^{\infty} \frac{\partial}{\partial z} F_X(z-y) \cdot f_Y(y)\, dy$$ ..... Leibinz Integration rule

$$= \int_{-\infty}^{\infty} f_X(z-y) \cdot f_Y(y)\, dy$$

$$= (f_X * f_Y)(z)$$

Convolution of f_X and f_Y

4) Conditionally Independant Random Variables

Definition: $(\Omega, \mathcal{A}, P)$ Let $\underline{X}_i : \Omega \longrightarrow \mathbb{R}^{n_i}$ be a $\mathbb{R}^{n_i}$ - valued random variable. $n_i \in \mathbb{Z}^+$ $\forall\, i = 1, \dots, k$.

Let $\underline{Y} : \Omega \longrightarrow \mathbb{R}^n$ be a $\mathbb{R}^n$ - valued random variable with n a positive integer.

Given joint cdf of $F_{\underline{X}_1 \underline{X}_2 \dots \underline{X}_k \underline{Y}}$, The rv's $\{\underline{X}_i\}_{i=1}^k$ are conditionally mutually independant conditioned on a given $\underline{Y}$, if

$$F_{\underline{X}_1 \dots \underline{X}_k | \underline{Y}} (\underline{x}_1, \dots, \underline{x}_k | \underline{y}) = \prod_{i=1}^{k} F_{\underline{X}_i | \underline{Y}} (\underline{x}_i | \underline{y})$$

$\forall\, \underline{x}_i \in \mathbb{R}^{n_i}$ $i = 1, \dots, k$ and $\forall\, \underline{y} \in \mathbb{R}^{n_0}$

with $p_{\underline{Y}}(y) > 0$ (if $\underline{Y}$ is discrete valued)

$f_{\underline{Y}}(y) > 0$ (if $\underline{Y}$ is continuos valued)

Remark: $X_1 \perp\!\!\!\perp X_2 \not\Longleftrightarrow X_1 \perp\!\!\!\perp X_2 \mid Y$

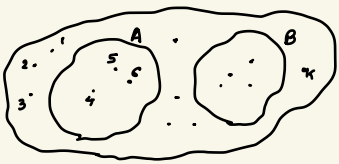
$\not\Longrightarrow$ ▸ Consider $X_1 \perp\!\!\!\perp X_2$ are $\{0,1\}$ valued with $P_X(1) = P_X(0) = 1/2$ and Let $Y = X_1 \oplus X_2$, show that $X_1 \not\perp\!\!\!\perp X_2 \mid Y$

$\not\Longleftarrow$ ▸ Consider $X_1 = X_2 = Y = X$ and $P(X = \text{constant}) = 0$. Then show $\underline{X}_1 \perp\!\!\!\perp \underline{X}_2 \mid \underline{Y}$. But $\underline{X}_1 \not\perp\!\!\!\perp \underline{X}_2$ since the condition $P(X = \text{constant}) = 0$

Exercise: Show that $\underline{X} \perp\!\!\!\perp \underline{X} \iff \underline{X}$ is a constant c with $P(X = c) = 1$

Exercise

1) Let $\underline{X}_1, \ldots, \underline{X}_k$ be conditionally mutually $\perp\!\!\!\perp$ given $\underline{Y}$.



Suppose that

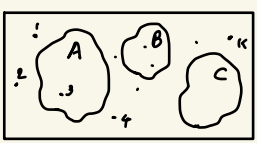
$A \subset \{1, 2, \ldots, k\}$

$B \subset \{1, 2, \ldots, k\}$

$A \neq \emptyset \quad B \neq \emptyset \quad A \cap B = \emptyset$

If $X_A \triangleq \{X_i, i \in A\}$ and $X_B \triangleq \{X_i; i \in B\}$ Then show $X_A \perp\!\!\!\perp X_B \mid \underline{Y}$

2) Suppose that $\underline{X}_1, \ldots, \underline{X}_k$ are mutually independant.



$A \subset \{1, 2, .., k\} \quad A \neq \emptyset$

$B \subset \{1, 2, \ldots, k\} \quad B \neq \emptyset$

$C \subset \{1, 2, \ldots, k\} \quad C \neq \emptyset$

And A, B, C are pairwise disjoint. Show that $\underline{X}_A \perp\!\!\!\perp \underline{X}_B \mid \underline{X}_C$

Conditional Expectations

Earlier we had X - $\mathbb{R}^n$ - valued rvs, Y - $\mathbb{R}^m$ - valued rvs. Now we assume $n=1$.

$$P(\cdot \mid \underline{Y} \in B_{\underline{Y}}) \quad B_{\underline{Y}} \in \mathcal{R}^m$$

$$P_{\underline{X}|\underline{Y}}(\cdot \mid B_{\underline{Y}})$$

$$F_{\underline{X}|\underline{Y}}(\cdot \mid B_{\underline{Y}})$$

Goal: Study expectations of X under the conditional probability measure,

$$\mathbb{E}[\underline{X} \mid \underline{Y} \in B_{\underline{Y}}] = \text{Conditional expectation of rv } X \text{ given that } \underline{Y} \in B_{\underline{Y}}$$

This depends only on $F_{\underline{X}|\underline{Y}}(\cdot \mid B_{\underline{Y}})$

1) Special Cases

1) $\underline{Y}$ is of discrete type and X is also of discrete type.

$$\mathbb{E}[X \mid \underline{Y} = y_i] = \sum_{k \in K} x_k \cdot P(X = x_k \mid \underline{Y} = y_i)$$

- a function of a single variable y_i

K - Index set $\quad y_i \in S_{\underline{Y}}$

2) $\underline{Y}$ is of continuos type. $(X, \underline{Y})$ is the absolute continuos type.

$$\mathbb{E}[X \mid \underline{Y} = y] = \int_{-\infty}^{\infty} x \cdot f_{X|\underline{Y}}(x \mid y_i)\, dx \qquad y \in \mathbb{R}^m$$

- function of a single variable $\mathbb{R}^m$

This defines a mapping : $\mathbb{R}^m \longrightarrow \mathbb{R}$

$$: y \longrightarrow \int_{-\infty}^{\infty} x \cdot f_{X|\underline{Y}}(x \mid y_i)\, dx \triangleq \mathbb{E}[X \mid \underline{Y} = y]$$

Theorem: The above mapping from $\mathbb{R}^m \longrightarrow \mathbb{R}$ is a Borel mapping, denoted by $\mathbb{E}[X \mid \underline{Y} = \cdot]$

Observation:

- Can compose $\mathbb{E}[X \mid \underline{Y} = \cdot]$ (deterministic mapping) with $\underline{Y}$ (rv)
- $\mathbb{E}[X \mid \underline{Y}]$ - can be obtained from conditional expectation of X given $\underline{Y}$
- $\mathbb{E}[X \mid \underline{Y} = \cdot]$ vs $\mathbb{E}[X \mid \underline{Y}]$
  - $\mathbb{E}[X \mid \underline{Y} = \cdot]$: Deterministic Borel mapping
  - $\mathbb{E}[X \mid \underline{Y}]$: $\mathbb{R}$-valued rv function of $\underline{Y}$
- Thus $\mathbb{E}[X \mid \underline{Y}]$ is a $\mathbb{R}$ valued random variable. (a Borel function of $\underline{Y}$). It's value when $\underline{Y} = y$ is $\mathbb{E}[X \mid \underline{Y} = y]$

2) Computation of $\mathbb{E}[X \mid \underline{Y}]$

For each $y \in \mathbb{R}^m$

$$\Rightarrow \mathbb{E}[X \mid \underline{Y} = y] = \int_{-\infty}^{\infty} x \cdot f_{X|\underline{Y}}(x|y) \cdot dx$$

$$= h(y) \quad \text{- a function of } y$$

$$\Rightarrow \mathbb{E}[X \mid \underline{Y}] = \mathbb{E}[X \mid \underline{Y} = y]\Big|_{y=\underline{Y}} = h(y)\Big|_{y=\underline{Y}} = h(\underline{Y})$$

Remark: If $n \neq 1$ and $X \in \mathbb{R}^n$ $\quad \mathbb{E}[\underline{X} \mid \underline{Y}] = \begin{bmatrix} \mathbb{E}[X_1 \mid \underline{Y}] \\ \vdots \\ \mathbb{E}[X_n \mid \underline{Y}] \end{bmatrix}_{n \times 1}$

3) Orthogonality Principle

- A fundemental property of conditional expectations
- Suppose $(X, \underline{Y})$ has the joint CDF of $F_{X\underline{Y}}$ (Absolute Continuous Type) and join pdf $f_{X\underline{Y}}$

$$\mathbb{E}[X \mid \underline{Y} = y] = h(y) \qquad y \in \mathbb{R}^m$$

$\therefore$ For all Borel mapping $g: \mathbb{R}^m \longrightarrow \mathbb{R}$

$$\mathbb{E}[g(\underline{Y}) \cdot \mathbb{E}[X \mid \underline{Y}]] = \mathbb{E}[g(\underline{Y}) \cdot h(\underline{Y})]$$

Claim: $\forall\, g: \mathbb{R}^m \longrightarrow \mathbb{R}$ is a Borel mapping such that the right hand side exists

$$\boxed{\mathbb{E}[g(\underline{Y}) \cdot \mathbb{E}[X \mid \underline{Y}]] = \mathbb{E}[X \cdot g(\underline{Y})]} \quad \cdots (*)$$

OP Orthogonality Principle

- For each $y \in \mathbb{R}^m$

$$\mathbb{E}[X \mid \underline{Y} = y] = \int_{-\infty}^{\infty} x \cdot f_{X|\underline{Y}}(x|y) \cdot dx = h(y) \quad \cdots (1)$$

$$\Rightarrow \mathbb{E}[g(\underline{Y}) \cdot h(\underline{Y})] = \int_{\mathbb{R}^m} g(y) \cdot h(y) \cdot f_{\underline{Y}}(y)\, dy$$

How RHS can exist

$\mathbb{E}[X] < \infty$, g bounded or $\mathbb{E}[|X|^2] < \infty$, $\mathbb{E}[|g(Y)|^2] < \infty$

from (1)

$$= \int_{\mathbb{R}^m} g(y) \cdot \left[\int_{\mathbb{R}} x \cdot f_{X|\underline{Y}}(x|y) \cdot dx\right] f_{\underline{Y}}(y)\, dy$$

from Fubini's Theorem

$$= \int_{\mathbb{R}} x \left[\int_{\mathbb{R}^m} g(y) \cdot f_{X\underline{Y}}(x, y) \cdot dy\right] dx$$

$$= \int_{\mathbb{R} \times \mathbb{R}^m} x \cdot g(y) \cdot f_{X\underline{Y}}(x, y) \cdot dy\, dx$$

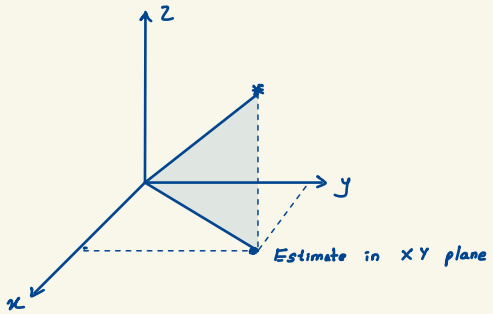
$$= \mathbb{E}[X \cdot g(\underline{Y})]$$

■

Using the observation $\underline{Y}$ to estimate X. ∴ We are looking for the error in estimating X. (ie $X - E[X|\underline{Y}]$)

$$E\left[(X - E[X|\underline{Y}]) \cdot g(\underline{Y})\right] = 0 \quad \forall\ g: \mathbb{R}^m \to \mathbb{R} \text{ a Borel mapping}$$

OP ⊛⊛



Definition: Random variables X_1 and X_2 are orthogonal if their correlation $E[X_1 X_2] = 0$

- If $E[X_1] = 0$ or $E[X_2] = 0$ then $X_1 \perp X_2 \iff X_1, X_2$ are Orthogonal

  $Cov(X_1, X_2) = E[X_1 X_2] - E[X_1] \cdot E[X_2]$

- Equivalent wording of OP ⊛⊛

  $X - E[X|\underline{Y}]$ is orthogonal to $g(\underline{Y})$ $\forall\ g: \mathbb{R}^m \to \mathbb{R}$ Borel mappings.

  OP in another form

Uniqueness

Q: Is there another mapping $\tilde{h}: \mathbb{R}^m \longrightarrow \mathbb{R}$ which is also Borel, where $\tilde{h} = h$, $h(y) = \mathbb{E}[x \mid \underline{Y} = y]$ $y \in \mathbb{R}^m$ that satisfies OP

A: No. In fact $h(y) = \mathbb{E}[x \mid \underline{Y} = y]$ $y \in \mathbb{R}^m$ is 'unique' in satisfying OP.

Proof: Suppose we have two Borel mappings $h: \mathbb{R}^m \longrightarrow \mathbb{R}$ and $\tilde{h}: \mathbb{R}^m \longrightarrow \mathbb{R}$ that satisfies OP. From OP we have

$$\Rightarrow \quad \mathbb{E}[g(\underline{Y}) \cdot h(\underline{Y})] = \mathbb{E}[x \cdot g(\underline{Y})]$$
$$\mathbb{E}[g(\underline{Y}) \cdot \tilde{h}(\underline{Y})] = \mathbb{E}[x \cdot g(\underline{Y})]$$

$$\Rightarrow \quad \mathbb{E}[g(\underline{Y}) \cdot h(\underline{Y})] = \mathbb{E}[g(\underline{Y}) \cdot \tilde{h}(\underline{Y})] \quad \forall\ g: \mathbb{R}^m \longrightarrow \mathbb{R} \text{ Borel}$$

$$\Rightarrow \quad \mathbb{E}[g(\underline{Y})\ (h(\underline{Y}) - \tilde{h}(\underline{Y}))] = 0 \quad \forall\ g$$

Constructive proof, pick $g(\underline{Y}) = \operatorname{Sgn}[h(y) - \tilde{h}(y)]$ .. $\operatorname{Sgn}(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$

Then above becomes,

$$\Rightarrow \quad \mathbb{E}[\,|h(\underline{Y}) - \tilde{h}(\underline{Y})|\,] = 0$$

$$\Rightarrow \quad \int_{-\infty}^{\infty} \underbrace{|h(y) - \tilde{h}(y)|}_{\geq 0} \cdot \underbrace{f_{\underline{Y}}(y)}_{\geq 0}\, dy = 0$$

$$\Rightarrow \quad P_{\underline{Y}}(h(y) \neq \tilde{h}(y)) = 0$$

$$\Rightarrow \quad h: \mathbb{R}^m \longrightarrow \mathbb{R} \text{ that satisfy OP } \circledast \text{ or } \circledast\circledast \text{ is unique}$$

∎

Extension of OP

when $k: \mathbb{R} \times \mathbb{R}^m \longrightarrow \mathbb{R}$ is Borel.

$$\mathbb{E}\left[\left(X - \underbrace{\mathbb{E}[k(X,\underline{Y}) \mid \underline{Y}]}_{\text{function of } \underline{Y}}\right) g(\underline{Y})\right] = 0 \qquad \forall\ g: \mathbb{R}^m \longrightarrow \mathbb{R} \text{ Borel}$$

Proof: Homework.

Corollary: OP (*) or (**) holds even when $X, \underline{Y}$ do not have a joint pdf.

Theorem: For $X: \Omega \longrightarrow \mathbb{R}$ with $\mathbb{E}[X] < \infty$ there always exists a $h: \mathbb{R}^m \longrightarrow \mathbb{R}$ Borel set s.t.

$$\left.\begin{array}{l} \mathbb{E}[g(\underline{Y}) \cdot h(\underline{Y})] = \mathbb{E}[g(\underline{Y}) \cdot X] \\ \Updownarrow \\ \mathbb{E}\left[(X - h(\underline{Y}))\, g(\underline{Y})\right] = 0 \end{array}\right\} \forall\ g: \mathbb{R}^m \longrightarrow \mathbb{R} \text{ Borel}$$

h is essentially unique $\sim \mathbb{E}_{P_{X|Y=y}}[X]$

Fact: $\mathbb{E}[X \mid \underline{Y} = \cdot]$ can be defined as satisfying OP ((*) $\Leftrightarrow$ (**)) and is characterised by the following two conditions.

- $\mathbb{E}[X \mid \underline{Y}]$ is a $\mathbb{R}$-valued rv that is a function of $\underline{Y}$
- $\mathbb{E}[X \mid \underline{Y}]$ satisfies (OP) in the role of $h(\underline{Y})$

4) Further Properties of $E[X|Y]$

- All the properties are established using OP, and hold with probability 1. $\Leftrightarrow$ a.e (almost everywhere)
- X, Y, Z has joint cdf F_{XYZ}

(i) Linearity

$$E[(aX+bZ)\,|\,Y] = a\cdot E[X|Y] + b\cdot E[Z|Y] \qquad a,b \in \mathbb{R}$$

Proof: By the Principal of orthogonality by assuming $aX+bZ$ as a rv,

$$E[g(Y)\cdot E[aX+bZ\,|\,Y]]$$

$$= E[g(Y)\cdot(aX+bZ)]$$

$$= a\cdot E[g(Y)\cdot X] + b\cdot E[g(Y)\,Z] \quad \cdots \text{ linearity of expectation}$$

$$= a\cdot E[g(Y)\cdot E[X|Y]] + b\cdot E[g(Y)\cdot E[Z|Y]] \quad \cdots \text{ OP}$$

$$= E[g(Y)\,(a\cdot E[X|Y] + b\cdot E[Z|Y])] \quad \cdots \text{linearity of expectation}$$

$$\Rightarrow E\Big[g(Y)\cdot\underbrace{\{E[aX+bZ\,|\,Y] - (a\cdot E[X|Y] + b\cdot E[Z|Y])\}}_{l(Y)}\Big] = 0$$

Choose $g(Y)$ s.t $g(y) = \operatorname{sgn}[l(Y)]$ $y \in \mathbb{R}^m$. Then

$$\Rightarrow E[|l(Y)|] = 0$$
$$\Rightarrow P(|l(Y)| = 0) = 1$$
$$\Rightarrow l(Y) = 0 \quad \text{with probability } 1$$

■

(ii) For $A \in \mathcal{A}$ Let $X = \mathbb{1}_A$ (i.e $X(\omega) = \mathbb{1}_A(\omega)$), Then

$$\mathbb{E}[X|\underline{Y}] = \mathbb{E}[\mathbb{1}_A|\underline{Y}] = P(A|\underline{Y})$$ = Conditional Probability of A given $\underline{Y}$

(iii) Unbiasness

$$\mathbb{E}[\mathbb{E}[X|\underline{Y}]] = \mathbb{E}[X]$$

Proof: Take $g(y) = 1 \quad y \in \mathbb{R}^m$ in OP.

(iv) Minimum Mean Squared Error (MMSE) Estimator

Property: Assume $\mathbb{E}[|X|^2] < \infty$, $\mathbb{E}[|h(\underline{Y})|^2] < \infty$ $\left(h(\underline{Y}) \longleftrightarrow \mathbb{E}[X|\underline{Y}]\right)$

$\mathbb{E}[X|\underline{Y}]$ is the best (minimum) MSE approximation to X among all functions $\tilde{h}: \mathbb{R}^m \longrightarrow \mathbb{R}$, Borel of $\underline{Y}$. (ie among all the estimations of X on the basis of $\underline{Y}$)

$$\mathbb{E}\left[|X - \mathbb{E}[X|\underline{Y}]|^2\right] \leq \mathbb{E}\left[|X - \tilde{h}(\underline{Y})|^2\right] \quad \forall\, \tilde{h}$$

Proof: $\mathbb{E}\left[|X - \tilde{h}(\underline{Y})|^2\right] = \mathbb{E}\left[\left|\underbrace{X - \mathbb{E}[X|\underline{Y}]}_{} + \underbrace{\mathbb{E}[X|\underline{Y}] - \tilde{h}(\underline{Y})}_{}\right|^2\right]$

$$= \mathbb{E}\left[|X - \mathbb{E}[X|\underline{Y}]|^2\right] + 2\,\mathbb{E}\left[\left(X - \mathbb{E}[X|\underline{Y}]\right)\cdot\left(\mathbb{E}(X|\underline{Y}) - \tilde{h}(\underline{Y})\right)\right] + \mathbb{E}\left[\left|\mathbb{E}[X|\underline{Y}] - \tilde{h}(\underline{Y})\right|^2\right]$$

0 from OP ⊛⊛

$$= \mathbb{E}\left[|X - \mathbb{E}[X|\underline{Y}]|^2\right] + \underbrace{\mathbb{E}\left[\left|\mathbb{E}[X|\underline{Y}] - \tilde{h}(\underline{Y})\right|^2\right]}_{\geq 0}$$

$$\geq \mathbb{E}\left[|X - \mathbb{E}[X|\underline{Y}]|^2\right]$$

With equality iff $\mathbb{E}\left[\left|\mathbb{E}[X|\underline{Y}] - \tilde{h}(\underline{Y})\right|^2\right] = 0 \iff \tilde{h}(\underline{Y}) = \mathbb{E}[X|\underline{Y}]$ with probability 1

■

Corollary:- $\mathbb{E}\left[(X - \mathbb{E}[X])^2\right] \leq \mathbb{E}\left[(X - c)^2\right] \quad \forall\, c \in \mathbb{R}$

Class of 'square integrable' estimators of X on the basis of $\underline{Y}$.

Theorem: Orthogonality Principle in MSE Estimation

Let $h \in G_2 = \left\{ g: \mathbb{R}^m \to \mathbb{R},\ \text{Borel},\ \mathbb{E}[|g(\underline{Y})|^2] < \infty \right\}$ Then

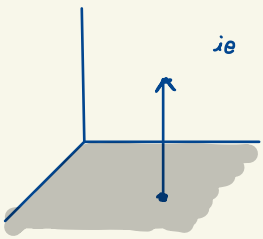
$$\mathbb{E}[\, g(\underline{Y})\,(X - h(\underline{Y}))\,] = 0 \iff \mathbb{E}[\,|X - h(\underline{Y})|^2\,] \leq \mathbb{E}[\,|X - g(\underline{Y})|^2\,]$$

$\forall\ g \in G_2$ — ie h satisfies OP

$\forall\ g \in G_2$ — ie h has the smallest MSE among all estimates in G_2

Proof: $\Longrightarrow$ Done already

$\Longleftarrow$ Homework



(v) If $X \perp\!\!\!\perp \underline{Y} \implies \mathbb{E}[X|\underline{Y}] = \mathbb{E}[X]$

Property: If $X \perp\!\!\!\perp \underline{Y} \implies \mathbb{E}[X|\underline{Y}] = \mathbb{E}[X]$

Proof: $\mathbb{E}[\, g(\underline{Y}) \cdot \mathbb{E}[X|\underline{Y}]\,] = \mathbb{E}[\, g(\underline{Y}) \cdot X\,]$ .... from Orthogonality Principle

$= \mathbb{E}[\, g(\underline{Y})\,] \cdot \mathbb{E}[X]$ .... $X \perp\!\!\!\perp \underline{Y}$

$= \mathbb{E}[\, g(\underline{Y}) \cdot \mathbb{E}[X]\,]$ .... since $\mathbb{E}[X]$ is a constant

$$\implies \mathbb{E}[\, g(\underline{Y}) \cdot \underbrace{\left(\mathbb{E}[X|\underline{Y}] - \mathbb{E}[X]\right)}_{l(\underline{Y})}\,] = 0 \quad \forall\ g$$

pick $g(\underline{Y}) = \operatorname{sgn}(l(\underline{Y}))$ and do the standard steps. ∎

Corollary: If $X = c$ (constant) with probability 1 $\implies \mathbb{E}[X|\underline{Y}] = c$

$\because X (= c) \perp\!\!\!\perp \underline{Y}$

$\lambda(\underline{Y})$ is a function of $\underline{Y}$.
$\underline{Y}$ - measurable

(vi) For any $\lambda: \mathbb{R}^m \longrightarrow \mathbb{R}$, Borel

$$\mathbb{E}[\lambda(\underline{Y})X \mid \underline{Y}] = \lambda(\underline{Y}) \cdot \mathbb{E}[X|\underline{Y}]$$

Property: For any $\lambda: \mathbb{R}^m \longrightarrow \mathbb{R}$, Borel $\mathbb{E}[\lambda(\underline{Y})X \mid \underline{Y}] = \lambda(\underline{Y}) \cdot \mathbb{E}[X|\underline{Y}]$

Proof: $\mathbb{E}\left[g(\underline{Y}) \cdot \mathbb{E}[\lambda(\underline{Y})\, X|\underline{Y}]\right] = \mathbb{E}[g(\underline{Y}) \cdot \lambda(\underline{Y})\, X]$ ... from OP

$= \mathbb{E}[g(\underline{Y}) \cdot \lambda(\underline{Y}) \cdot \mathbb{E}[X|\underline{Y}]]$ .... from OP

$= \mathbb{E}[g(\underline{Y}) \cdot \lambda(\underline{Y}) \cdot \mathbb{E}[X|\underline{Y}]]$

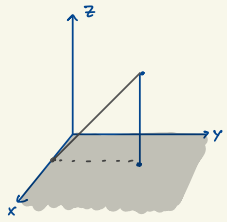
$\Rightarrow \mathbb{E}\left[g(\underline{Y})\left(\mathbb{E}[\lambda(\underline{Y})X|\underline{Y}] - \lambda(\underline{Y}) \cdot \mathbb{E}[X|\underline{Y}]\right)\right] = 0$

Proceed the proof in some way. ∎

Corollary: $k: \mathbb{R} \times \mathbb{R}^m \longrightarrow \mathbb{R}$ Borel of the form $k(x,y) = \sum_i r_i(x) \cdot \lambda_i(y)$
r_i, λ_i Borel. Then

$$\mathbb{E}[k(X,\underline{Y})|\underline{Y}] = \mathbb{E}\left[\sum_i r_i(X) \cdot \lambda_i(\underline{Y}) \,\middle|\, \underline{Y}\right] = \sum_j \lambda_j(\underline{Y}) \cdot \mathbb{E}[r_i(X)|\underline{Y}]$$

(vii) Smoothing Property / Law of Iterated Conditioning



Property - For any $\lambda: \mathbb{R}^m \to \mathbb{R}^p$ Borel

$$E\left[E[X|\lambda(Y)]\,|\,Y\right] \overset{a}{=} E[X|\lambda(Y)] \overset{b}{=} E\left[E[X|Y]\,|\,\lambda(Y)\right]$$

Projecting to smaller space like X axis

Projecting it to a larger space like X,Y plane

Proof:

@ Substitute $\lambda(Y) = E[X|\lambda(Y)]$, $X = 1$ to property (vi)

$$E[E[X|\lambda(Y)]\cdot 1\,|\,Y] = E[X|\lambda(Y)]\cdot E[1|Y] = E[X|\lambda(Y)]$$

(b) Substitute $Y = \lambda(Y)$, $X = E[X|Y]$ to OP (*)

$$\Rightarrow E\{g(\lambda(Y))\cdot E[E[X|Y]\,|\,\lambda(Y)]\} = E\{E[X|Y]\cdot g(\lambda(Y))\} = E\{X\cdot g(\lambda(Y))\} \quad \text{..... by picking } g = g \text{ in OP, } g(Y) = g(\lambda(Y))$$

$$= E\{g(\lambda(Y))\cdot E[X|\lambda(Y)]\} \quad \text{.... by } Y = \lambda(Y) \text{ in OP}$$

$$\Rightarrow E\{g(\lambda(Y))\cdot E[E[X|Y]\,|\,\lambda(Y)]\} - E\{g(\lambda(Y))\cdot E[X|\lambda(Y)]\} = 0$$

$$\Rightarrow E[g(\lambda(Y))]\cdot E\{\underbrace{E[E[X|Y]|\lambda(Y)] - E[X|\lambda(Y)]}_{m(\lambda(Y))}\} = 0$$

Take $g(\lambda(Y)) = \text{Sgn}(\lambda(Y))$.

$$\Rightarrow E[|m(\lambda(Y))|] = 0$$

$$\Rightarrow P(m(\lambda(Y)) = 0) = 1$$

$$\Rightarrow m(\lambda(Y)) = 0 \text{ with probability 1.}$$

$$\Rightarrow E[E[X|Y]|\lambda(Y)] - E[X|\lambda(Y)] = 0$$

$$\Rightarrow E[E[X|Y]|\lambda(Y)] = E[X|\lambda(Y)]$$

■

Special Case: For any rv $\underline{Z} : \Omega \to \mathbb{R}^p$ given joint cdf $F_{\underline{X}\underline{Y}\underline{Z}}$

$$\mathbb{E}\left[\, \mathbb{E}[\underline{X} \mid \underline{Y}] \,\middle|\, \underline{Y}, \underline{Z} \,\right] \overset{(a')}{=} \mathbb{E}[\underline{X} \mid \underline{Y}] \overset{(b')}{=} \mathbb{E}\left[\, [\underline{X} \mid \underline{Y}, \underline{Z}] \,\middle|\, \underline{Y} \,\right]$$

Proof: In (vii) set $\underline{Y} \to (\underline{Y}, \underline{Z})$

$$\mathcal{L}(\underline{Y}) \to \mathcal{L}(\underline{Y}, \underline{Z}) = \underline{Y}$$

(viii) $\mathcal{L} : \mathbb{R}^p \to \mathbb{R}$ Borel map

$$\mathbb{E}\left[\, \tilde{\mathcal{L}}(\underline{Z})\, \underline{X} \mid \underline{Y} \,\right] = \mathbb{E}\left[\, \tilde{\mathcal{L}}(\underline{Z}) \cdot \mathbb{E}[\underline{X} \mid \underline{Y}, \underline{Z}] \,\middle|\, \underline{Y} \,\right]$$

Property :- $\mathbb{E}\left[\, \tilde{\mathcal{L}}(\underline{Z})\, \underline{X} \mid \underline{Y} \,\right] = \mathbb{E}\left[\, \tilde{\mathcal{L}}(\underline{Z}) \cdot \mathbb{E}[\underline{X} \mid \underline{Y}, \underline{Z}] \,\middle|\, \underline{Y} \,\right]$

Proof: $\mathbb{E}\left[\, \tilde{\mathcal{L}}(\underline{Z})\, \underline{X} \mid \underline{Y} \,\right] = \mathbb{E}\left[\, \mathbb{E}\left[\tilde{\mathcal{L}}(\underline{Z})\, \underline{X} \mid \underline{Y}, \underline{Z}\right] \,\middle|\, \underline{Y} \,\right]$

$$= \mathbb{E}\left[\, \tilde{\mathcal{L}}(\underline{Z}) \cdot \mathbb{E}[\underline{X} \mid \underline{Y}, \underline{Z}] \,\middle|\, \underline{Y} \,\right]$$

(ix) Property IX

Property :- If $\underline{Y} : \Omega \to \mathbb{R}^m$, $\underline{Z} : \Omega \to \mathbb{R}^m$ are both $\mathbb{R}^m$ valued rvs and $\underline{Y} = \mathcal{L}(\underline{Z})$, $\underline{Z} = k(\underline{Y})$ for some $\mathcal{L}, k : \mathbb{R}^m \to \mathbb{R}^m$ Borel mappings.

$$\mathbb{E}[\underline{X} \mid \underline{Y}] = \mathbb{E}[\underline{X} \mid \underline{Z}]$$

Proof: $\mathbb{E}\left[\, \mathbb{E}[\underline{X} \mid \underline{Y}] \mid \underline{Z} \,\right] = \mathbb{E}\left[\, \mathbb{E}[\underline{X} \mid \underline{Y}] \mid k(\underline{Y}) \,\right]$

$$= \mathbb{E}\left[\, \underline{X} \mid k(\underline{Y}) \,\right]$$

$$= \mathbb{E}[\underline{X} \mid \underline{Z}] \quad \cdots\cdots (1)$$

$$\mathbb{E}\left[\, \mathbb{E}[\underline{X} \mid \underline{Y}] \mid \underline{Z} \,\right] = \mathbb{E}\left[\, \mathbb{E}[\underline{X} \mid \mathcal{L}(\underline{Z})] \,\middle|\, \underline{Z} \,\right]$$

$$= \mathbb{E}[\underline{X} \mid \mathcal{L}(\underline{Z})]$$

$$= \mathbb{E}[\underline{X} \mid \underline{Y}] \quad \cdots (2)$$

ie $\mathbb{E}[\underline{X} \mid \underline{Y}] = \mathbb{E}[\underline{X} \mid \underline{Z}]$

Remark: If $E[X|\underline{Y}] = a(\underline{Y})$

$E[X|\underline{Z}] = b(\underline{Z})$

(ix) does not claim that deterministic mappings $a: \mathbb{R}^m \to \mathbb{R}$ $b: \mathbb{R}^m \to \mathbb{R}$ are the same. (ie $a(\cdot) \neq b(\cdot)$). Rather (ix) says that

$$a(\underline{Y}) = b(\underline{Z}) \quad \text{as rvs}$$

$$a(\underline{Y}(\omega)) = b(\underline{Z}(\omega)) \quad \forall \ \omega \in \Omega$$

$$E[X|\underline{Y}=y] = E[X|\underline{Z}=k(y)]$$

5) Application to Estimation

Will see more on this in ENEE 621

Setting: Given $(\Omega, \mathcal{A}, P)$.

$\underline{X}: \Omega \to \mathbb{R}^n$: state of systems / signals — not observed directly

$\underline{Y}: \Omega \to \mathbb{R}^m$: is measured

Assume joint cdf $F_{\underline{X}\underline{Y}}$ is known, from mathematical modelling or emperical data.

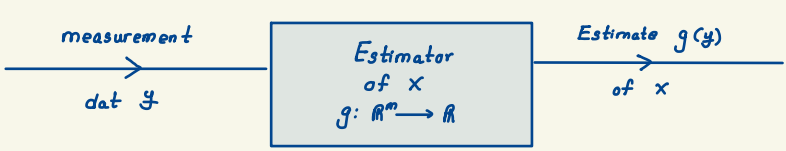
Ex: $X: \Omega \to \mathbb{R}$, $(Y = \mathcal{A}(X) + N)$ where N- noise. Y is observed. X is the original signal that undergo attenuation.

$$\{F_{XN}, \mathcal{A}\} \implies \{F_{XY}\}$$

Known — Find

Estimation Problem: How "best to approximate" $X: \Omega \to \mathbb{R}$ on the basis of Y so as to minimize a suitable estimation error?

Assume $E[|X|^2] < \infty$



Estimator - Definition An estimator of X on the basis of $\underline{Y}$ is any Borel mapping $g: \mathbb{R}^m \longrightarrow \mathbb{R}$

$g(\underline{Y})$ - $\mathbb{R}$ valued rv
$g(\underline{y})$ - $\mathbb{R}$ valued deterministic
} Estimate of X

g - deterministic } Estimator

Estimation Performance Measure

For an estimator $g: \mathbb{R}^m \longrightarrow \mathbb{R}$ Mean Square Error (MSE):

$$\mathbb{E}\left[|X - g(\underline{Y})|^2\right]$$

Estimation Optimization Problem

Find optimal $g^*: \mathbb{R}^m \longrightarrow \mathbb{R}$ in G (suitable class of estimators) such that

$$g^* = \arg\min_{g \in G} \mathbb{E}\left[|X - g(\underline{Y})|^2\right]$$

ie $$\mathbb{E}\left[|X - g^*(\underline{Y})|^2\right] \leq \mathbb{E}\left[|X - g(\underline{Y})|^2\right]$$

Important Choices for G (Estimator)

1) Square Integrable Estimators

$$\mathcal{G} = \mathcal{G}_2 = \left\{ g: \mathbb{R}^m \longrightarrow \mathbb{R}, \text{ Borel}, \ \mathbb{E}\left[|g(\underline{Y})|^2\right] < \infty \right\}$$

$$g^*(\underline{y}) = \mathbb{E}\left[X \mid \underline{Y} = \underline{y}\right], \quad \underline{y} \in \mathbb{R}^m$$

$g^* \triangleq$ minimum mean squared error estimator (MMSE estimator) of X on the basis $\underline{Y}$

2) Affine or Linear Estimators

$$\mathcal{G} = \mathcal{G}_L = \left\{ g : \mathbb{R}^m \to \mathbb{R},\ \text{affine},\ g(\underline{y}) = \underline{a}^T \underline{y} + \underline{b} \quad \underline{a} \in \mathbb{R}^m,\ b \in \mathbb{R} \right\}$$

$$g^*(\underline{y}) = \underline{a}^{*T} \underline{y} + \underline{b}^* \triangleq \hat{E}[x \mid \underline{Y} = \underline{y}]$$

$g^* \triangleq$ Linear MMSE estimator of X on the basis of $\underline{Y}$

Properties of Affine/Linear MMSE Estimator - $\hat{E}[x|\underline{y}]$

- In general, for most of the properties regarding $\hat{E}[x|\underline{y}]$, gaussianity is required.
- Let's consider the following generalized case of G_L where we are trying to estimate 'n dimensional' X-space, rather than a 1 dimensional one.

$$\mathcal{G} = \mathcal{G}_L = \left\{ g : \mathbb{R}^m \to \mathbb{R}^n,\ \text{affine},\ g(\underline{y}) = \underline{A}\,\underline{y} + \underline{b} \quad A \in \mathbb{R}^{n \times m},\ b \in \mathbb{R}^n \right\}$$

- $\hat{E}$ minimizes $E[\| \underline{X} - g(\underline{Y}) \|^2]$ over $\forall\, g \in \mathcal{G}_L$

$$\hat{E}[\underline{X} \mid \underline{Y} = \underline{y}] = \underline{A}^* \underline{y} + \underline{b}^* \quad (\underline{A}^*, \underline{b}^* \text{ to be decided})$$

- Following will be prooved in ENEE 621

1) $$\hat{E}[\underline{X} \mid \underline{Y} = \underline{y}] = \underbrace{\text{Cov}[\underline{X}, \underline{Y}]}_{n \times m} \cdot (\text{Cov}[\underline{Y}])^{-1} \cdot \underline{y} + \left(\underline{\mu}_{\underline{X}} - \text{Cov}[\underline{X}, \underline{Y}] \underbrace{(\text{Cov}[\underline{Y}])^{-1}}_{n \times n} \underline{\mu}_{\underline{Y}} \right)$$

$$\text{Cov}[\underline{X}, \underline{Y}] = E\left[\underbrace{(\underline{X} - \underline{\mu}_{\underline{X}})}_{n \times 1} \underbrace{(\underline{Y} - \underline{\mu}_{\underline{Y}})^T}_{1 \times m} \right] \qquad \text{Cov}[\underline{Y}] = E\left[(\underline{X} - \underline{\mu}_{\underline{X}})(\underline{X} - \underline{\mu}_{\underline{X}})^T \right]$$

2) Orthogonality Principle

$$\mathbb{E}\left[\underbrace{\left(\underline{X} - \hat{\mathbb{E}}[\underline{X}|\underline{Y}]\right)^T}_{1\times n} \underbrace{g(\underline{Y})}_{n\times 1}\right] = 0 \qquad \forall g \in G_2$$

$$\mathbb{E}\left[\underbrace{\left(\underline{X} - \hat{\mathbb{E}}[\underline{X}|\underline{Y}]\right)}_{n\times 1} \underbrace{g(\underline{Y})^T}_{1\times m}\right] = \underbrace{\underline{0}}_{n\times m} \qquad \forall g \in G_2$$

3) Unbiassness

$$\mathbb{E}\left[\hat{\mathbb{E}}[\underline{X}|\underline{Y}]\right] = \mathbb{E}[\underline{X}]$$

Gaussian Random Variables

Carl Fredrick Gauss
1777 - 1855

Definition of Gaussian RV : Given $(\Omega, \mathcal{A}, P)$. A random variable $X : \Omega \to \mathbb{R}$ is called Gaussian rv (Normal rv, Normally distributed, of a normal distribution) if its characteristic function is of form

$$\phi_X(u) \triangleq \mathbb{E}[e^{iuX}] = e^{iu\mu - \frac{1}{2}\sigma^2 u^2}$$

for some $\mu \in \mathbb{R}$, $\sigma > 0$

- ϕ_X ($\Leftrightarrow F_X$) is uniquely specified by two parameters : μ, σ^2
- From definition of Characteristic function

$$\mathbb{E}[X^K] = \frac{1}{i^K} \frac{d^K}{du^K} \phi_X(u)\Big|_{u=0} \qquad K = 1, 2, \ldots$$

$\therefore$ Expectation of X ($K=1$) $= \mathbb{E}[X] = \frac{1}{i} \frac{d}{du} \phi_X(u)\Big|_{u=0}$ (steps not shown)

$$= \frac{1}{i} iu$$

$$= \mu$$

$\therefore$ Second Moment of X ($K=2$) $= \mathbb{E}[X^2] = \frac{1}{i^2} \frac{d^2}{du^2} \phi_X(u)\Big|_{u=0}$

$$= \mu^2 + \sigma^2$$

$\therefore$ Variance of X = Var (X) = $\mathbb{E}[X^2] - \mu^2 = \sigma^2$

1) Two Cases of Interest

I) $\sigma^2 = 0 \implies \phi_X(u) = \mathbb{E}[e^{iuX}] = e^{iu\mu - \frac{1}{2}\cdot 0 \cdot \mu^2} = e^{iu\mu}$

$$\int_{-\infty}^{\infty} |\phi_X(u)|\, du = \int_{-\infty}^{\infty} |e^{iu\mu}|\, du = \int_{-\infty}^{\infty} 1\, du = \infty.$$

hence this is not absolutely integrable. $\therefore$ This does not have an inverse. $\therefore$ $\sigma^2 = 0$ is not exciting.

By Chebyshev's Inequality $P(|X-\mu|^2 > \varepsilon) \leq \frac{\text{Var}(X)}{\varepsilon^2} = 0 \quad \forall \varepsilon > 0$

$$\Rightarrow P(\{\omega \mid X(\omega) = \mu\}) = 1$$

(ii) $\sigma^2 > 0 : \Rightarrow \phi_X(u)$

$$\int_{-\infty}^{\infty} |\phi_X(u)|\, du = \int_{-\infty}^{\infty} e^{-\frac{1}{2}\sigma^2 u^2} du = \sqrt{2\pi\sigma^2} < \infty$$

F_X is absolute continuos type

Note :- $\therefore \sigma^2 > 0$ is the interesting case and it is assumed hereafter.

2) PDF of Gaussian RV

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2} \quad \cdots (*)$$

Proof: Due to absolute integrability, take inverse FT of characteristic function.

$$f_X(x) = \left(\mathcal{F}^{-1}\phi_X\right)(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \phi_X(u)\cdot e^{-iux}\, du$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\frac{1}{2}\cdot\sigma^2 u^2 + iu\mu}\, e^{-iux}\, du$$

Note

$$-\tfrac{1}{2}\sigma^2 u^2 + iu(\mu - x) = -\tfrac{1}{2}\sigma^2 u^2 + i\left(\frac{\mu-x}{\sigma}\right)\sigma u + \frac{(\mu-x)^2}{2\sigma^2} - \frac{1}{2}\left(\frac{\mu-x}{\sigma}\right)^2$$

$$= \frac{1}{2}\left[\frac{\mu-x}{\sigma} + i\sigma u\right]^2 - \frac{1}{2}\left(\frac{\mu-x}{\sigma}\right)^2$$

From substitution

$$\therefore f_X(x) = \frac{1}{2\pi} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \underbrace{\int_{-\infty}^{\infty} e^{\frac{1}{2}\left[\frac{\mu-x}{\sigma} + i\sigma u\right]^2} du}_{= \frac{\sqrt{2\pi}}{\sigma} \text{ by Contour Integration}}$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2} \qquad x \in \mathbb{R}$$

■

3) Jointly Gaussian RV

Definition:- A $\mathbb{R}^n$- valued rv $\underline{X}: \Omega \longrightarrow \mathbb{R}^n$ $\left(\text{ie } X: \omega \to \underline{X}(\omega) = (X_1(\omega), \ldots, X_n(\omega))\right)$ is a Gaussian rv, or $X_1, X_2, \ldots, X_n$ are jointly Gaussian if for every $\underline{\alpha} \in \mathbb{R}^n$,

$$\underline{\alpha}^T \underline{X} = \sum_{i=1}^{n} \alpha_i X_i$$

is a $\mathbb{R}$ valued Gaussian rv.

Expectation

$$\mathbb{E}[\underline{X}] = \underline{\mu} = \begin{bmatrix} \mu_{X_1} \\ \vdots \\ \mu_{X_n} \end{bmatrix}$$ - $n \times 1$ matrix with $\mathbb{R}$ valued entries

Covariance Matrix

$$\underline{R} \triangleq \mathbb{E}\left[\underset{n\times 1}{(\underline{X} - \underline{\mu})} \underset{n\times 1}{(\underline{X} - \underline{\mu})^T} \right]$$ - Covariance matrix of $\underline{X} \triangleq \text{Cov}[\underline{X}]$ $n \times n$ matrix with $\mathbb{R}$ valued entries

where

$$R_{ij} = (\underline{R})_{ij} = \mathbb{E}[(X_i - \mu_{X_i})(X_j - \mu_{X_j})] = \text{Cov}[X_i, X_j] \quad 1 \le i,j \le n$$

- $\underline{R}$ is a positive semidefinite matrix
  - $\underline{R}$ is symmetric (ie $R_{ij} = R_{ji}$)
  - $\underline{u}^T \underline{R}\, \underline{u} \geq 0 \quad \forall$ $\mathbb{R}^n$ valued vector $\underline{u}$

Definition - A Positive Definite Matrix is a matrix (R) where $\underline{u}^T \underline{R}\, \underline{u} > 0 \ \forall \ \underline{u} \neq 0$

Characteristic Function of Jointly Gaussian RV

For $\underline{X}$ to be a $\mathbb{R}^n$- valued Gaussian rv $\sim N(\underline{\mu}, \underline{R})$ a necessary and sufficient condition is the joint characteristic function to be of form

$$\phi_{\underline{X}}(\underline{u}) = \mathbb{E}\left[e^{i\underline{u}^T\underline{X}}\right] = e^{i\underline{u}^T\underline{\mu} - \frac{1}{2}\underline{u}^T\underline{R}\,\underline{u}} \qquad \underline{u} \in \mathbb{R}^n,\ \underline{R} = \mathrm{cov}[\underline{X}]$$

Proof:-

- Let's start with the joint Gaussian definition with starts with $\underline{\alpha}^T\underline{X}$. What it mean for $\underline{\alpha}^T\underline{X}$ to be a $\mathbb{R}$ valued Gaussian. For $\underline{\alpha}^T X$ to $\mathbb{R}$ valued Gaussian,

$$\mathbb{E}\left[\underbrace{\underline{\alpha}^T}_{1\times n}\underline{X}\right] = \underline{\alpha}^T\mathbb{E}[\underline{X}] = \underbrace{\underline{\alpha}^T\underline{\mu}}_{(\times 1)} \quad \cdots\cdots (\phi)'$$

$$\mathrm{Var}\left[\underline{\alpha}^T\underline{X}\right] = \mathbb{E}\left[\left(\underline{\alpha}^T\underline{X} - \underline{\alpha}^T\underline{\mu}\right)^2\right]$$

$$= \mathbb{E}\left[\underbrace{\underline{\alpha}^T}_{1\times n}\underbrace{(\underline{X}-\underline{\mu})}_{1\times n}\underbrace{(\underline{X}-\underline{\mu})^T}_{1\times n}\underbrace{\underline{\alpha}}_{n\times 1}\right] \overset{\text{also}}{=} \mathrm{Cov}\left(\underline{\alpha}^T\underline{X}\right)$$

$$= \underbrace{\underline{\alpha}^T}_{1\times n}\cdot\mathbb{E}\left[\underbrace{(\underline{X}-\underline{\mu})\cdot(\underline{X}-\underline{\mu})^T}_{n\times n}\right]\cdot\underbrace{\underline{\alpha}}_{n\times 1}$$

$$= \underline{\alpha}^T\underline{R}\,\underline{\alpha} \quad \cdots \quad \underline{R} = \mathrm{Cov}[\underline{X}] \quad \cdots (\phi)^2$$

- $\therefore (\phi)'$ and $(\phi)^2$ shows that R is a positive semidefinite.

- $\underline{X}$ Gaussian $\xrightarrow[\text{Definition}]{\text{From}}$ $\underline{\alpha}^T\underline{X}$ is $\mathbb{R}$- valued Gaussian $\forall\ \underline{\alpha} \in \mathbb{R}^n$ $\longrightarrow$ $\underline{\alpha}^T\underline{X}$ should be able to be denoted using $\mathbb{E}[\underline{\alpha}^T\underline{X}]$ and $\mathrm{Var}(\underline{\alpha}^T\underline{X})$

$$\Longrightarrow \quad \mathbb{E}\left[e^{iu\underline{\alpha}^T\underline{X}}\right] = e^{iu(\underline{\alpha}^T\underline{\mu}) - \frac{1}{2}(\underline{\alpha}^T\underline{R}\,\underline{\alpha})u^2} \qquad u \in \mathbb{R}$$

$$\Longrightarrow \quad \text{Denote } \underline{u} = u\underline{\alpha} \in \mathbb{R}^n$$
$$\mathbb{E}\left[e^{iu\underline{\alpha}^T\underline{X}}\right] = e^{i\underline{u}^T\underline{\mu} - \frac{1}{2}\underline{u}^T\underline{R}\,\underline{u}} \qquad \underline{u} \in \mathbb{R}^n$$

- Is a necessary for $\underline{X}$ to be Gaussian. Also is sufficient by backtracking to the earlier definition of a Gaussian rv $\underline{X}$. ■

Example: $X_1, \ldots, X_n$ mutually $\perp\!\!\!\perp$ $\mathbb{R}$-valued rvs $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ $\sigma_i \geq 0$
then $\underline{X} = (X_1, \ldots, X_n)$ is Gaussian rv.

4) Properties of Gaussian RVs

I) $\underline{X} : \Omega \to \mathbb{R}^n$ $\underline{A}$ - $p \times n$ matrix with $\mathbb{R}$-valued entries (Linear Transformation)
Define $\underline{Z} \triangleq \underline{A}\,\underline{X}$, then $\underline{Z} : \Omega \to \mathbb{R}^p$. If

$\underline{X}$ is Gaussian $\iff$ $\underline{Z}$ is Gaussian

Proof: Characteristic function of $\underline{Z}$ $= \phi_{\underline{Z}}(\underline{u}) = \mathbb{E}\left[e^{i\underline{u}^T\underline{Z}}\right]$

$= \mathbb{E}\left[e^{i\underline{u}^T\underline{A}\underline{X}}\right]$ $\quad \underline{u} \in \mathbb{R}^p$

$= \mathbb{E}\left[e^{i(\underline{u}^T\underline{A})\underline{X}}\right]$

$= \mathbb{E}\left[e^{i(\underline{A}^T\underline{u})^T\underline{X}}\right]$

$= \phi_{\underline{X}}(\underline{A}^T\underline{u})$

If $\underline{X}$ is Gaussian we know $\phi_{\underline{X}}$ ..... $= e^{i(\underline{A}^T\underline{u})^T\underline{\mu}_{\underline{X}} - \frac{1}{2}(\underline{A}^T\underline{u})^T\underline{R}_{\underline{X}}(\underline{A}^T\underline{u})}$

$= e^{i\underline{u}^T(\underline{A}\underline{\mu}_{\underline{X}}) - \frac{1}{2}\underline{u}^T(\underline{A}\underline{R}_{\underline{X}}\underline{A}^T)\underline{u}}$ ..... (1)

$\underline{Z} = \underline{A}\,\underline{X} \Rightarrow \underline{\mu}_{\underline{Z}} = \underline{A}\,\underline{\mu}_{\underline{X}}$ .... (2)

$\Rightarrow \mathrm{Cov}(\underline{Z}) = \underline{R}_{\underline{Z}} = \mathbb{E}\left[(\underline{Z} - \underline{\mu}_{\underline{Z}})(\underline{Z} - \underline{\mu}_{\underline{Z}}^T)\right]$

$= \mathbb{E}\left[(\underline{A}\underline{X} - \underline{A}\underline{\mu}_{\underline{X}})(\underline{A}\underline{X} - \underline{A}\underline{\mu}_{\underline{X}})^T\right]$

$= \underline{A} \cdot \mathbb{E}\left[(\underline{X} - \underline{\mu}_{\underline{X}})(\underline{X} - \underline{\mu}_{\underline{X}})^T\right]\underline{A}^T$

$= \underline{A}\,\underline{R}_{\underline{X}}\,\underline{A}^T$ .... (3)

Substitute (2), (3) to (1)

$\phi_{\underline{Z}}(\underline{u}) = e^{i\underline{u}\underline{\mu}_{\underline{Z}} - \frac{1}{2}\underline{u}^T\underline{R}_{\underline{Z}}\underline{u}}$

This is sufficient condition for Joint Gaussianness. Hence $\underline{Z}$ is a $\mathbb{R}^p$ valued Gaussian rv

■

Corollary: $\underline{X} : \Omega \longrightarrow \mathbb{R}^n$ $\quad \underline{A}$ - $p \times n$ matrix with $\mathbb{R}$ - valued entries

Define $\underline{Z} \triangleq \underline{A}\,\underline{X} + \underline{b}$ then $\underline{Z} : \Omega \longrightarrow \mathbb{R}^p$. If

$\underline{X}$ is Gaussian $\iff$ $\underline{Z}$ is Gaussian

$$\underline{\mu}_{\underline{Z}} = \underline{A}\,\underline{\mu}_{\underline{X}} + \underline{b}$$

$$\underline{R}_{\underline{Z}} = \underline{A}\,\underline{R}_{\underline{X}}\,\underline{A}^T$$

II) If $\underline{X} = (X_1, \ldots, X_n)$ is Gaussian with covariance matrix $\underline{R}_{\underline{X}}$ then

$X_1, \ldots, X_n$ is mutually Independant $\iff$ $\underline{R}_{\underline{X}}$ diagonal

Proof: $X_1, \ldots, X_n$ are mutually Independant

$X_1, \ldots, X_n$ are mutually Independant $\implies$ $X_1, \ldots, X_n$ are pairwise Independant

$\implies X_i \perp\!\!\!\perp X_j \quad \forall \; 1 \leq i \neq j \leq n$

$\implies X_i \perp X_j \quad \forall \; 1 \leq i \neq j \leq n$ .... uncorrelated

$\implies (\underline{R}_{\underline{X}})_{ij} = E[(X_i - \mu_i)(X_j - \mu_j)] = 0 \quad \forall \; 1 \leq i \neq j \leq n$

$\implies \underline{R}_{\underline{X}}$ is diagonal.

This part is always true and does not need Gaussianity.

$\underline{R}_{\underline{X}}$ is Diagonal

From the Gaussianity of $\underline{U} = (X_1, X_2, \ldots, X_n)$ we have,

$$\phi_{\underline{X}}(\underline{u}) = E\left[e^{i\underline{u}^T\underline{X}}\right] = e^{i\underline{u}^T\underline{\mu}_{\underline{X}} - \frac{1}{2}\underline{u}^T\underline{R}_{\underline{X}}\underline{u}}$$

$$= e^{i\sum_{j=1}^{n} u_j \mu_{X_j} - \frac{1}{2}\sum_{j=1}^{n} (\underline{R}_{\underline{X}})_{ij} u_j^2} \quad \text{... From Diagonality of } R_X$$

$$= \prod_{j=1}^{n} e^{i u_j \mu_{X_j} - \frac{1}{2}(R_X)_{ij} u_j^2}$$

$$= \prod_{j=1}^{n} \phi_{X_j}(u_j)$$

$\therefore$ $X_1, X_2, \ldots, X_n$ are mutually $\perp\!\!\!\perp$.

Hence proved from part 1 and 2 ∎

Corollary: If $\underline{X} = (X_1, X_2)$ is Jointly Gaussian then

$$X_1 \perp\!\!\!\perp X_2 \iff X_1 \perp X_2$$

Proof: Assume $X_1 \perp\!\!\!\perp X_2$

Without even having the Gaussianity $X_1 \perp\!\!\!\perp X_2 \Rightarrow X_1 \perp X_2$

Assume $X_1 \perp X_2$

$X_1 \perp X_2 \Rightarrow R_{\underline{X}}$ is Diagonal $\Rightarrow X_1 \perp\!\!\!\perp X_2$ from property II

■

III) If $\underline{X} = (X_1, \ldots, X_n)$ is Gaussian,

$\{X_1, \ldots, X_n\}$ mutually Independant $\iff$ $\{X_1, \ldots, X_n\}$ Pairwise Independant

($\Leftarrow$ is not true without Gaussianity)

Proof: $\Rightarrow$ direction is true by definition of mutual independance even without Gaussianity.

$\{X_1, \ldots, X_n\}$ Pairwise Independant

Pairwise Independant $\Rightarrow X_i \perp\!\!\!\perp X_j \quad \forall \ 1 \leq i \neq j \leq n$

$\Rightarrow X_i \perp X_j \quad \forall \ 1 \leq i \neq j \leq n$

$\Rightarrow \mathrm{Cov}[X_i, X_j] = (R_X)_{ij} = 0$

$\Rightarrow R_X$ is diagonal

$\Rightarrow$ By property II $\{X_1, \ldots, X_n\}$ are mutually Independant

■

Compilation and Slight Extension

Consider k random variables $\{\underline{X}_1, \ldots, \underline{X}_K\}$ $\underline{X}_i : \Omega \longrightarrow \mathbb{R}^{n_i}$ $i = 1, 2, \ldots, k$ (Each is a vector)

If rvs $\underline{X}_1, \ldots, \underline{X}_K$ are Jointly Gaussian, then the following statements are equivalent.

a) $\{\underline{X}_1, \ldots, \underline{X}_K\}$ mutually $\perp\!\!\!\perp$

b) $\{\underline{X}_1, \ldots, \underline{X}_K\}$ pairwise $\perp\!\!\!\perp$

c) $\{\underline{X}_1, \ldots, \underline{X}_K\}$ pairwise $\perp$

d) Covariance matrix of $[\underline{X}_1, \ldots, \underline{X}_K]$ is block diagonal.

$\underline{X}_1 = [\ ]\ n_1$

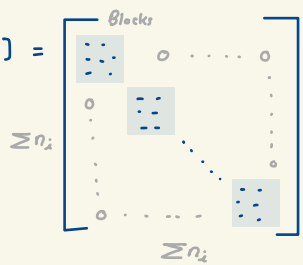
$\underline{X}_2 = [\ ]\ n_2$

$\vdots$

$\underline{X}_K = [\ ]\ n_K$

} Stack them to get $n_1 + \ldots + n_K$ vector

$\text{Cov}[\underline{X}_1, \ldots, \underline{X}_K] =$



Note: Assumption of $\{\underline{X}_1, \ldots, \underline{X}_K\}$ is Jointly Gaussian is crucial for (b) $\Rightarrow$ (a), (c) $\Rightarrow$ (b), (d) $\Rightarrow$ (a)

(Minimum Mean Square Estimate)

IV) MMSE and Linear MMSE Estimation

Property :- If $\underline{X} : \Omega \to \mathbb{R}^n$ and $\underline{Y} : \Omega \to \mathbb{R}^m$ are jointly Gaussian Then

$$\begin{bmatrix} E[x_1 | \underline{Y}] \\ \vdots \\ E[x_n | \underline{Y}] \end{bmatrix} = E[\underline{X} | \underline{Y}] = \hat{E}[\underline{X} | \underline{Y}] \triangleq \begin{bmatrix} \hat{E}[x_1 | \underline{Y}] \\ \vdots \\ \hat{E}[x_n | \underline{Y}] \end{bmatrix}$$

Proof :- $E[\underline{X} | \underline{Y}] = E\left[\underline{X} - \hat{E}[\underline{X}|\underline{Y}] + \hat{E}[\underline{X}|\underline{Y}] \;\middle|\; \underline{Y}\right]$

$$= E\left[\underline{X} - \hat{E}[\underline{X}|\underline{Y}] \middle| \underline{Y}\right] + E\left[\hat{E}[\underline{X}|\underline{Y}] \middle| \underline{Y}\right] \quad \cdots (1)$$

- $E\left[\hat{E}[\underline{X}|\underline{Y}] \middle| \underline{Y}\right] = E\left[\hat{E}[\underline{X}|\underline{Y}] \cdot 1 \middle| \underline{Y}\right]$

$$= \hat{E}[\underline{X}|\underline{Y}] \cdot \underbrace{E[1 | \underline{Y}]}_{1} \quad \cdots \text{ Since linear estimator is constant}$$

$$= \hat{E}[\underline{X}|\underline{Y}] \quad \cdots\cdots (2)$$

- **Claim :-** $(\underline{X} - \hat{E}[\underline{X}|\underline{Y}], \underline{Y})$ is Jointly Gaussian

Proof :- From Linear MMSE estimator property.

$$\Rightarrow \hat{E}[\underline{X}|\underline{Y}] = \underline{A}^* \underline{Y} + \underline{b}^*$$

$$\Rightarrow \begin{bmatrix} \underline{X} - \hat{E}[\underline{X}|\underline{Y}] \\ \underline{Y} \end{bmatrix} = \underbrace{\begin{bmatrix} \underline{I}_n & -\underline{A}^* \\ \underline{O}_m & \underline{I}_m \end{bmatrix}}_{n \times m} \begin{bmatrix} \underline{X} \\ \underline{Y} \end{bmatrix} + \underbrace{\begin{bmatrix} -\underline{b}^* \\ \underline{O} \end{bmatrix}}_{m+1}$$

$\Rightarrow (\underline{X} - \hat{E}[\underline{X}|\underline{Y}], \underline{Y})$ is jointly Gaussian.

Claim Proved ■

- From the Unbiasness Property (3rd property of Linear MMSE estimators under "Application to Estimation")

$$\Rightarrow E\left[\hat{E}[\underline{X}|\underline{Y}] \middle| \underline{Y}\right] = E[\underline{X}]$$

$$E[\underline{X} - \hat{E}[\underline{X}|\underline{Y}]] = E[\underline{X}] - E[\hat{E}[\underline{X}|\underline{Y}]]$$

$$= E[\underline{X}] - E[\underline{X}] = 0 \quad \cdots\cdots (3)$$

By the orthogonality principle on Linear Affine estimators (2nd property under "Application to Estimation")

$$\Rightarrow \underline{X} - \hat{E}[\underline{X}|\underline{Y}] \text{ is orthogonal to } \underline{Y} \quad \cdots\cdots ④$$

From ③ and ④

$$\Rightarrow \underline{X} - \hat{E}[\underline{X}|\underline{Y}] \perp \underline{Y}$$

$$\Rightarrow \underline{X} - \hat{E}[\underline{X}|\underline{Y}] \perp\!\!\!\perp \underline{Y} \quad \cdots\cdots$$

From Claim proved above. & From Property II - Corollary of Gaussian rvs, $\underline{X} \perp \underline{Y} \Leftrightarrow \underline{X} \perp\!\!\!\perp \underline{Y}$ when jointly Gaussian

$$\Rightarrow E\left[\underline{X} - \hat{E}[\underline{X}|\underline{Y}] \,\middle|\, \underline{Y}\right] = E\left[\underline{X} - \hat{E}[\underline{X}|\underline{Y}]\right] \quad \cdots \text{ orthogonality}$$

$$\Rightarrow E\left[\underline{X} - \hat{E}[\underline{X}|\underline{Y}] \,\middle|\, \underline{Y}\right] = 0 \quad \cdots\cdots ④ \text{ from } ③$$

Substituting ②, ④ to ①

$$\Rightarrow E[\underline{X}|\underline{Y}] = \hat{E}[\underline{X}|\underline{Y}]$$

QED ∎

Corollary 1 - If $\underline{X}$ and $\underline{Y}$ are jointly Gaussian,

$$E\left[(\underline{X} - E[\underline{X}|\underline{Y}])(\underline{X} - E[\underline{X}|\underline{Y}])^T \,\middle|\, \underline{Y}\right] = E\left[(\underline{X} - E[\underline{X}|\underline{Y}])(\underline{X} - E[\underline{X}|\underline{Y}])^T\right]$$

Proof: Howe Work 5.

Note - LHS is conditioned on $\underline{Y}$. ∴ It's a rv of $\underline{Y}$.
RHS is deterministic constant.

Corollary 2 - If $\underline{X}$ and $\underline{Y}$ are jointly Gaussian, The conditional probability distribution of $\underline{X}$ given $\underline{Y}$ (or $\underline{Y} = \underline{y}$) is Gaussian with mean $E[\underline{X}|\underline{Y}]$ (or $E[\underline{X}|\underline{Y} = \underline{y}]$) and covariance $\text{Cov}[\underline{X} - E[\underline{X}|\underline{Y}]]$.

Proof: Home Work 5.

(V) Every Jointly Gaussian collection of rvs can be represented as a linear transformation of mutually independant Gaussian rvs.

Property - Let $\underline{X} : \Omega \to \mathbb{R}^n$ be a Gaussian rv with $E[\underline{X}] = \underline{\mu}_X$ and $Cov[\underline{X}] = \underline{R}$ Then $\underline{X}$ can be represented as a linear transformation of n mutually Independant $\mathbb{R}$-valued Gaussian rvs $Z_1, Z_2, \ldots, Z_n$.

Results from Linear Algebra

▶ Theorem :- Every symmetric positive semi-definite matrix $\underline{R}$ can be written as

$$\underline{R} = \underline{P} \begin{bmatrix} \underline{I}_r & \underline{O}_{r,n-r} \\ \underline{O}_{n-r,r} & \underline{O}_{n-r,n-r} \end{bmatrix} \underline{P}^T \quad \cdots\cdots \; 1 \leq r \leq n$$

for some $n \times n$ orthogonal and non-singular matrix $\underline{P}$

▶ Orthogonal - A matrix $\underline{P}$ is called orthogonal if $\underline{P}^T\underline{P} = \underline{P}\,\underline{P}^T = \underline{I}_n$ and $\underline{P}^{-1} = \underline{P}^T$

Proof: ▸ $\underline{X} = (X_1, \ldots, X_n)$ is Jointly Gaussian. $\Longrightarrow$ Covariance matrix $R_{\underline{X}}$ should be positive semi-definite

$$\Longrightarrow \quad \underline{R}_{\underline{X}} = \underline{\rho} \begin{bmatrix} \underline{I}_r & \underline{O}_{r,n-r} \\ \underline{O}_{n-r,r} & \underline{O}_{n-r,n-r} \end{bmatrix} \underline{\rho}^T \quad \ldots \textcircled{1} \quad \ldots \quad 1 \leq r \leq n$$

$n \times n$ orthogonal and non-singular matrix $\underline{\rho}$

▸ Take above matrix $\underline{\rho}$ and set

$$\underline{Z} \triangleq \underline{\rho}^{-1} \underline{X}$$

$\Rightarrow$ $\underline{Z}$ is a linear transformation of $\underline{X}$ where transformation matrix $\underline{A} = \underline{\rho}^{-1}$

$\Rightarrow$ $\underline{Z}$ is Gaussian $\ldots\ldots\ldots$ ($\because$ Property I)

$\Rightarrow$ $\underline{X} \triangleq \underline{\rho}\,\underline{Z}$ is a linear transformation of n Gaussian variables $Z_1, \ldots, Z_n$ $\ldots\ldots \textcircled{2}$

▸ By Property I, we know the mean and the variance of $\underline{Z}$.

- $\underline{\mu}_{\underline{Z}} = \underline{A}\,\underline{\mu}_{\underline{X}} = \underline{\rho}^{-1}\underline{\mu}_{\underline{X}} \quad \ldots\ldots \textcircled{3}$
- $\mathrm{Cov}[\underline{Z}] = \underline{A}\,\underline{R}_{\underline{X}}\,\underline{A}^T$

$$= (\underline{\rho}^{-1}) \cdot \underline{R}_{\underline{X}} \, (\underline{\rho}^{-1})^T$$

$$= (\underline{\rho}^{-1}) \cdot \underline{R}_{\underline{X}} \, (\underline{\rho}^T)^{-1} \quad \ldots\ldots \left(\underline{\rho}^{-1} = \underline{\rho}^T \text{ since } \underline{\rho} \text{ is orthogonal} \right)$$

$$\Rightarrow \quad \underline{\rho} \cdot \mathrm{Cov}[\underline{Z}] \cdot \underline{\rho}^T = \underbrace{\underline{\rho} \cdot (\underline{\rho}^{-1})}_{\underline{I}} \cdot \underline{R}_{\underline{X}} \cdot \underbrace{(\underline{\rho}^T)^{-1} \cdot \underline{\rho}^T}_{\underline{I}}$$

$$= \underline{R}_{\underline{X}}$$

$$= \underline{\rho} \begin{bmatrix} \underline{I}_r & \underline{O}_{r,n-r} \\ \underline{O}_{n-r,r} & \underline{O}_{n-r,n-r} \end{bmatrix} \underline{\rho}^T \quad \ldots \quad [\because \textcircled{0}]$$

$\Rightarrow$ Comparing LHS and RHS

$$\text{Cov}[\underline{Z}] = \underline{R}_{\underline{Z}} = \left[\begin{array}{c|c} \underline{I}_r & \underline{O}_{r,n-r} \\ \hline \underline{O}_{n-r,r} & \underline{O}_{n-r,n-r} \end{array}\right] \quad \cdots\cdots (4)$$

- Since $\underline{Z}$ is jointly Gaussian, by using the definition with (3) and (4), we can write the characteristic equation of $\underline{Z}$ as,

$$\phi_{\underline{Z}}(\underline{u}) = E\left[e^{i\underline{u}^T\underline{Z}}\right] = e^{i\underline{u}^T\underline{\mu}_{\underline{Z}} - \frac{1}{2}\underline{u}'\underline{R}_{\underline{Z}}\underline{u}} \qquad \underline{u} \in \mathbb{R}^n$$

$$= \exp\left(i\underline{u}^T\underline{\mu}_{\underline{Z}} - \frac{1}{2}\underline{u}^T\underline{R}_{\underline{Z}}\underline{u}\right)$$

$$= \exp\left(i\sum_{j=1}^{r} u_j(\underline{P}^{-1}\underline{\mu}_{\underline{X}})_j - \frac{1}{2}\sum_{j=1}^{r} u_j^2\right) \cdot \exp\left(i\sum_{j=r+1}^{n} u_j(\underline{P}^{-1}\underline{\mu}_{\underline{X}})_j\right) \quad \text{from (3) and (4)}$$

$$= \prod_{j=1}^{r} \exp\left(i u_j \underbrace{(\underline{P}^{-1}\underline{\mu}_{\underline{X}})_j}_{\text{mean}} - \frac{1}{2}\cdot 1 \cdot u_j^2\right) \cdot \prod_{j=r+1}^{n}\left(i u_j \underbrace{(\underline{P}^{-1}\underline{\mu}_{\underline{X}})_j}_{\text{mean}} - \frac{1}{2}\cdot 0 \cdot u_j^2\right)$$

$\Rightarrow$ Gaussian rv with mean $(\underline{P}^{-1}\underline{\mu}_{\underline{X}})_j$ and variance 1

$$Z_j \sim N\left((\underline{P}^{-1}\underline{\mu}_{\underline{X}})_j, 1\right) \quad j \in [1, r]$$

$\Rightarrow$ Gaussian rv with mean $(\underline{P}^{-1}\underline{\mu}_{\underline{X}})_j$ and variance 0

$$Z_j \sim N\left((\underline{P}^{-1}\underline{\mu}_{\underline{X}})_j, 0\right) \quad j \in [r+1, n]$$

↳ $r < j \leq n$ Z_j are constants with probability 1.

$$= \prod_{j=1}^{r} E\left[e^{iu_jZ_j}\right] \cdot \prod_{j=r+1}^{n} E\left[e^{iu_jZ_j}\right]$$

$$= \prod_{j=1}^{n} E\left[e^{iu_jZ_j}\right]$$

$$= \prod_{j=1}^{n} \phi_{Z_j}(u_j)$$ · product of characteristic functions of $Z_1, .., Z_n$

$\Rightarrow$ $Z_1, \ldots, Z_n$ are Independant $\cdots\cdots$ (5)

From (2) and (5) we have

$\Rightarrow$ $\underline{X}$ can be represented as a linear transformation of n mutually Independant $\mathbb{R}$-valued Gaussian rvs $Z_1, Z_2, \ldots, Z_n$.

■

(vi) PDF of a Gaussian RV

Property: Let $\underline{X} : \Omega \longrightarrow \mathbb{R}^n$ be a Gaussian random variable with $\mathcal{N}(\underline{\mu}_{\underline{X}}, \underline{R})$

(i) If $\text{cov}[\underline{X}] = \underline{R}$ is positive semidefinite (ie invertible / non-singular). Then $F_{\underline{X}}$ is of the absolute continuous type and admits a PDF $f_{\underline{X}}$ which is

$$f_{\underline{X}}(\underline{x}) = \frac{1}{(2\pi)^n \cdot \det \underline{R}} \exp\left(-\frac{1}{2} (\underline{x} - \underline{\mu}_{\underline{X}})^T \underline{R}^{-1} (\underline{x} - \underline{\mu}_{\underline{X}}) \right) \qquad \underline{x} \in \mathbb{R}^n$$

(ii) If $\text{cov}[\underline{X}] = \underline{R}$ is non invertible / singular. Then $F_{\underline{X}}$ can be obtained from $\phi_{\underline{X}}$. But $\underline{X}$ does not have a PDF

Proof of (i): Note :- To avoid having to do contour integrations to get PDF of $\underline{X}$ from the characteristic functions, let's use the same $\underline{Z} = \underline{P}^{-1}\underline{X}$ transformation used in property (v).

From ①

$$\Rightarrow \underline{R} = \underline{P} \begin{bmatrix} \underline{I}_r & \underline{O}_{r,n-r} \\ \underline{O}_{n-r,r} & \underline{O}_{n-r,n-r} \end{bmatrix} \underline{P}^T$$

$\Rightarrow$ Since $\underline{R}$ is assumed to be invertible the matrix in the middle should be an identity. $\therefore r = n$ ..... ⑥

$$\Rightarrow \underline{R} = \underline{P}\,\underline{I}\,\underline{P}^T$$
$$\Rightarrow \underline{R} = \underline{P}\,\underline{P}^T$$
$$\Rightarrow \underline{R}^{-1} = (\underline{P}\,\underline{P}^T)^{-1}$$
$$\Rightarrow \underline{R}^{-1} = (\underline{P}^T)^{-1}\,\underline{P}^{-1} \quad \text{.... due to squareness of matrix}$$
$$\Rightarrow \underline{R}^{-1} = (\underline{P}^{-1})^T\,\underline{P}^{-1} \quad \text{.... ⑦ ..... from orthogonality of } \underline{P}$$

From ④ since $r = n$ from ⑥

$$\Rightarrow \quad \underline{R}_{\underline{z}} = \begin{bmatrix} \underline{I}_r & \underline{O}_{r,n-r} \\ \underline{O}_{n-r,r} & \underline{O}_{n-r,n-r} \end{bmatrix} = I_n \qquad \cdots\cdots (8)$$

From ⑤ since $r = n$ from ⑥

$$\Rightarrow \quad \phi_{\underline{z}}(\underline{u}) = \prod_{j=1}^{n} \phi_{z_j}(u_j) \qquad z_j \sim N\left(\underbrace{(\underline{\rho}^{-1}\underline{\mu}_{\underline{x}})_j}_{\underline{\mu}_{\underline{z}}}, 1\right) \quad 1 \le j \le n$$

$\Rightarrow$ $z_1, \ldots, z_n$ are mutually Independant with $z_j \sim N\left((\underline{\rho}^{-1}\underline{\mu}_{\underline{x}})_j, 1\right)$ $1 \le j \le n$

$\Rightarrow$ Due to the independantness and the existance of a PDF for $\underline{z}$, we can write

$$f_{\underline{z}}(\underline{z}) = f_{z_1,\ldots,z_n}(z_1, \ldots, z_n) = \prod_{j=1}^{n} f_{z_j}(z_j)$$

$$\Rightarrow \quad f_{\underline{z}}(\underline{z}) = \prod_{j=1}^{n} \frac{1}{\sqrt{2\pi}} \exp\left[- \frac{(z_j - (\underline{\mu}_{\underline{z}})_j)^2}{2} \right] \qquad \underline{z} \in \mathbb{R}^n$$

$$= \frac{1}{\sqrt{(2\pi)^n}} \exp\left[- \frac{1}{2} (\underline{z} - \underline{\mu}_{\underline{z}})^T (\underline{z} - \underline{\mu}_{\underline{z}}) \right] \cdots (9)$$

To change the basis of pdf from $\underline{z}$ to $\underline{x}$, let's use Jacobian variable transformations.

$f: \mathbb{R}^n \longrightarrow \mathbb{R}^m$ one to one and invertible

f & f^{-1} have continuos partial derivatives w.r.t co-ordinates

$g \triangleq f^{-1}$, $g: \mathbb{R}^m \longrightarrow \mathbb{R}^n$ $\underline{y} = f(\underline{x})$ has pdf $f_{\underline{y}}$ given by

$$f_{Y_1,\ldots,Y_n}(y_1, \ldots, y_n) = f_{X_1 \ldots X_n}\left(g^{-1}(y)\right) \cdot |\det J_g(y)|$$

$$J_g(y) = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_2}{\partial y_1} & \cdots & \frac{\partial x_n}{\partial y_1} \\ \frac{\partial x_1}{\partial y_2} & \frac{\partial x_2}{\partial y_2} & \cdots & \frac{\partial x_n}{\partial y_2} \\ \vdots & \vdots & & \vdots \\ \frac{\partial x_1}{\partial y_n} & \frac{\partial x_2}{\partial y_n} & \cdots & \frac{\partial x_n}{\partial y_n} \end{vmatrix}$$

$$\underline{z} = \underline{\rho}^{-1}\underline{x} \quad \Longrightarrow \quad \underline{x} = \underline{\rho}\,\underline{z}$$

Consider the one-to-one mapping $f: \mathbb{R}^n \to \mathbb{R}^n$ $f(\underline{z}) = \underline{\rho}\,\underline{z} = \underline{x}$. Hence $g \triangleq f^{-1}$ is $g(\underline{x}) = \underline{\rho}^{-1}\underline{x} = \underline{z}$. Since $\underline{\rho}^{-1}$ is a matrix, $\underline{\rho}^{-1}\underline{x}$ is a linear mapping. Hence f and g are boath continuosly differentiable.

$$\therefore f_{x_1,\ldots,x_n} = f_{z_1,\ldots,z_n}\Big(\underbrace{g^{-1}(\underline{z})}_{\underline{\rho}^{-1}\underline{x}}\Big) \cdot \left|\det J_g(\underline{x})\right|$$

Claim:- $\left|\det J_g(\underline{x})\right| = \dfrac{1}{\sqrt{\det \underline{R}}}$

Proof:- Since g is $\underline{\rho}^{-1}\underline{x}$ $\Rightarrow$ $J_g(\underline{x}) = \left|\det \underline{\rho}^{-1}\right|$

From ⑦ $\Rightarrow$ $\underline{R} = \underline{\rho}\,\underline{\rho}^T$

$\Rightarrow$ $\det \underline{R} = (\det \underline{\rho})\cdot(\det \underline{\rho}^T) = (\det \underline{\rho})^2$ $\cdots$ $\det \underline{\rho} = \det \underline{\rho}^T$

$\Rightarrow$ $\det \underline{\rho} = \sqrt{\det \underline{R}}$

$\Rightarrow$ $\det \underline{\rho}^{-1} = \dfrac{1}{\det \underline{\rho}} = \dfrac{1}{\sqrt{\det \underline{R}}}$ ..... ⑩ ← * How? Show?

$$\therefore f_{x_1,\ldots,x_n} = f_{z_1,\ldots,z_n}\Big(\underbrace{g^{-1}(\underline{z})}_{\underline{\rho}^{-1}\underline{x}}\Big) \cdot \left|J_g(\underline{x})\right|$$

$$= f_{\underline{z}}\left(\underline{\rho}^{-1}\underline{x}\right) \cdot \left|J_g(\underline{x})\right|$$

$$= \frac{1}{\sqrt{(2\pi)^n}} \exp\left[-\frac{1}{2}\left(\underline{z}-\underline{\mu}_{\underline{z}}\right)^T\left(\underline{z}-\underline{\mu}_{\underline{z}}\right)\right]_{\underline{z}=\underline{\rho}^{-1}\underline{x}} \left|J_g(\underline{x})\right| \quad \cdots [\because ⑨]$$

$$= \frac{1}{\sqrt{(2\pi)^n}} \exp\left[-\frac{1}{2}\left(\underline{\rho}^{-1}\underline{x}-\underline{\rho}^{-1}\underline{\mu}_{\underline{x}}\right)^T\left(\underline{\rho}^{-1}\underline{x}-\underline{\rho}^{-1}\underline{\mu}_{\underline{x}}\right)\right]\left|J_g(\underline{x})\right| \quad \cdots [\because ③]$$

$$= \frac{1}{\sqrt{(2\pi)^n}} \exp\left[-\frac{1}{2}\left(\underline{x}-\underline{\mu}_{\underline{x}}\right)^T\underbrace{\left(\underline{\rho}^{-1}\right)^T\underline{\rho}^{-1}}_{\underline{R}^{-1}\text{ from ⑦}}\left(\underline{x}-\underline{\mu}_{\underline{x}}\right)\right]\left|J_g(\underline{x})\right|$$

$$= \frac{1}{\sqrt{(2\pi)^n}} \exp\left[-\frac{1}{2}\left(\underline{x}-\underline{\mu}_{\underline{x}}\right)^T\underline{R}^{-1}\left(\underline{x}-\underline{\mu}_{\underline{x}}\right)\right]\left|J_g(\underline{x})\right| \quad \cdots \text{from ⑦}$$

$$= \frac{1}{\sqrt{(2\pi)^n}} \exp\left[-\frac{1}{2}\left(\underline{x}-\underline{\mu}_{\underline{x}}\right)^T\underline{R}^{-1}\left(\underline{x}-\underline{\mu}_{\underline{x}}\right)\right]\frac{1}{\sqrt{\det \underline{R}}} \quad \cdots \text{from ⑩}$$

■

Stochastic Processes

Definition - Given $(\Omega, \mathcal{A}, P)$ probability measure and an arbitary set T, a collection $\{\underline{X}_t, t \in T\}$ of $\mathbb{R}^n$ valued random variables $\underline{X}_t : \Omega \longrightarrow \mathbb{R}^n$ is a $\mathbb{R}^n$ valued **stochastic process** with parameter space T.

Notation - $\{\underline{X}_t, t \in T\}$ or $\{\underline{X}(t), t \in T\}$

Special Cases

1) Random Field
   - When T is a subset in m dimensional real space (ie $T \subseteq \mathbb{R}^m$)
   - Eg :- Markov Random Field (MRF)

2) Random Sequence
   - When T is a countable set
   - Also called "Discrete Parametric Process".
   - Eg :- $T = \mathbb{N} = \{0, 1, \ldots\}$ or $\{1, 2, \ldots\}$ or $\{t_1, t_2, \ldots\}$

3) Continuos Parametric Process
   - When T is an interval in $\mathbb{R}$
   - Eg :- $T = \mathbb{R}$ or $T = [0, +\infty)$ or $T = (0, \tau)$

Note ▸ In case (2) and (3), T usually represents time and rv $\underline{X}_t(\omega)$ is interpreted as a "state" at time t, corresponding to outcome $\omega \in \Omega$

Definition - Corresponding to a sample value $\omega \in \Omega$, the mapping created by $\underline{X}$ between t and $\underline{X}_t$. (ie $t :\longrightarrow \underline{X}_t(\omega)$ / $T :\longrightarrow \mathbb{R}^n$) is termed as a **Path**, **Sample Path** or **Trajectory** of that process.

Note

- For a given process, if we fixed the observation corresponding to a single outcome $\omega \in \Omega$, only a single path or sample function can be observed.
- A stochastic process $\{\underline{X}_t(\omega)\}$ is a function of two variables $\omega \in \Omega$ and $t \in T$.
- Only constraints -

$$\forall t \in T \quad \forall \underline{a} \in \mathbb{R}^n \quad \{\omega \mid \omega \in \Omega, \underline{X}_t(\omega) < \underline{a}\} \in \mathcal{A}$$

Random Sequences

- A sequence of random variables that is in a countable family is a random sequence.
- Typically : $\{\underline{X}_t, t = 1, 2, \ldots\}$
- To define a random sequence we need the probability distribution functions.

$$F_{\underline{X}}^{(k)}(\underline{x}_1, \ldots, \underline{x}_k) = P(\underline{X}_1 < \underline{x}_1, \ldots, \underline{X}_k < \underline{x}_k) \quad \forall\ k = 1, 2, \ldots$$

- Above countable family of distribution functions required to define a random sequence is called "family of finite-dimensional distributions."

$$\{F_{\underline{X}}^{(k)}, k = 1, 2, \ldots\} \triangleq \text{Family of finite dimensional distributions of } \{\underline{X}_t\}_{t=1}^{\infty}$$

- $\{F_{\underline{X}}^{(k)}\}_{k=1}^{\infty}$ is highly redundant collection since $F_{\underline{X}}^{(k+1)}$ has info about $F_{\underline{X}}^{(k)}$.

$$F_{\underline{X}}^{(k)}(\underline{x}_1, \ldots, \underline{x}_k) = F_{\underline{X}}^{(k+1)}(\underline{x}_1, \ldots, \underline{x}_k, \infty)$$

- Usually we don't write all $F_{\underline{X}}^{(k)}$. Rather we specify a set of rules to uniquely determine all the $F_{\underline{X}}^{(k)}$ s. $k = 1, 2, \ldots$

Ex 1 - $F_{\underline{X}}^{(k)}$ for iid sequence

Consider an Independant and Identically Distributed i.i.d sequence $\{\underline{X}_t,\ t = 1, 2, \ldots\}$

Identically Distributed $\Rightarrow F_{\underline{X}_t} = F_{\underline{X}} \qquad t = 1, 2, \ldots, k$

Independant $\Longrightarrow F_{\underline{X}}^{(k)}(\underline{x}_1, \ldots, \underline{x}_k) = \prod_{t=1}^{k} F_{\underline{X}_t}(\underline{x}_t)$

$= \prod_{t=1}^{k} F_{\underline{X}}(\underline{x}_t) \qquad \forall\ (\underline{x}_1, \ldots, \underline{x}_k)$

Ex 2 - $F_{\underline{X}}^{(k)}$ for a Simple form of a "Markov Chain"

Consider an Independant and Identically Distributed i.i.d sequence $Z_1, \ldots, Z_n$ where $Z_i \sim N(0,1)$. Define $\{X_t\ t = 1, 2, \ldots, n\}$ by

$$X_1 \triangleq Z_1$$
$$X_2 \triangleq X_1 + Z_1$$
$$X_3 \triangleq X_2 + Z_2$$
$$\vdots$$
$$X_n \triangleq X_{n-1} + Z_n \qquad n \geq 2$$

Then $\{X_t\}_{t=1}^{\infty}$ is a simple form of a "Markov Chain".

- X_n (State at $t=n$) $= X_{n-1}$ (State at $t=n-1$) $+ Z_n$ (Independant Noise)
- $\begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} = \underline{A} \begin{bmatrix} Z_1 \\ \vdots \\ Z_n \end{bmatrix}$ where $\underline{A} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 1 & 0 & \cdots & 0 \\ \vdots & & & & & \\ 1 & 1 & 1 & 1 & \cdots & 1 \end{bmatrix}$

$$\underline{X} = \underline{A}\,\underline{Z} \quad \cdots\cdots (*)$$

Claim: $\underline{Z}$ is Gaussian with $\mathcal{N}(\underline{0}, \underline{I}_n)$

Proof: Since $\underline{Z}$ is gaussian from the definition itself, $(\underline{Z} = z_1, \ldots, z_n)$

$$\phi_{\underline{Z}}(\underline{u}) = \mathbb{E}\left[e^{i\underline{u}^T\underline{Z}}\right] = \mathbb{E}\left[\exp\left(i\,\underline{u}^T\underline{Z}\right)\right] \qquad \underline{u} = (u_1, \ldots, u_n)$$

$$= \mathbb{E}\left[\exp\left(i(u_1 z_1 + u_2 z_2 + \ldots + u_n z_n)\right)\right]$$

$$= \mathbb{E}\left[\prod_{t=1}^{n} \exp\left(i u_t z_t\right)\right]$$

$$= \prod_{t=1}^{n} \mathbb{E}\left[\exp\left(i u_t z_t\right)\right] \quad \cdots \text{ mutuality of } z_i\text{'s}$$

$$= \prod_{t=1}^{n} e^{i u_t \mu_z - \frac{1}{2}\,\mathrm{Cov}(z)\cdot u_t^2}$$

$$= \prod_{t=1}^{n} e^{i\cdot u_t \cdot 0 - \frac{1}{2}\cdot 1 \cdot u_t^2} \quad \cdots \quad z_t \sim \mathcal{N}(0,1) \quad t = 1, \ldots, n$$

$$= e^{i\underline{u}^T\underline{0} - \frac{1}{2}\underline{u}^T\underline{I}_n\underline{u}}$$

(mean: $i\underline{u}^T\underline{0}$; variance: $\frac{1}{2}\underline{u}^T\underline{I}_n\underline{u}$)

$$\therefore \underline{\mu}_z = \underline{0} \quad \text{and} \quad \mathrm{Cov}(z) = \underline{I}_n \Rightarrow \underline{Z} \sim \mathcal{N}(0, I_n)$$

- Since from (*) $\underline{X} = \underline{A}\underline{Z}$, by using Property I of Gaussian distributions,

$$\Rightarrow \underline{X} \sim \mathcal{N}\left(\underline{A}\underline{\mu}_z,\ \underline{A}\,\underline{R}_z\,\underline{A}^T\right)$$

$$\Rightarrow \underline{X} \sim \mathcal{N}\left(\underline{0},\ \underline{A}\,\underline{A}^T\right) \quad \cdots \text{ from claim}$$

- Exercise: Show $f_{\underline{X}}(\underline{x}) = \prod_{t=1}^{n} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(x_t - x_{t-1})^2\right] \quad \forall (x_1, \ldots, x_n)$

$x_0 = 0$

Extending Finite Random Sequences to Infinite Subspaces

- As seen earlier in Sigma fields, given the family of finite dimensional distributions $\{F_{\underline{X}}^{(n)}\}_{n=1}^{\infty}$ of a $\mathbb{R}$-valued sequence $\{X_t\}_{t=1}^{\infty}$ can be extended to a larger class of events by,

  i) $P(A^c) = 1 - P(A)$
  ii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $\quad A, B \in \mathcal{A}$
  iii) Sequential Continuity

- In Random Sequences we can use same sequential continuity to extend $F_{\underline{X}}^{(k)}$ from finite k to infinity.

Ex:- $\{X_n\}_{n=1}^{\infty}$ is i.i.d with $N(0,1)$. Calculate the probability of event $S \triangleq \{\omega \mid \omega \in \Omega \; X_n(\omega) \geq 0 \; \forall n\}$

$$S = \{\omega \mid \omega \in \Omega \quad X_n(\omega) \geq 0 \quad \forall n\}$$

$$= \bigcap_{n=1}^{\infty} \{\omega \mid X_n(\omega) \geq 0\}$$

$$= \lim_{N \to \infty} \underbrace{\bigcap_{n=1}^{N} \{\omega \mid X_n(\omega) \geq 0\}}$$

Russian doll set as N increases. ∴ We can use sequential Continuity

$$P(S) = P\left(\lim_{N \to \infty} \bigcap_{n=1}^{N} \{\omega \mid X_n(\omega) \geq 0\}\right)$$

$$= \lim_{N \to \infty} P\left(\bigcap_{n=1}^{N} \{\omega \mid X_n(\omega) \geq 0\}\right) \quad \cdots \text{ Sequential continuity}$$

$$= \lim_{N \to \infty} \prod_{n=1}^{N} P\left(\{\omega \mid X_n(\omega) \geq 0\}\right) \quad \cdots \text{ mutually independant}$$

$$= \lim_{N \to \infty} \prod_{n=1}^{N} \underbrace{P(X_n \geq 0)}_{1/2} = \lim_{N \to \infty} (1/2)^N = 0 \; //$$

Convergence Concepts

- All probabilities are on given $(\Omega, \mathcal{A}, P)$ probability measure.
- Convergence concepts for a sequence of rvs correspond to the same $\omega \in \Omega$.
- Application of convergence concepts are on
  (i) Calculus of probability and stochastic process.
  (ii) Validation of probability models.

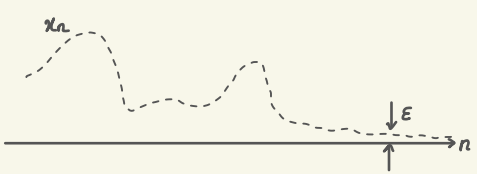
1) Prerequisite Preliminaries

Convergence

Definition :- Sequence $\{x_n\}_{n=1}^{\infty}$ of real numbers are said to be converges (or is convergent) if $\forall \varepsilon > 0$ $\exists\, x \in \mathbb{R}$ such that $\exists\, N = N(\varepsilon)$ such that $\forall\, n \geq N(\varepsilon)$, we have

$$|x_n - x| < \varepsilon$$

- This x is said to be the limit of $\{x_n\}_{n=1}^{\infty}$
- Notation :- $\lim_{n\to\infty} x_n = \lim_{n} x_n = x$ or $x_n \xrightarrow[n\to\infty]{} x$
- If such limit exists ie $x_n \xrightarrow[n\to\infty]{} x$ then x is unique.



- The definition of convergence above relies on the knowledge of limit $x \in \mathbb{R}$. ∴ We are lost if the limit $x \in \mathbb{R}$ is not available.

↓ Use Chauchy Criterion

Cauchy Criterion for Convergence

$\{x_n\}_{n=1}^{\infty}$ is convergent $\iff$ (i) $|x_{n+k} - x_n| \xrightarrow[n\to\infty]{} 0$ uniformly in $k \geq 1$

$\Updownarrow$

(ii) $\forall \varepsilon > 0 \quad \exists N = N(\varepsilon)$ s.t. $\forall n \geq N(\varepsilon)$ it holds that

$$|x_{n+k} - x_n| < \varepsilon \quad \forall k$$

- If $\{x_n\}_{n=1}^{\infty}$ is monotone

  (i) Non-decreasing ($x_n \leq x_{n+1} \ \forall n$) either
  - x_n is bounded (above) and a real - $\mathbb{R}$ limit always exists.

    or
  - x_n is unbounded $\lim_n x_n = +\infty$

  (ii) Non-increasing ($x_n \geq x_{n+1} \ \forall n$) either
  - x_n is bounded (below) and a real - $\mathbb{R}$ limit always exists.

    or
  - x_n is unbounded $\lim_n x_n = -\infty$

- In the case (ii) $\{-x_n\}_{n=1}^{\infty}$ is non decreasing and the case (i) can be applied.

Lim Sup and Lim Inf

- Assume for a sequence $\{x_1, x_2, \ldots\}$ we define $A_n = \{x_n, x_{n+1}, \ldots\}$ Since the set $A_n \supseteq A_{n+1}$, the inferior of A_n is non decreasing and Superior is non increasing.

Limit Superior	Limit Inferior
Lim Sup	Lim Inf
$\overline{\lim}$	$\underline{\lim}$

$$\Rightarrow \quad \inf\{A_n\} \leq \sup\{A_n\}$$

and non decreasing in n — and non increasing in n

$$\Rightarrow \quad \lim_n \inf\{A_n\} \leq \lim_n \sup\{A_n\}$$

$$\Rightarrow \quad \lim_n \left(\inf_{k \geq n} \{x_k\} \right) \leq \lim_n \left(\sup_{k \geq n} \{x_k\} \right)$$

$$\Rightarrow \quad \sup_n \left(\inf_{k \geq n} x_k \right) \leq \inf_n \left(\sup_{k \geq n} x_k \right)$$

$$\Rightarrow \quad \underline{\lim_n} \; x_n \leq \overline{\lim_n} \; x_n$$

- In this notation we can consider that for a sequence to converge, the equality case of above inequality should hold.

$$\lim_n x_n \text{ exists} \iff \lim_n \inf x_n = \lim_n \sup x_n = x$$

- When $\lim_n x_n$ exists, the limit 'x' is that common value in RHS

- $\lim_n \sup x_n$ and $\lim_n \inf x_n$ exists regardless of whether or not $\lim_n x_n$ exists.

2) Types of Convergence in Sequences of RVs

- Let $\{X_n\}_{n=1}^{\infty}$ be a sequence of $\mathbb{R}$-valued random variables all defined on the same $(\Omega, \mathcal{A}, P)$. /

  $\{X_n\}_{n=1}^{\infty}$ converges to a random variable $X: \Omega \to \mathbb{R}$

- Following are types of convergences.

X is Available	X is not Available
The converging r.v X is known for the sequence $\{X_n\}$	The converging r.v X is not known for the sequence $\{X_n\}$. ∴ Use the Cauchy criterion.

1) Almost Surely (a.s) / With Probability 1 (w.p.1)

$\{X_n\}_{n=1}^{\infty}$ converges to a rv X, **almost surely** iff $\exists\, A \in \mathcal{A}$ s.t $P(A) = 0$ and $\forall\, \omega \notin A$ (ie $\omega \in A^c = \Omega \setminus A$) $\{X_n(\omega)\}_{n=1}^{\infty}$ converges to $X(\omega)$.

- Notation - $\lim_{n\to\infty} X_n = X$ a.s
  $X_n \longrightarrow X$ a.s
  $X_n \longrightarrow X$ w.p.1
  $X_n \xrightarrow{a.s} X$
  $X_n \xrightarrow{w.p.1} X$

- $\forall\, \varepsilon > 0 \quad \forall\, \omega \in A^c \quad \exists\, N = N(\omega, \varepsilon)$ s.t.
  $|X_n(\omega) - X(\omega)| < \varepsilon \quad \forall\, n \geq N(\omega, \varepsilon)$

$\{X_n\}_{n=1}^{\infty}$ converges (to some rv X), **almost surely** iff $\exists\, A \in \mathcal{A}$ s.t $P(A) = 0$ and $\forall\, \omega \notin A$ (ie $\omega \in A^c = \Omega \setminus A$) $\{X_n(\omega)\}_{n=1}^{\infty}$ is Cauchy. ie

$$\lim_n \left| X_{n+k}(\omega) - X_n(\omega) \right| = 0 \quad \text{uniformly in } k$$

2) In Probability

$\{X_n\}_{n=1}^{\infty}$ converges to a rv X, in Probability iff $\forall \varepsilon > 0$

$$\lim_n P(|X_n - X| \geq \varepsilon) = 0$$

- Notation $X_n \xrightarrow[n\to\infty]{P} X$
  $X_n \xrightarrow[n\to\infty]{} X$ in probability

- i.e $\lim_n P(\{\omega \mid \omega \in \Omega, |X_n(\omega) - X(\omega)| \geq \varepsilon\})$ is 0.

- $\forall \varepsilon > 0 \quad \forall \delta > 0 \quad \exists N = N(\varepsilon, \delta)$ s.t
  $P(\{\omega \mid \omega \in \Omega, |X_n(\omega) - X(\omega)| \geq \varepsilon) < \delta$
  $\forall n \geq N(\varepsilon, \delta)$

$\{X_n\}_{n=1}^{\infty}$ converges (to some rv X), in Probability iff $\forall \varepsilon > 0$

$$\lim_n P(|X_{n+k} - X_n| \geq \varepsilon) = 0 \quad \text{uniformly in } k$$

3) In the ν^{th} - Mean, $\nu > 0$

$\{X_n\}_{n=1}^{\infty}$ converges to a rv X, in the ν^{th} mean, iff

$$\mathbb{E}[|X_n|^\nu] < \infty \qquad n = 1, 2, \ldots$$

and

$$\lim_n \mathbb{E}[|X_n - X|^\nu] = 0$$

- Notation - $X_n \xrightarrow[n\to\infty]{\text{r.m.}} X$

- $\forall \varepsilon > 0 \quad \exists N = N(\varepsilon, \nu)$ s.t
  $\mathbb{E}[|X_n - X|^\nu] < \varepsilon$
  $\forall n \geq N(\varepsilon, \nu)$

X_n is said to be convergin in Quadratic Mean (q.m) when $\nu = 2$

- Notation :- $X_n \xrightarrow[n\to\infty]{\text{q.m.}} X$

$\{X_n\}_{n=1}^{\infty}$ converges (to some rv X), in the ν^{th} mean, iff

$$\mathbb{E}[|X_n|^\nu] < \infty \qquad n = 1, 2, \ldots$$

and

$$\lim_n \mathbb{E}[|X_{n+k} - X_n|^\nu] = 0 \quad \text{uniformly in } k$$

4) In Distribution / Weak Convergence / Converges Weakly

$\{X_n\}_{n=1}^{\infty}$ converges to a rv X, **in distribution**, iff at $\forall x \in \mathbb{R}$ points where F_X (cdf of X) is continuous

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$$

- Notation: $X_n \xrightarrow[n\to\infty]{\mathcal{D}} X$
- $\forall \varepsilon > 0 \quad \exists N = N(\varepsilon)$ s.t
  $|F_{X_n}(x) - F_X(x)| < \varepsilon \quad \forall n \geq N(\varepsilon)$
  at every point of continuity of F_X
- 1) 2) 3) $\Rightarrow$ 4)

$\{X_n\}_{n=1}^{\infty}$ converges (to some rv X), **in distribution**, iff at $\forall x \in \mathbb{R}$ points where F_X (cdf of X) is continuous

$$\lim_{n\to\infty} |F_{X_{n+k}}(x) - F_{X_n}(x)| = 0$$

- Since to know the points 'x', where F_X is continuous, we should have an idea about X in the first place. But X is not know a priori in this column. ∴ Not useful

---

Ex: Use of Cauchy Criterion

Assume $\{X_n\}_{n=1}^{\infty}$ is $\mathbb{R}$-valued iid sequence of rvs. with $\mathbb{E}[X_n] = 0$, $\text{var}[X_n] = \mathbb{E}[X_n^2] - 0 = \mathbb{E}[X_n^2] = 1 \quad n = 1, 2, \ldots$

Let

$$S_n = \sum_{t=1}^{n} \frac{X_t}{t}$$

Does $\{S_n\}_{n=1}^{\infty}$ converge in Quadratic Mean.?

$$\forall\ k \geq 1 \qquad S_{n+k} - S_n = \sum_{t=1}^{n+k} \frac{X_t}{t} - \sum_{t=1}^{n} \frac{X_t}{t} = \sum_{t=n+1}^{n+k} \frac{X_t}{t}$$

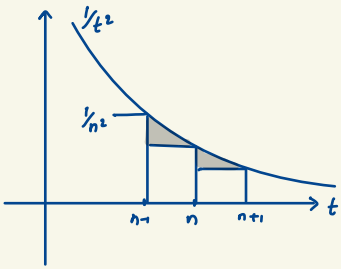
$$\Rightarrow \mathbb{E}\left[|S_{n+k} - S_n|^2\right] = \mathbb{E}\left[\left(\sum_{t=n+1}^{n+k} X_t/t\right)^2\right]$$

Steps are ommited

$$= \sum_{t=n+1}^{n+k} 1/t^2$$

$$\leq \sum_{t=n+1}^{\infty} 1/t^2$$

$$\Rightarrow \mathbb{E}\left[|S_{n+k} - S_n|^2\right] \leq \sum_{t=n+1}^{\infty} 1/t^2 \quad \cdots\cdots (1)$$



By looking at the plot

$\Rightarrow$ Discrete area $\leq$ Continuous area

$$\Rightarrow \sum_{t=n+1}^{\infty} 1/t^2 \leq \int_{n-1}^{\infty} 1/t^2 \, dt$$

$$= \left[\frac{-1}{t}\right]_{n-1}^{\infty}$$

$$= \frac{1}{n-1} \quad \cdots (2)$$

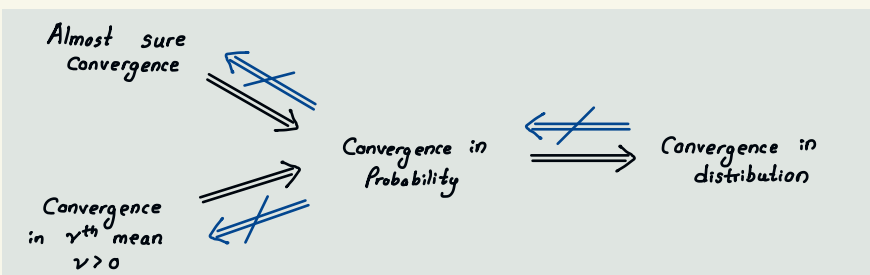
$(1), (2) \Rightarrow \mathbb{E}\left[|S_{n+k} - S_n|^2\right] \leq 1/n-1$

$$\Rightarrow \lim_{n} \mathbb{E}\left[|S_{n+k} - S_n|^2\right] \leq \lim_{n} 1/n-1 = 0$$

$$\Rightarrow \lim_{n} \mathbb{E}\left[|S_{n+k} - S_n|^2\right] = 0 \quad \text{uniformly in } \mathbb{R}$$

$\therefore \{S_n\}_{n=1}^{\infty}$ Converges in q.m. ∎

3) Relations Among Types of Convergence



Theorem : Almost Surely (a.s) Convergence $\Longrightarrow$ Convergence in Probability

If $\{X_n\}_{n=1}^{\infty}$ converges a.s. (ie w.p.1) to some rv X, then it converges in probability to the same rv X.

Proof: $\{X_n\}_{n=1}^{\infty}$ converges a.s. (ie w.p.1) to 'some' rv X $\Longrightarrow$ $\exists\ A \in \mathcal{A}$ s.t $P(A) = 0$ $\forall\ \omega \notin A$ $\cdots (1)$, $\{X_n(\omega)\}_{n=1}^{\infty}$ converges to $X(\omega)$

- Define $B_n(\varepsilon)$ as the Batch set for any given n and any $\varepsilon > 0$ as

$$B_n(\varepsilon) = \left\{ \omega \;\middle|\; \omega \in \Omega, \; |X_k(\omega) - X(\omega)| \geq \varepsilon \quad \text{for some } k \geq n \right\}$$

This set is the set of 'ω's where, for a given n we can find a future value k in the sequence

$$(X_1, X_2, \ldots, X_n, \ldots, X_k, \ldots)$$

where X_k is away from at 'ε' amount by the converging sequence X.

- When $n \uparrow \longrightarrow$ Options for X_k rv's reduces $\longrightarrow B_n(\varepsilon)$ set is a contained russian doll set.

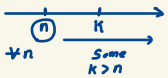
$$B_n \supseteq \left\{ \omega \;/\; \omega \in \Omega, \; |X_n(\omega) - X(\omega)| \geq \varepsilon \right\}$$

$$\left\{ |X_n(\omega) - X(\omega)| \geq \varepsilon \right\} \quad \cdots \cdot (3)$$

$$B(\varepsilon) = \bigcap_{n=1}^{\infty} B_n(\varepsilon) = \lim_{n \to \infty} B_n(\varepsilon) \cdots (3) \cdots \text{Since its a contained set}$$

$$B(\varepsilon) = \bigcap_{n=1}^{\infty} B_n(\varepsilon)$$

$$= \bigcap_{n=1}^{\infty} \left\{ \omega \;\middle|\; \omega \in \Omega, \; |X_k(\omega) - X(\omega)| \geq \varepsilon \quad \text{for some } k \geq n \right\}$$



$$= \left\{ \omega \;\middle|\; \omega \in \Omega, \; |X_k(\omega) - X(\omega)| \geq \varepsilon \quad \text{for som } k \geq n \;\; \forall n \right\}$$

$$= \left\{ \omega \;\middle|\; \omega \in \Omega, \; |X_m - X| \geq \varepsilon \quad \text{for infinitely often (i.o) } m \text{ values} \right\}$$

∴ only k's appearing infinitely many times be considered for this infinite intersection

When $|X_m - X| \geq \varepsilon$ for i.o m values, X_m does not converge to X at that ω. ∴ For all such ω's X_m does not converge

$$\therefore \omega \in B(\varepsilon) \Longrightarrow \omega \text{ does not converge} \Rightarrow \omega \in A$$

$$\Rightarrow B(\varepsilon) \subseteq A \quad \cdots \cdots (4)$$

$$\therefore \quad 0 = P(A) \quad \cdots\cdots \text{ from } ①$$

$$\geq P(B(\varepsilon)) \quad \cdots \text{ from } ④$$

$$= P\left(\lim_{n\to\infty} B_n(\varepsilon)\right) \quad \cdots \text{ from } ③$$

$$= \lim_{n\to\infty} P(B_n(\varepsilon)) \quad \cdots \text{ from Sequential Continuity Theorem}$$

$$\geq \lim_{n\to\infty} P\left(\left\{|X_n(\omega) - X(\omega)| \geq \varepsilon\right\}\right) \quad \cdots \text{ from } ②$$

$$\geq 0 \quad \cdots \text{ Probabilities are non negative}$$

$$\Rightarrow \quad 0 \geq \lim_n P(|X_n - X| \geq \varepsilon) \geq 0$$

$$\Rightarrow \quad \forall\, \varepsilon > 0 \qquad \lim_n P(|X_n - X| \geq \varepsilon) = 0$$

$$\Rightarrow \quad X_n \xrightarrow[n\to\infty]{P} X$$

■

Example: Convergence in Probability $\not\Longrightarrow$ Almost Surely (a.s.) Convergence

$\Omega = [0,1]$ Let all the intervals be events in $\mathcal{A}$ and let $P(\text{interval}) = \text{length of interval}$. Consider the sequence of rvs.

$$X_{nk} \quad - \quad \begin{array}{l} n \text{ - group - } \{1,2,\ldots\} \\ k \text{ - rv number - } \{1,2,\ldots,2^n\} \end{array}$$

such that

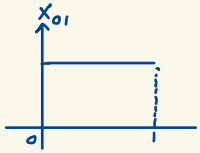
$$X_{nk}(\omega) = \begin{cases} 1 & \frac{k-1}{2^n} \leq \omega \leq \frac{k}{2^n} \\ 0 & \text{o.w.} \end{cases}$$

Renumber the above sequence to get $\{X_m\}_{m=1}^{\infty}$ as below

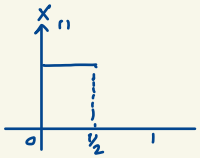
$$X_1 = X_{01} \quad X_2 = X_{11} \quad X_3 = X_{12} \quad X_4 = X_{21} \quad \cdots$$

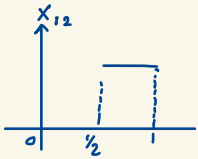
Show that X_m converge in probability, but not almost surely.

Group 0

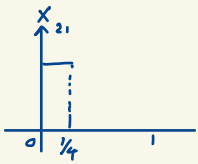


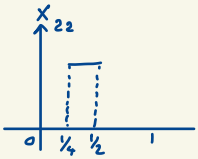
Group 1

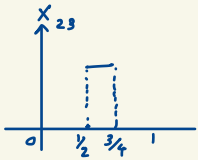


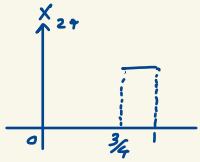


Group 2









- By looking at the above plots for any n,k

$$P(|X_{n,k} - 0| > 0) = 1/2^n$$

$$\Rightarrow \lim_{n\to\infty} P(|X_{n,k} - \underbrace{0}_{X}| > 0) = 0$$

$$\Rightarrow \lim_{n\to\infty} P(|X_m - \underbrace{X}_{0}| > 0) = 0$$

$\therefore$ renumbered sequence $\{X_m\}_{m=1}^{\infty}$ converges in probability to $X = 0$.

- From the previous theorem, if this sequence converges almost surely, that should also be to $X = 0$ ie.

$$\Rightarrow \forall \varepsilon > 0 \quad \forall \omega \in A^c \quad \exists N = N(\omega, \varepsilon) \text{ s.t. } |X_m(\omega) - 0| < \varepsilon \quad \forall m \geq N(\omega, \varepsilon)$$

* Since for any ω, we can find infinitely often n,k s.t rv $X_{n,k}(\omega) = 1$. $\therefore$ We won't have the condition $|X_m - 0| < \varepsilon \quad \forall m > N$. $\therefore$ Does not converge a.s.

■

Theorem :- Convergence in ν^{th} mean for Any $\nu > 0$ $\Longrightarrow$ Convergence in Probability

If $\{X_n\}_{n=1}^{\infty}$ converges in ν^{th} mean for any $\nu > 0$, to some rv X, then it converges in probability to the same rv X.

Proof:- $\{X_n\}_{n=1}^{\infty}$ converges in ν^{th} mean $\Longrightarrow$ $\forall\, \varepsilon > 0$ $\exists\, N = N(\varepsilon, \nu)$ s.t

$$\mathbb{E}\left[|X_n - X|^{\nu}\right] < \varepsilon \quad \cdots \quad (1)$$

$$\forall\, n \geq N(\varepsilon, \nu)$$

By applying the Markov Inequality $\forall\, \varepsilon > 0$

$$P(|X_n - X| \geq \varepsilon) \leq \frac{\mathbb{E}\left[|X_n - X|^{\nu}\right]}{\varepsilon^{\nu}} \qquad \forall\, \nu > 0$$

$$(1) \Rightarrow P(|X_n - X| \geq \varepsilon) \leq \frac{\mathbb{E}\left[|X_n - X|^{\nu}\right]}{\varepsilon^{\nu}} < \frac{\varepsilon}{\varepsilon^{\nu}} \qquad \forall\, n \geq N(\varepsilon, \nu)$$

$$\Rightarrow \lim_n P(|X_n - X| \geq \varepsilon) < \lim_n \frac{\varepsilon}{\varepsilon^{\nu}} = 0$$

$$\Rightarrow \lim_n P(|X_n - X| \geq \varepsilon) = 0$$

$$\Longrightarrow \{X_n\}_{n=1}^{\infty} \text{ converges in probability to rv } X$$

■

Example :- Convergence in Probability $\not\Longrightarrow$ Convergence in ν^{th} mean for Any $\nu > 0$

Same example setting as before where $\Omega = [0, 1]$ and all the intervals are events. $P(\text{Interval}) = \text{length of interval}$. Define a sequence $\{X_n\}_{n=1}^{\infty}$ as

$$X_n(\omega) = \begin{cases} 2^n & 0 \leq \omega \leq 1/n \\ 0 & \text{o.w} \end{cases}$$

Show that this converges in Probability, but not in ν^{th} mean.

- Since $\{\omega \mid X_n(\omega) \neq 0\} = [0, 1/n]$

$$\Rightarrow \quad P(|X_n| > 0) = 1/n$$

$$\Rightarrow \quad \lim_n P(|X_n| > 0) = \lim_n 1/n$$

$$\Rightarrow \quad \lim_n P(|X_n - \underset{\parallel}{X}_0| > \varepsilon) = 0$$

$$\Rightarrow \quad X_n \xrightarrow[n\to\infty]{P} X = 0$$

- Since $X_n(\omega) = 2^n$ in the interval $[0, 1/n]$ of length $1/n$

$$\Rightarrow \quad \mathbb{E}[|X_n|^\nu] = \left(\frac{1}{n}\right) \cdot [2^n]^\nu = 2^{n\nu}/n$$

$$\Rightarrow \quad \lim_n \mathbb{E}[|X_n|^\nu] = \lim_{n\to\infty} \left(2^{n\nu}/n\right) = \infty$$

$$\Rightarrow \quad \lim_n \mathbb{E}[|X_n - \underset{\parallel}{X}_0|^\nu] = \infty \qquad \forall\, \nu > 0$$

$$\Rightarrow \quad \{X_n\}_{n=1}^{\infty} \text{ does not converge to } 0 \text{ in the } \nu^{th} \text{ mean}$$

■

Example :- Convergence in Probability $\Longrightarrow$ Convergence in Distribution

If $\{X_n\}_{n=1}^{\infty}$ converges in probability, to some rv X, then it converges in distribution to the same rv X.

- Assume 'x' be a point of continuity of F_X

$$\cdot \quad F_X(x) = \lim_{\varepsilon\to 0} F_X(x+\varepsilon) = \lim_{\varepsilon\to 0} F_X(x-\varepsilon) \quad \cdots\cdots (1)$$

- For $\forall\, \varepsilon > 0 \quad \forall n$

$$\Rightarrow F_{X_n}(x) = P(X_n < x)$$
$$= P(X_n < x,\ X < x+\varepsilon) + P(X_n < x,\ X \geq x+\varepsilon)$$

Since $(X_n < x,\ X < x+\varepsilon) \subseteq (X < x+\varepsilon) \Rightarrow P(X_n < x,\ X < x+\varepsilon) \leq P(X < x+\varepsilon)$

Since $(X_n < x,\ X \geq x+\varepsilon) \Rightarrow X_n - X < -\varepsilon \Rightarrow (X_n < x,\ X \geq x+\varepsilon) \subseteq (X_n - X < \varepsilon)$

$$\leq P(X < x+\varepsilon) + P(X_n - X < -\varepsilon)$$

$$\leq F_X(x+\varepsilon) + P(|X_n - X| \geq \varepsilon) \quad \cdots\cdots \quad (X_n - X < -\varepsilon) \subseteq |X_n - X| \geq \varepsilon$$

$$\Rightarrow \lim_n \sup F_{X_n}(x) \leq \lim_n \sup \left(F_X(x+\varepsilon)\right) + \underbrace{\lim_n \sup \left(P(|X_n - X| \geq \varepsilon)\right)}_{= 0 \text{ from Convergence in probability}}$$

$$\leq F_X(x+\varepsilon) + 0$$

$$\Rightarrow \lim_n \sup F_{X_n}(x) \leq F_X(x+\varepsilon) \quad \cdots\cdots ②$$

▶ For $\forall\, \varepsilon > 0 \quad \forall n$

$$\Rightarrow F_X(x-\varepsilon) = P(X < x-\varepsilon)$$

$$= P(X < x-\varepsilon,\ X_n < x) + P(X < x-\varepsilon,\ X_n \geq x)$$

$$\leq P(X_n < x) + P(X_n - X < -\varepsilon)$$

$$\leq F_{X_n}(x) + P(|X_n - X| \geq \varepsilon) \quad \cdots\cdots \quad (X_n - X < -\varepsilon) \subseteq |X_n - X| \geq \varepsilon$$

$$\Rightarrow \lim_n \inf F_X(x-\varepsilon) \leq \lim_n \inf \left(F_{X_n}(x)\right) + \underbrace{\lim_n \inf \left(P(|X_n - X| \geq \varepsilon)\right)}_{= 0 \text{ from Convergence in probability}}$$

$$\Rightarrow F_X(x-\varepsilon) \leq \lim_n \inf \left(F_{X_n}(x)\right) + 0$$

$$\Rightarrow F_X(x-\varepsilon) \leq \lim_n \inf \left(F_{X_n}(x)\right) \quad \cdots\cdots ③$$

Since $\lim_n \inf F_{X_n}(x) \leq \lim_n \sup F_X(x)$ and from ②, ③

$$\Rightarrow F_X(x-\varepsilon) \leq \lim_n \inf F_{X_n}(x) \leq \lim_n \sup F_X(x) \leq F_X(x+\varepsilon)$$

Since this is $\forall\, \varepsilon$ send ε to 0

$$\Rightarrow \lim_{\varepsilon \to 0} F_X(x-\varepsilon) \leq \lim_n \inf F_{X_n}(x) \leq \lim_n \sup F_X(x) \leq \lim_{\varepsilon \to 0} F_X(x+\varepsilon)$$

from ① $\Rightarrow$

$$F_X(x) \leq \lim_n \inf F_{X_n}(x) \leq \lim_n \sup F_X(x) \leq F_X(x)$$

$\Rightarrow F_X(x) = \lim_n \inf F_{X_n}(x) = \lim_n \sup F_{X_n}(x)$

$\Rightarrow \lim_n F_{X_n}(x) = F_X(x) \quad \forall\, x$ points of continuity of F_X

$\Rightarrow \{X_n\}_{n=1}^{\infty}$ converges in distribution to X ∎

Facts

1) If ϕ_{X_n} and ϕ are characteristic functions of X_n $n=1,2,\ldots$ and X respectively. Then

$$X_n \xrightarrow{\mathcal{D}} X \iff \lim_n \phi_{X_n}(u) = \phi_X(u) \qquad u \in \mathbb{R}$$

2) Convergence in Probability $\Longrightarrow$ Convergence in Distribution is useful in characterizing limiting distributions.

For instance

If $\{X_n\}_{n=1}^{\infty}$ is a sequence of Gaussian rvs with $X_n \xrightarrow[n\to\infty]{P} X$ then X is Gaussian. Furthermore

$$\mathbb{E}[X_n] \xrightarrow{n\to\infty} \mathbb{E}[X]$$

$$\text{Var}[X_n] \xrightarrow{n\to\infty} \text{Var}[X]$$

Emile Borel - 1871 - 1956
Francesco Paolo Cantelli - 1875-1966

4) Borel - Cantelli Lemma

Lemma: For an arbitary sequence of events $\{A_n\}_{n=1}^{\infty}$ in a, if

1) $\sum_{n=1}^{\infty} P(A_n) < \infty \Longrightarrow P\left(\lim_n \sup A_n\right) = P\left(\bigcap_{n=1}^{\infty} \bigcup_{k\geq n} A_k\right) = 0$

This means ω does not belong to lot of A_n s.

If sum ω belongs to this set, that ω should belong to infinitely many A_n's

2) $\sum_{n=1}^{\infty} P(A_n) = \infty$ and $A_1, A_2 \ldots$ are mutually Independant $\Longrightarrow P\left(\lim_n \sup A_n\right) = 1$

There are lot of overlaps in A_n. $\therefore$ lot of ω's are in lot of A_ns

Proof:

1) Assume $\sum_{n=1}^{\infty} P(A_n) < \infty$

$$P\left(\lim_n \sup A_n\right) = P\left(\bigcap_{n=1}^{\infty} \bigcup_{k\geq n} A_k\right)$$

$$= P\left(\lim_n \bigcup_{k\geq n} A_k\right) \quad \ldots \text{ Since } \bigcup_{k\geq n} A_k \text{ is a collection of increasing contained set}$$

$$= \lim_n P\left(\bigcup_{k\geq n} A_k\right) \quad \ldots \text{ Sequential continuity}$$

$$\leq \lim_n \sum_{k=n}^{\infty} P(A_k) \quad \ldots \text{ union bound}$$

$$\leq \lim_n \left(\underbrace{\sum_{k=1}^{\infty} P(A_k)}_{<\infty} - \underbrace{\sum_{k=n}^{n-1} P(A_k)}_{\text{finite and approaches the value of infinite sum with } n\to\infty} \right)$$

$$\leq 0$$

$$\Rightarrow P\left(\lim_n \sup A_n\right) = 0$$

∎

2) Assume $\sum_{n=1}^{\infty} P(A_n) = \infty$

$$P\left(\lim_n \sup A_n\right) \leq 1$$

$$\Rightarrow \quad 0 \leq 1 - P(\limsup_n A_n) = 1 - P\left(\bigcap_{n=1}^{\infty} \bigcup_{k \geq n} A_k\right)$$

$$= 1 - P\left(\lim_n \bigcup_{k \geq n} A_k\right)$$

$$= 1 - \lim_n P\left(\bigcup_{k \geq n} A_k\right) \quad \text{. . as in part 1}$$

$$= 1 - \lim_n P\left(\left(\bigcap_{k \geq n} A_k^c\right)^c\right) \quad \cdots \text{Demorgan's}$$

$$= 1 - \lim_n \left(1 - P\left(\bigcap_{k \geq n} A_k^c\right)\right)$$

$$= \lim_n P\left(\bigcap_{k \geq n} A_k^c\right)$$

$$= \lim_n \prod_{k=n}^{\infty} P(A_k^c) \quad \cdots \quad A_1, A_2, \ldots \text{ are mutually independant}$$

$$= \lim_n \prod_{k=n}^{\infty} (1 - P(A_k))$$

$$\leq \lim_n \prod_{k=n}^{\infty} e^{-P(A_k)} \quad \cdots \quad 1 - x \leq e^{-x}$$

$$\leq \lim_n e^{-\sum_{k=n}^{\infty} P(A_k)}$$

$$\sum_{k=n}^{\infty} P(A_k) = \sum_{k=1}^{\infty} P(A_k) - \sum_{k=1}^{n-1} P(A_k) \quad \ldots$$
$$= \infty - \text{finite value}$$
$$= \infty$$

$$\leq \lim_n e^{-\infty}$$

$$\leq 0$$

$$\Rightarrow \quad 1 - P(\limsup_n A_n) = 0$$

$$\Rightarrow \quad P(\limsup_n A_n) = 1$$

$$\Rightarrow \quad P(\{\omega : \omega \text{ is in infinitely many of the } A_n\text{s}\}) = 1$$

■

- Borel Cantelli lemma is a powerful and standard tool for proving properties that are true with probability 1.

Applications of B-C Lemma

Ex 1:- Let $\{X_n\}_{n=1}^{\infty}$ be a sequence of $\{0,1\}$ valued rv's such that

$$P(X_n = 0) = \frac{1}{n^2} = 1 - P(X_n = 1) \qquad n = 1, 2, \ldots$$

Prove that this sequence converge almost surely (a.s).

Proof:- When n increase, the probability of X_n being '0' reduces. $\therefore$ If $\{X_n\}$ converges, it should be to '0'.

Let $A_n = \{\omega \in \Omega : X_n(\omega) = 0\}$

$$\Rightarrow P(A_n) = 1/n^2$$

$$\Rightarrow \sum_{n=1}^{\infty} P(A_n) = \lim_{n\to\infty} 1/n^2 = 0 < \infty$$

$\therefore$ By applying the first part of the Borel - Cantelli lemma,

$$\Rightarrow P(\limsup_n A_n) = 0$$

$$\Rightarrow P(\omega : \omega \in \Omega \text{ and } \omega \text{ is in infinitely often } A_n\text{s}) = 0$$

$$\Rightarrow P(\omega : \omega \in \Omega \text{ and } X_n(\omega) = 0 \text{ for infinitely often } n\text{'s}) = 0 \quad \cdots\cdots \text{definition of } A_n$$

$\therefore$ This means after some value $N(\omega)$, $X_n(\omega)$ stops being equal to '0'. ie they should be 1.

$$\Rightarrow P(\omega : \omega \in \Omega \text{ and } X_n(\omega) = 1 \ \forall\ n > N(\omega)) = 1$$

$$\Rightarrow X_n(\omega) \text{ converge to } X = 1 \text{ in a set } A \text{ with probability } 1$$

$$\Rightarrow X_n \xrightarrow[n\to\infty]{a.s.} 1$$

■

Ex 2- Let $\{X_n\}_{n=1}^{\infty}$ be mutually independant $\{0,1\}$ valued sequence of rvs. with

$$P(X_n = 1) = \frac{1}{n} \qquad P(X_n = 0) = 1 - \frac{1}{n} \qquad n = 1, 2, \ldots$$

(i) Does $\{X_n\}_{n=1}^{\infty}$ converges in Probability

(ii) Does $\{X_n\}_{n=1}^{\infty}$ converges almost surely.

(i) When n increase, the probability of X_n being '0' reduces. ∴ If $\{X_n\}$ converges, it should be to '1'.

$$\Rightarrow P(|X_n| > \varepsilon) = P(\{\omega : X_n(\omega) = 1\}) = 1/n$$

$$\Rightarrow \lim_n P(|X_n| \geq \varepsilon) = \lim_n 1/n$$

$$\Rightarrow \lim_n P(|X_n - \underset{0}{X}| \geq \varepsilon) = 0$$

$$\Rightarrow X_n \xrightarrow[n\to\infty]{P} 0$$

(ii) Since converging rv is unique, if it converges a.s, it should also be to $X = 0$.

Let $A_n = \{\omega : X_n(\omega) = 1\}$

$$\Rightarrow P(A_n) = P(\{\omega : X_n(\omega) = 1\}) = 1/n$$

$$\Rightarrow \sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} 1/n = \infty \quad \cdots \quad (1 + 1/2 + 1/3 + \ldots = \infty)$$

∴ By applying the second part of the Borel - Cantelli lemma,

$$\Rightarrow P(\lim_n \sup A_n) = 1$$

$$\Rightarrow P(\{\omega \in \Omega \quad \omega \text{ is in infintely often } A_n s\}) = 1$$

$$\Rightarrow P(\{\omega \in \Omega \quad X_n(\omega) = 1 \text{ for infintely many } n's\}) = 1$$

$$\Rightarrow X_n(\omega) \text{ converge to 1 in a set } A \text{ with } P(A) = 1$$

$$\Rightarrow X_n \xrightarrow[n\to\infty]{a.s} 1 \quad \# \text{ Since the converging value should be 0.}$$

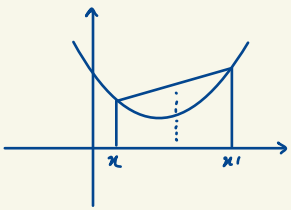
5) Convex / Concave Functions and Jensens Inequality

Let $g: \mathbb{R} \longrightarrow \mathbb{R} \cup \{+\infty\}$ be a mapping such that $g(x) < \infty$ for at least one $x \in \mathbb{R}$

Convex Function

g is convex $\iff$ $g(\alpha x + (1-\alpha)x') \leq \alpha g(x) + (1-\alpha) \cdot g(x')$

$\forall\ x < x' \in \mathbb{R}$ and $\forall\ \alpha \in [0,1]$



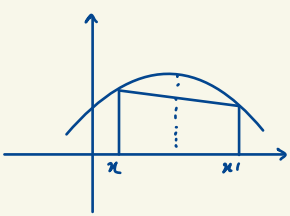
Chord lies above Curve

Concave Function

g is concave $\iff$ $g(\alpha x + (1-\alpha)x') \geq \alpha g(x) + (1-\alpha) \cdot g(x')$

$\forall\ x < x' \in \mathbb{R}$ and $\forall\ \alpha \in [0,1]$

g is concave if $-g$ is convex.



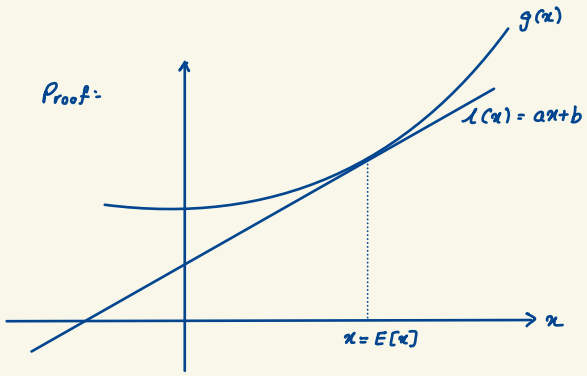
Chord lies below Curve

Jensens' Inequality

Theorem: Let $g: \mathbb{R} \longrightarrow \mathbb{R} \cup \{+\infty\}$ be a convex mapping. Let X be a $\mathbb{R}$-valued rv with $\mathbb{E}[|X|] < \infty$. Then

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$$

If g is concave $\mathbb{E}[g(X)] \leq g(\mathbb{E}[X])$



Drop a tangent at $E[x]$ point in the x axis. Assume tangent equation is $l(x) = ax+b$.

At the point where we drew the tangent, l and g are same

$\Rightarrow g(E[x]) = l(E[x]) \cdots (1)$

Convex $\Rightarrow$ tangent lies below the curve

$\Rightarrow g(x) \geq l(x) \quad \underset{(w.p.1)}{\forall x}$

$\Rightarrow E[g(x)] \geq E[l(x)] = E[ax+b]$

$= a\,E[x] + b$

$= l(E[x])$

$= g(E[x]) \quad \cdots$ from (1)

■

- Jensen's Inequality $\Rightarrow E[x^2] \geq (E[x])^2$

Limit Theorems and Sampling

1) SLLN and WLLN

Why we need Probability Estimation

- Basic assumptions in the axiomatic approach :
  - Only one 'ω' occurs as the outcome of an experiment.
  - If 'X' is a rv associated with this experiment, we have only one value $X(\omega)$ available.

- Q :- How to deduce anything about X from just $X(\omega)$?

- A :- Consider a sequence of rvs, all of which "share with X the property we seek to deduce".

At some level figuring out the mean of a rv is important and sufficient.

Converting Prob. Estimation ⟶ Mean Estimation

Ex:- Given $(\Omega, \mathcal{A}, P)$ Probability Measure. We have to decide P.
Let $A \in \mathcal{A}$. Define rv X such that

$$X(\omega) = \mathbb{I}_A(\omega)$$

Then how to assign $P(A)$ (is equivalent to how to assign $\mathbb{E}[X]$. since $P(A) = \mathbb{E}[X]$)

Remark :- In general, we seek to estimate probabilities and the expected values of functions of rvs. These estimation problems can be recast as that of estimating the mean of a suitably identified rv

⇓ W.L.O.G

New Problem - Mean Estimation

- Generate a parallel experiment under independant and identical conditions and compute the relative frequency of the occurance of A.

- ∴ For an experiment with large n,

$$\left(\text{Relative frequency of Occurance of } A\right)_n \cong \text{Good measure of } P(A)$$

- Axiamatic approach should be able to confirm this based on theory.

- Example :- Assume rv X is a noise voltage in a communication reciever circuit at $t=0$
  - Measurement of X at $t=0$ is a single number $X(\omega_0)$ for a fixed $\omega_0 \in \Omega$.
  - To find the average value of X, figure out a sequence of rvs $X_1, X_2, \cdots$ all with the same mean as X.

Mean Estimation

Idea :- Estimate the true mean $\mu = E[X]$ by a sequence of sampled or sample means μ_n defined by

$$\mu_n(\omega) \triangleq \frac{1}{n}\sum_{k=1}^{n} X_k(\omega) = \frac{S_n(\omega)}{n} \qquad n = 1, 2, \ldots$$

In these settings we can expect under certain conditions

$$\mu_n \xrightarrow[n\to\infty]{} \mu \text{ in some sense}$$

Assumption :- $\{X_n\}_{n=1}^{\infty}$ are rv's on the same Ω, and μ_n, S_n are also rv's on Ω.

- SLLN and WLLN are two laws that allows us to get these convergences under certain assumptions.

Special Case

- Before stating the WLLN and SLLN, we test the conditions with a simple sequence $\{X_n\}_{n=1}^{\infty}$ given below.

- Example - If $\{X_n\}_{n=1}^{\infty}$ is an $\{0,1\}$ valued i.i.d sequence with $P(X_1=1) = E[X_1] = p \quad 0<p<1$.

- Then we should expect

$$\mu_n(\omega) \xrightarrow[n\to\infty]{} p \quad \text{for most } \omega s$$

Bernoulli + Poisson

Weak Law of Large Numbers (WLLN) - For the special case

If $\{X_n\}_{n=1}^{\infty}$ is an $\{0,1\}$ valued i.i.d sequence with $P(X_1=1) = E[X_1] = p \quad 0<p<1$.

$$\mu_n(\omega) \triangleq \frac{1}{n}\sum_{k=1}^{n} X_k(\omega) = \frac{S_n(\omega)}{n}$$

Then,

$$\mu_n = \frac{S_n}{n} \xrightarrow[n\to\infty]{P} p$$

Convergence is to a degenerate rv which is a constant

Proof:

$$\text{Var}\left[\sum_{k=1}^{n} X_k\right] = E\left[\left|\sum_{k=1}^{n} X_k - E\left[\sum_{k=1}^{n} X_k\right]\right|^2\right] = E\left[|S_n - np|^2\right] \quad \cdots (1)$$

$$P\left(|\mu_n - p| \geq \varepsilon\right) = P\left(\left|\frac{S_n}{n} - p\right| \geq \varepsilon\right)$$

$$= P\left(|S_n - np| \geq n\varepsilon\right)$$

$$\leq \frac{P\left(|S_n - np|^2\right)}{n^2\varepsilon^2} \quad \cdots \text{ by Chebyshev Inequality}$$

$E[X_k] = p$
$E[X_k^2] = p$
$\text{Var}[X_k] = p - p^2$

$$= \frac{\text{Var}\left[\sum_{k=1}^{n} X_k\right]}{n^2\varepsilon^2} \quad \cdots \text{ from } (1)$$

$$= \frac{\sum_{k=1}^{n} \text{Var}[X_k]}{n^2\varepsilon^2} \quad \cdots \text{ iid}$$

$$= \frac{n \cdot \mathrm{Var}[X_1]}{n^2 \varepsilon^2} \quad \ldots \ldots \quad \mathrm{Var}[X_1] = \mathrm{Var}[X_k] = p$$

$$= \frac{p(1-p)}{n\varepsilon^2}$$

$$\Rightarrow P\left(|\mu_n - p| \geq \varepsilon\right) = \frac{p(1-p)}{n\varepsilon^2}$$

$$\Rightarrow \lim_n P\left(|\mu_n - p| \geq \varepsilon\right) = \lim_n \frac{p(1-p)}{n\varepsilon^2}$$

$$\Rightarrow \lim_n P\left(|\mu_n - \underset{p}{\mu}| \geq \varepsilon\right) = 0$$

$$\Rightarrow \mu_n \xrightarrow[n \to \infty]{P} p$$

■

Chebyshev, Markov, Borel, Cantelli, Kolmogorov, Kinchin

Strong Law of Large Numbers (SLLN) - For the special case

If $\{X_n\}_{n=1}^{\infty}$ is an $\{0,1\}$ valued i.i.d sequence with $P(X_1 = 1) = \mathbb{E}[X_1] = p \quad 0 < p < 1$.

$$\mu_n(\omega) \triangleq \frac{1}{n} \sum_{k=1}^{n} X_k(\omega) = \frac{S_n(\omega)}{n}$$

Then,

$$\mu_n = \frac{S_n}{n} \xrightarrow[n \to \infty]{a.s.} p$$

Proof: Needs more tools.

Weak Law of Large Numbers (WLLN)

Theorem: $\{X_k\}_{k=1}^{n}$ is a sequence of $\mathbb{R}$ valued uncorrelated rvs on the same $(\Omega, \mathcal{A}, P)$ probability measure with

$$\left.\begin{array}{ll} \text{mean} & \mathbb{E}[X_k] = \mu_k \\ \text{Variance} & \mathbb{E}[(X_k - \mu_k)^2] = \sigma_k^2 \end{array}\right\} \forall k$$

Assume $\mathbb{E}[X_k^2] < \infty \ \forall k$. Define a sequence of sample / sampled means as,

$$\mu_n(\omega) = \frac{1}{n} \sum_{k=1}^{n} X_k(\omega)$$

Then

$$\frac{1}{n^2} \sum_{k=1}^{n} \sigma_k^2 \xrightarrow[n\to\infty]{} 0 \quad \Longrightarrow \quad \mu_n - \mathbb{E}[\mu_n] \xrightarrow[n\to\infty]{\text{q.m.}} 0$$

(Bounded Variance)

Proof:

$$\mathbb{E}[\mu_n(\omega)] = \mathbb{E}\left[\frac{1}{n}\sum_{k=1}^{n} X_k(\omega)\right] = \frac{1}{n}\sum_{k=1}^{n} \mathbb{E}[X_k(\omega)] = \frac{1}{n}\sum_{k=1}^{n} \mu_k \quad \cdots (1)$$

When $\{X_n\}_{n=1}^{\infty}$ is uncorrelated $\Longrightarrow$ $\mathrm{Cov}[X_l, X_m] = \sigma_l^2 \delta_{lm}$

where $\delta_{lm} = \begin{cases} 1 & l = m \\ 0 & \text{o.w} \end{cases}$

$$\mathrm{Var}[\mu_n] = \mathbb{E}\left[(\mu_n - \mathbb{E}[\mu_n])^2\right]$$

$$= \mathbb{E}\left[\left(\frac{1}{n}\sum_{k=1}^{n} X_k(\omega) - \frac{1}{n}\sum_{k=1}^{n} \mu_k\right)^2\right] \quad \cdots \text{ from } (1)$$

$$= \frac{1}{n^2}\, \mathbb{E}\left[\left(\sum_{k=1}^{n} (X_k(\omega) - \mu_k)\right)^2\right]$$

$$= \frac{1}{n^2} \cdot \sum_{k=1}^{n} \mathbb{E}\left[(X_k - \mu_k)^2\right] + \frac{1}{n^2}\left[\sum_{1 \le l \ne m \le n} \mathbb{E}[(X_l - \mu_l)(X_m - \mu_m)]\right]$$

$$= \frac{1}{n^2}\sum_{k=1}^{n} \mathrm{Var}[X_k] + \frac{1}{n^2}\left[\sum_{1 \le l \ne m \le n} \mathrm{Cov}[X_l, X_m]\right] \quad \cdots (*)$$

$$= \frac{1}{n^2}\sum_{k=1}^{n} \sigma_k^2 + 0 \quad \cdots \quad X_i \text{ is uncorrelated}$$

$$\Rightarrow \therefore \lim_n \mathrm{Var}[\mu_n] = \lim_n \mathbb{E}\left[(\mu_n - \mathbb{E}[\mu_n])^2\right] = \lim_n \frac{1}{n^2}\sigma_k^2 = 0 \quad \cdots \text{ from bounded Variance hypothesis}$$

$$\Rightarrow \lim_n \mathbb{E}\left[(\mu_n - \mathbb{E}[\mu_n])^2\right] = 0$$

$$\Rightarrow \mu_n - \mathbb{E}[\mu_n] \xrightarrow[n\to\infty]{\text{q.m}} 0$$

■

Corollary 1 - $\frac{1}{n^2}\sum_{k=1}^{n}\sigma_k^2 \xrightarrow[n\to\infty]{} 0 \implies \mu_n - E[\mu_n] \xrightarrow[n\to\infty]{P} 0$

(Bounded Variance)

Proof: Since Converge in r^{th} mean for any $r>0$ $\implies$ Converge in Probability proof follows.

Corollary 2 - If $E[X_k] = \mu \ \forall k$

$\frac{1}{n^2}\sum_{k=1}^{n}\sigma_k^2 \xrightarrow[n\to\infty]{} 0 \implies \mu_n \xrightarrow[n\to\infty]{q.m.} \mu$

(Bounded Variance)

Corollary 3 - If $E[X_k] = \mu \ \forall k$

$\frac{1}{n^2}\sum_{k=1}^{n}\sigma_k^2 \xrightarrow[n\to\infty]{} 0 \implies \mu_n \xrightarrow[n\to\infty]{P} \mu$

(Bounded Variance)

- $\therefore$ The sequence of sample means converges to the actual/true mean in q.m (and also in probability).

Strong Law of Large Numbers (SLLN)

Theorem :- $\{X_k\}_{k=1}^{n}$ is a sequence of $\mathbb{R}$ valued uncorrelated rvs on the same $(\Omega, \mathcal{A}, P)$ probability measure with

mean $E[X_k] = \mu$

Variance $E[(X_k - \mu_k)^2] = \sigma_k^2$ $\forall k$

Not to μ_k. $\mu_k = \mu$ Constant

Assume $E[X_k^2] < \infty \ \forall k$. Define a sequence of sample / sampled means as,

$$\mu_n(\omega) = \frac{1}{n}\sum_{k=1}^{n} X_k(\omega)$$

Then

$\frac{1}{n^2}\sum_{k=1}^{n}\sigma_k^2 \xrightarrow[n\to\infty]{} 0 \implies \mu_n - E[\mu_n] \xrightarrow[n\to\infty]{a.s} 0$

(Bounded Variance)

2) Central Limit Theorem(s) - CLT(s)

Theorem: Let $\{X_k\}_{k=1}^{\infty}$ be a sequence of mutually independent ($\perp\!\!\!\perp$) $\mathbb{R}$-valued random variables on some (and the same) $(\Omega, \mathcal{A}, P)$ probability measure. Assume $\mathbb{E}[|X_k|^2] < \infty$ $k = 1, 2, \ldots$ guarantees variances are finite.

Define B_n, μ_n $n = 1, 2, \ldots$ as (Assumption, the variance of X_i are equal)

$$B_n \triangleq \sum_{k=1}^{n} \mathbb{E}\left[(X_k - \mathbb{E}[X_k])^2\right]$$

$$\mu_n = \frac{1}{n} \sum_{k=1}^{n} X_k$$

only randomness in this exercise is from μ_n.

Define a sequence of normalized random variable $\{Z_n\}_{n=1}^{\infty}$ as

$$Z_n \triangleq \frac{n(\mu_n - \mathbb{E}[\mu_n])}{\sqrt{B_n}}$$

Then,

$$F_{Z_n}(z) \xrightarrow{n \to \infty} \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, du$$

Most Elementary form of CLT

Proof:

$$B_n \triangleq \sum_{k=1}^{n} \sigma_k^2 = \sum_{k=1}^{n} \mathbb{E}\left[(X_k - \mathbb{E}[X_k])^2\right] = \sum_{k=1}^{n} \mathbb{E}\left[(X_k - \mu_k)^2\right]$$

$$Z_n \triangleq \frac{n(\mu_n - \mathbb{E}[\mu_n])}{\sqrt{B_n}} = \frac{\mu_n - \mathbb{E}[\mu_n]}{\sqrt{B_n/n}}$$

$$= \frac{n\left(\frac{1}{n}\sum_{k=1}^{n} X_k - \mathbb{E}\left[\frac{1}{n}\sum_{k=1}^{n} X_k\right]\right)}{\sqrt{B_n}}$$

$$= \frac{\sum_{k=1}^{n} (X_k - \mathbb{E}[X_k])}{\sqrt{B_n}} \qquad (\mathbb{E}[X_k] = \mu_k)$$

$$= \frac{1}{\sqrt{B_n}} \sum_{k=1}^{n} (X_k - \mu_k) \qquad (\sqrt{B_n} = \sqrt{\textstyle\sum_{k=1}^{n} \sigma_k^2})$$

Observation: Z_n has '0' mean and variance 1 $n = 1, 2, \ldots$

Let F_{Z_n} = cdf of Z_n $\Longrightarrow$ $F_{Z_n}(z) = P(Z_n \leq z)$ $z \in \mathbb{R}$

We wish to examine: F_{Z_n} as $n \longrightarrow \infty$

Since $\{X_k\}_{k=1}^{\infty}$ is an independant sequence.

$$\phi_{Z_n}(u) = \mathbb{E}\left[e^{iuZ_n}\right]$$

$$= \mathbb{E}\left[e^{iu\frac{1}{\sqrt{B_n}}\sum_{k=1}^{n}(X_k-\mu_k)}\right]$$

$$= \mathbb{E}\left[\prod_{k=1}^{n} e^{iu\left(\frac{X_k-\mu_k}{\sqrt{B_n}}\right)}\right]$$

$$= \prod_{k=1}^{n} \mathbb{E}\left[e^{iu\left(\frac{X_k-\mu_k}{\sqrt{B_n}}\right)}\right] \quad \ldots \text{ from independance}$$

heuristically if B_n is large we can use Taylor series expansion around zero, Using the taylor series expansion of e^y around $y=0$, $y \in \mathbb{R}$

$$\Rightarrow e^y = 1 + y + y^2/2! + y^3/3! + \ldots$$

$$= \sum_{k=0}^{\infty} y^k/k! \quad \forall y$$

$\therefore$ For $y = iux$, $x = \frac{X_k-\mu_k}{\sqrt{B_n}}$

$$\Rightarrow e^{iux} = 1 + iux + i^2u^2x^2/2! + i^3u^3x^3/3! + \ldots$$

$$= 1 + iux - u^2x^2/2! + iu^3x^3/3! + \ldots$$

$$\Rightarrow \phi_{Z_n}(u) = \prod_{k=1}^{n}\left[1 - \frac{u^2\sigma_k^2}{2B_n} + o\left(\frac{\sigma_k^2}{B_n}\right)\right]$$

- If the variance of rv's are equal (which is the case in most practical cases) $\sigma_k^2 = \sigma^2 \; \forall k$

$$Z_n = \frac{\sum_{k=1}^{n}(X_k-\mu_k)}{\sqrt{n\sigma^2}}$$

$\therefore$ characteristic function becomes,

$$\phi_{Z_n}(u) = \prod_{k=1}^{n}\left[1 - \frac{u^2\sigma^2}{2n\sigma^2} + o\left(\frac{\sigma_k^2}{B_n}\right)\right]$$

$$\simeq \prod_{k=1}^{n} \left[1 - \frac{u^2 \sigma^2}{2n\sigma^2} \right]$$

$$\simeq \left(1 - \frac{u^2}{2n} \right)^n$$

$$\Rightarrow \lim_n \phi_{Z_n}(u) \simeq \lim_n \left(1 - \frac{u^2}{2n} \right)^n = \underset{N(0,1)}{e^{-u^2/2}} \qquad u \in \mathbb{R}$$

$$\Rightarrow \phi_{Z_n}(u) \xrightarrow[n \to \infty]{} e^{-u^2/2} = \phi(u) \sim N(0,1) \qquad u \in \mathbb{R}$$

$$\text{ie} \Rightarrow F_{Z_n}(z) \xrightarrow[n \to \infty]{} \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} \, du$$

Q.E.D

Remarks on CLT -

Above ⊛ holds under general conditions

I) σ_k's do not have to be the same, as long as $(X_k - \mu_k)$ $k = 1, 2, \ldots, n$ all contribute roughly equally to Z_n $\forall n$

II) The independant assumption $\{X_n\}_{n=1}^{\infty}$ can be relaxed some what.

III) CLT identifies rate of convergence $O(1/n)$ of the LLN to the limit.

Central Limit Theorem (CLT) states the sum of a large number of (nearly) Independant rvs of comparable magnitudes (but not necessarily identically distributed), suitably normalized is often approximately Gaussian.

Back To Stochastic Processes

Scalar rvs are used to avoid Stochastic Calculus
But vectors will be used in Markov Processes.

- $\mathbb{R}$ valued stochastic process $\{X_t, t \in T\}$ $\quad T = \begin{cases} \text{Countable set} \\ \text{or} \\ \text{Interval in } \mathbb{R} \end{cases}$

- Given $(\Omega, \mathcal{A}, P)$ probability measure, for every finite subset $\{t_1, \ldots, t_n\} \in T$

$$F_{X_{t_1} \ldots X_{t_n}}(x_1, \ldots, x_n) \triangleq P(X_{t_1} < x_1, \ldots, X_{t_n} < x_n) \qquad \forall\, x_1, \ldots, x_n \in \mathbb{R}$$

Family of finite dimensional distributions for $\{X_t, t \in T\}$ $\triangleq$ Collection of $F_{X_{t_1} \ldots X_{t_n}}$ $\forall$ finite subsets $\{t_1, \ldots, t_n\}$ of T $n \geq 1$

- Can compute the probability of every event in $\mathcal{A}$ of the form

$$\{\omega \in \Omega : (X_{t_1}(\omega), \ldots, X_{t_n}(\omega)) \in A\} \qquad A \in \mathcal{B}^n$$

- Taking the limits using the sequential Continuity Theorem, we can compute the probabilities of events involving a countable number of X_t s.

- However if events in $\mathcal{A}$ involve an uncountable no. of rvs X_t, then the probability of such an event may not be computable using the (countable) family of finite dimensional distributions.

$\Downarrow$

Additional structure is exploited or needed.

- Attributes of $\{X_t, t \in T\}$
  - for each fixed $t \in T$, X_t is a rv on $(\Omega, \mathcal{A}, P)$
  - $\{X_t, t \in T\}$ is a function on $T \times \Omega$ with values $X_t(\omega)$
  - for each fixed $\omega \in \Omega$ $X_t(\omega)$ is a function on T ⟵ Sample path corresponding to $\omega \in \Omega$

Notations

$\mu_{X_t} = \mathbb{E}[X_t]$ $t \in T$ = mean at time t

$C_X(s,t) = C(s,t) = \mathbb{E}[X_s, X_t]$ = Correlation function at $s,t \in T$

$R_X(s,t) = R(s,t) = Cov[X_s, X_t]$
$= \mathbb{E}[(X_s - \mathbb{E}[X_s])(X_t - \mathbb{E}[X_t])]$
= Covariance function at $s, t \in T$

$F_{X_{t_1}} \dots F_{X_{t_n}} = P(X_{t_1} < x_1, \dots, X_{t_n} < x_n)$ n^{th} order cdf $n \geq 1$

Second order Processes

Definition :- $\{X_t, t \in T\}$ is a second order processes if $\mathbb{E}[|X_t|^2] < \infty$ $\forall\ t \in T$

Remark:- Verify that for a second order process, the mean, covariance and correlation exists and are finite

1) Independance Concepts

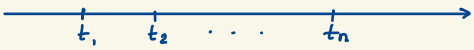
Definition : $\{X_t,\ t \in T\}$ is independant if for every finite collection $\{X_{t_1}, \ldots, X_{t_n}\}$ of rvs are mutully independant. $\forall\ n \geq 2$

Definition : $\{X_t,\ t \in T\}$ is identically distributed if $F_{X_t} = F \quad \forall\ t \in T$

Definition : $\{X_t,\ t \in T\}$ is independant and identically distributed (i.i.d) if for every finite collection $\{X_{t_1}, \ldots, X_{t_n}\}$ of rvs are mutully independant and identically distributed $\forall\ n \geq 2$

Remark : i.i.d is of particular interest when T is countable.

Definition : $\{X_t,\ t \in T\}$ is an Independant Increments process if $\forall\ n \geq 2$ and $\forall\ t_1 < \ldots < t_n$ in T, the rvs $X_{t_2} - X_{t_1}$, $X_{t_3} - X_{t_2}$, $\ldots$, $X_{t_n} - X_{t_{n-1}}$ are mutually independant



If T contains a smaller element t_0, then we take X_{t_0}, $X_{t_2 - t_1}$, $\ldots$, $X_{t_n} - X_{t_{n-1}}$

Examples of Independant Increment Processes

a) **Poisson Process** $\{N_t,\ t \geq 0\}$ is a $\mathbb{N}$-valued independant increment process, with $N_0 \equiv 0$ w.p. 1 and $\forall\ h > 0$

$$P(N_{t+h} - N_t = k) = \frac{(\lambda h)^k e^{-\lambda h}}{k!} \qquad k = 0, 1, \ldots \quad = \text{Poisson}(\lambda h)$$

b) **Wiener process** : $\{W_t,\ t \geq 0\}$ is a $\mathbb{R}$-valued independant increment process, with $W_0 \equiv 0$ w.p. 1 and $\forall\ h > 0$

$$W_{t+h} - W_t \sim \mathcal{N}(0, \sigma^2 h)$$

2) Continuity Concepts

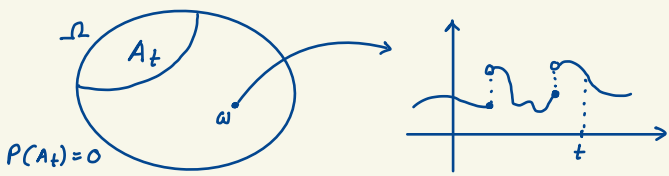
$\{X_t,\ t \in T\}$ is continuos at t

1) Almost Sure Continuity (a.s.)

$\{X_t,\ t \in T\}$ is continuos at t almost surely if

- $P\left(\lim_{s \to t} |X_s - X_t| = 0 \right) = 1$

i.e. - if $A_t = \left\{ \omega \in \Omega : \lim_{s \to t} |X_s(\omega) - X_t(\omega)| \neq 0 \right\}$ then $P(A_t) = 0$



Complement set will show continuity at t for rv X

2) Continuity In Probability

$\{X_t,\ t \in T\}$ is continuos at t in Probability if

$$\forall \varepsilon > 0 \quad P\left(|X_s - X_t| \geq \varepsilon \right) \xrightarrow[s \to t]{} 0$$

3) Continuity in r^{th} Mean ($r > 0$)

$\{X_t,\ t \in T\}$ is continuos at t in r^{th} Mean if

$$E\left[|X_s - X_t|^r \right] \xrightarrow[s \to t]{} 0$$

$\{X_t,\ t \in T\}$ is continuos

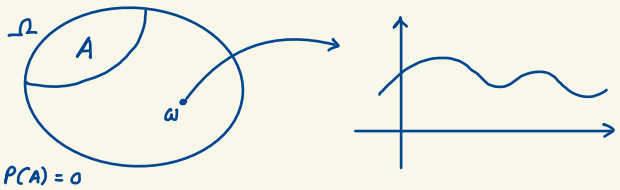
- The process $\{X_t,\ t \in T\}$ has continuity property / properties above (without 't') if it/they are valid $\forall\, t \in T$
- Ex: $\{X_t,\ t \in T\}$ is almost surely continuos (withot at 't'), i.e a.s. continuos at every $t \in T$, then $P(A_t) = 0 \quad \forall\, t \in T$

4) Sample Continuity

$\{X_t,\ t \in T\}$ is sample continuous if,

- With probability 1, every sample function / path $X(\omega, \cdot)$ is continuous on T.

i.e. ▸ Except for $A \in \mathcal{A}$ set with $P(A) = 0$, $X(\omega, \cdot)$ is continuous $\forall\, \omega \in A^c$



Complement set will show continuity at $\forall\, t \in T$

i.e $X(\omega, t)$ is continuous in t $\forall\, t \in T$

Sample Continuity $\Longrightarrow$ Almost Sure Continuity at every $t \in T$.

i.e. $P(A_t) = 0$ $\forall\, t \in T$

$A \triangleq \bigcup_{t \in T} A_t$

hence $P(A) = P\left(\bigcup_{t \in T} A_t\right) = 0$

3) Martingales

Definition:- Martingales

A $\mathbb{R}^n$ valued process $\{X_t,\ t \in T\}$ is a martingale if for every $t_1 < t_2 < \dots < t_K$ in T and $\forall\ n \geq 2$

$$\mathbb{E}\left[X_{t_K} \mid X_{t_1} \dots X_{t_{K-2}}\ X_{t_{K-1}} \right] = X_{t_{K-1}}$$

almost surely (P- a.s.)

Next rv — Previous rvs — Current rv

- If $\mathbb{E}\left[X_{t_K} \mid X_{t_1} \dots X_{t_{K-2}}\ X_{t_{K-1}} \right] \geq X_{t_{K-1}}$ - submartingale
- If $\mathbb{E}\left[X_{t_K} \mid X_{t_1} \dots X_{t_{K-2}}\ X_{t_{K-1}} \right] \leq X_{t_{K-1}}$ - supermartingale
- Expected value of the next r.v. given all previous values (including the current value), is equal to the current value.
- This represents a "fair game" in which best predicted value of the future is the current value.
- Applications ⟶ Mathematical finance
  Gambling
  Estimation / filtering in stochastic processes.
- Examples ⟶ Brownian motion
  Poisson process - $\{N_t,\ t \geq 0\}$ with $N_0 \equiv 0$ w.p. 1
  $\{X_t,\ t \in T\}$ Independant increment process with $X_0 = 0$ w.p. 1 and $\mathbb{E}[X_t] = c$

Ex:- Show that $\{X_t,\ 0 \leq t < \infty\}$ independant increment process with $X_0 = 0$ with probability 1 and $\mathbb{E}[X_t] = c$ $0 \leq t < \infty$ is a Martingale.

Suppose $0 < t_1 < t_2 < \dots < t_{n-1} < t_n$ $\forall\ n \geq 2$

Assume $X_{t_1}, \dots, X_{t_{n-1}}$ is known $\iff$ Since $X_0 = 0$
(i.e we are conditioning on them)

$X_{t_1} - X_{t_0} = X_{t_1}$ is known
$X_{t_2} - X_{t_1} = X_{t_0}$ is known
$\vdots$
$X_{t_{n-1}} - X_{t_{n-2}} = X_{t_{n-1}}$ is known

$\therefore$ Hence conditioning on $X_{t_1}, \dots, X_{t_{n-1}}$ is equivalent to conditioning on $X_{t_1} - X_{t_0}, X_{t_2} - X_{t_1}, \dots, X_{t_{n-1}} - X_{t_{n-2}}$ ..... (*)

$$\mathbb{E}[X_{t_n} \mid X_{t_1}, \dots, X_{t_{n-1}}]$$

$$= \mathbb{E}[X_{t_n} - X_{t_{n-1}} + X_{t_{n-1}} \mid X_{t_1}, \dots, X_{t_{n-1}}]$$

$$= \mathbb{E}[X_{t_{n-1}} \mid X_{t_1}, \dots, \underbrace{X_{t_{n-1}}}] + \mathbb{E}[X_{t_n} - X_{t_{n-1}} \mid X_{t_1}, \dots, X_{t_{n-1}}]$$

Since it is conditioned so that $X_{t_{n-1}}$ is known

$$= X_{t_{n-1}} + \mathbb{E}[X_{t_n} - X_{t_{n-1}} \mid X_{t_1}, \dots, X_{t_{n-1}}]$$

From (*)

$$= X_{t_{n-1}} + \mathbb{E}[X_{t_n} - X_{t_{n-1}} \mid X_{t_1} - X_{t_0}, X_{t_2} - X_{t_1}, \dots, X_{t_{n-1}} - X_{t_{n-2}}]$$

Since this is independant increment proces, $X_{t_n} - X_{t_{n-1}}$ is independant of all conditioned values

$$= X_{t_{n-1}} + \mathbb{E}[X_{t_n} - X_{t_{n-1}}]$$

$$= X_{t_{n-1}} + \underbrace{\mathbb{E}[X_{t_n}]}_{c} - \underbrace{\mathbb{E}[X_{t_{n-1}}]}_{c}$$

$$= X_{t_{n-1}} \quad P\text{-a.s}$$

$\Rightarrow$ This is a martingale

■

4) Markov Processes

Definition: Markov Processes

A $\mathbb{R}^n$ valued process $\{X_t,\ t \in T\}$ is a Markov Process (or has the Markov Property) if for every $t_1 < t_2 < \dots < t_K < t$ and $\forall\ k \geq 2$

$$P\left(X_t \in B \mid X_{t_1}, \dots, X_{t_K} \right) = P\left(X_t \in B \mid X_{t_K} \right)$$

Present — Past — Most recent past

Almost Surely (P-a.s.)

$\forall\ B \in \mathcal{B}^n$

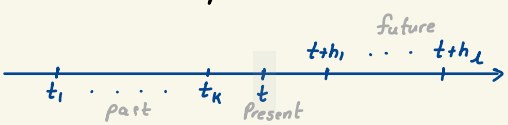
i.e.

$\forall\ x_1, \dots, x_K \in \mathbb{R}^n$

$$P\left(X_t \in B \mid X_{t_1} = x_1, \dots, X_{t_K} = x_K \right) = P\left(X_t \in B \mid X_{t_K} = x_K \right)$$

Conditional Independence

Theorem: Past and Future are conditionally independant given the present in a markov process.



$\forall\ t \in T \quad \forall\ t_1 < \dots < t_K \quad \forall\ k \geq 1$

$\forall\ 0 < h_1 < \dots < h_L \quad \forall\ L \geq 1$

$$P\left(X_{t_1} \in B_1, \dots, X_{t_K} \in B_K,\ X_{t+h_1} \in A_1, \dots, X_{t+h_L} \in A_L \mid X_t \right)$$

$$= P\left(X_{t_1} \in B_1, \dots, X_{t_K} \in B_K \mid X_t \right) + P\left(X_{t+h_1} \in A_1, \dots, X_{t+h_L} \in A_L \mid X_t \right)$$

almost surely

$\forall\ B_1, \dots, B_K, A_1, \dots, A_K \in \mathcal{B}^n$

Proof: Homework.

Proof suffices to have
$\underline{X}_0 = C$ w.p.1 and
$E[\underline{X}_t] = C$ - constant

Ex: $\{\underline{X}_t, \ 0 \leq t < \infty\}$ is an independant increment process with $\underline{X}_0 = 0$ w.p 1 and $E[\underline{X}_t] = 0 \quad 0 \leq t \leq \infty$. Show that this is a Markov process.

Proof: Consider $\forall \ 0 < t_1 < t_2 < \ldots < t_{k-1} < t_k \quad k \geq 2$. If $\{\underline{X}_t, t \in T\}$ let's drop the underscore and assume $\underline{X}_i$'s are one dimensional. $\{X_t, t \in T\}$ is $\mathbb{R}$- valued.

Assume $X_{t_1}, \ldots, X_{t_{k-1}}$ is known $\iff$ Since $X_0 = 0$
(i.e we are conditioning on them)

$X_{t_1} - X_{t_0} = X_{t_1}$ is known
$X_{t_2} - X_{t_1} = X_{t_0}$ is known
$\vdots$
$X_{t_{n-1}} - X_{t_{n-2}} = X_{t_{k-1}}$ is known

$\therefore$ Hence conditioning on $X_{t_1}, \ldots, X_{t_{k-1}}$ is equivalent to conditioning on $X_{t_1} - X_{t_0}, X_{t_2} - X_{t_1}, \ldots, X_{t_{k-1}} - X_{t_{k-2}}$ ..... (*)

$$P\left(X_{t_k} < x_k \mid X_{t_1} = x_1, \ldots, X_{t_{k-1}} = x_{k-1}\right)$$

$$= P\left(X_{t_k} - X_{t_{k-1}} < x_k - x_{k-1} \mid X_{t_1} = x_1, \ldots, X_{t_{k-1}} = x_{k-1}\right)$$

from (*)

$$= P\left(X_{t_k} - X_{t_{k-1}} < x_k - x_{k-1} \mid \underline{X}_{t_1} - \underline{X}_{t_0}, \underline{X}_{t_2} - \underline{X}_{t_1}, \ldots, \underline{X}_{t_{k-1}} - \underline{X}_{t_{k-2}}\right)$$

$$= P\left(X_{t_k} - X_{t_{k-1}} < x_k - x_{k-1}\right)$$

from independant increment ness.

$$= P\left(X_{t_k} - X_{t_{k-1}} < x_k - x_{k-1} \mid X_{t_{k-1}} - X_{t_0} = x_{k-1} - \underbrace{x_0}_{0}\right)$$ .. from independant increment ness.

$$= P\left(X_{t_k} - X_{t_{k-1}} < x_k - x_{k-1} \mid X_{t_{k-1}} = x_{k-1}\right)$$

$$= P\left(\boxed{X_{t_k} < x_k} \mid X_{t_{k-1}} = x_{k-1}\right)$$

$\therefore$ This is a Markov Process ■

Ideally we should prove for all Borel sets. But Sigma field generated from $(-\infty, x)$ include all Borel sets

Transition Probability Function

- Based on the definition of a Markov Processes all-important term of interest $P(\underline{X}_t \in B \mid \underline{X}_{t_k} = \underline{x}_k)$, transition probability function is defined.

Definition:- Transition Probability Function $= P(s, \underline{x}; t, B) \triangleq P(\underline{X}_t \in B \mid \underline{X}_s = \underline{x})$

$\forall\, s < t \in T \qquad B \in \mathcal{B}^n \qquad \underline{x} \in \mathbb{R}^n$

denotes a - conditional probability

Theorem :- $\{\underline{X}_t, t \in T\}$ is a $\mathbb{R}^n$-valued Markov process with transition probability function

$P(s, \underline{x}; t, B) \qquad s < t \in T \qquad \underline{x} \in \mathbb{R}^n \qquad B \in \mathcal{B}^n$

Then $\forall\, t_1 < \ldots < t_k$ in T, $k \geq 2$ the joint cdf of $\underline{X}_{t_1}, \ldots, \underline{X}_{t_k}$ is completely determined by the cdf $\underline{X}_{t_1}$ and the transition probability function.

Special Cases

1) Time Homogenous Transition Probability Function

$$P(s, \underline{x}; t, \underline{B}) = P(t-s; \underline{x}, B)$$

2) Markov Chain

$\exists$ a countable subset S in $\mathbb{R}^n$ with

$$P(\underline{X}_t \in S) = 1 \quad \forall\, t \in T$$

$S \triangleq$ "State Space" of Markov chain

- The transition probability function can be denoted by a $\#(S) \times \#(S)$ matrix $P(s,t)$ $\forall\, s < t$ where

$$(P(s,t))_{\underline{x},\underline{y}} = P(s, \underline{x}; t, \underline{y}) = P(\underline{X}_t = \underline{y} \mid \underline{X}_s = \underline{x}) \qquad \forall\, \underline{x}, \underline{y} \in S,\ \forall\, s < t \in T$$

Ex: $\forall\ t_1 < \dots < t_K \quad K \geq 2$ in T

$$P(\underline{X}_{t_1} = \underline{x}_1, \dots, \underline{X}_{t_K} = \underline{x}_K) = P(\underline{X}_{t_1} = \underline{x}_1) \cdot \prod_{i=1}^{k-1} P(\underline{X}_{t_{i+1}} = \underline{x}_{i+1} \mid \underline{X}_{t_i} = \underline{x}_i)$$

3) The transition probability function has a conditional pdf.

$\forall\ s < t \in T \quad \forall\ \underline{x} \in \mathbb{R}^n \quad \exists\ \rho(s, \underline{x}; t, \cdot) : \mathbb{R}^n \longrightarrow \mathbb{R}^+ \cup \{0\}$ such that

$$P(s, \underline{x}; t, B) = \int_B \rho(s, \underline{x}; t, y)\, dy \qquad B \in \mathbb{R}^n$$

In this case, the joint cdf of $\underline{X}_{t_1}, \dots, \underline{X}_{t_K}$ admits a joint pdf if $\underline{X}_{t_1}$ has a pdf. Specifically

$$f_{\underline{X}_{t_1}, \dots, \underline{X}_{t_K}}(\underline{x}_1, \dots, \underline{x}_k) = f_{\underline{X}_1}(\underline{x}_1) \prod_{i=1}^{k-1} \rho(t_i, \underline{x}_i; t_{i+1}, \underline{x}_{i+1})$$

Theorem :- $\{\underline{X}_t, t \in T\}$ is a $\mathbb{R}^n$ valued Markov process. Let $g : \mathbb{R}^n \to \mathbb{R}$ Borel with property $\mathbb{E}[|g(\underline{X}_t)|] < \infty\ \forall\ t \in T$. For $t_1 < \dots < t_K < t$ in T.

$$\mathbb{E}[g(\underline{X}_t) \mid \underline{X}_{t_1}, \dots, \underline{X}_{t_K}] = \mathbb{E}[g(\underline{X}_t) \mid \underline{X}_{t_K}]$$

Stationarity Concepts

Second Order Processes

Definition :- If $\{\underline{X}_t, t \in T\}$ is a $\mathbb{R}^n$-valued second order process then, $\mathbb{E}[\|\underline{X}_t\|^2] < \infty \quad \forall t \in T$.

- Under this assumption :
  - Mean function : $\mu_{\underline{X}_t} = \mu_{\underline{X}}(t) = \mathbb{E}[\underline{X}_t]$
  - Correlation function : $C_{\underline{X}}(s,t) = \mathbb{E}[\underline{X}_s \underline{X}_t^T]$ nxn matrix
  - Covariance function : $R_{\underline{X}}(s,t) = \text{Cov}[\underline{X}_s, \underline{X}_t]$
    $= \mathbb{E}\left[(\underline{X}_s - \mathbb{E}[\underline{X}_s])(\underline{X}_t - \mathbb{E}[\underline{X}_t])^T\right]$

$\mu_{\underline{X}_t}$, $C_{\underline{X}}(s,t)$ and $R_{\underline{X}}(s,t)$ are finite and well defined.

Linear Index Set

Definition :- Let T be such that $t, s \in T \implies t+s \in T$
Such a set is called "linear index set".

Ex :- $T = \mathbb{N}$
$T = \mathbb{Z}$
$T = \mathbb{R}$
$T = [0, \infty)$

$\mathbb{R}^n$- valued stochastic process $\{\underline{X}_t,\ t \in T\}$ is

1) Stationary of Order K

Definition:- The $\mathbb{R}^n$- valued stochastic process $\{\underline{X}_t,\ t \in T\}$ is **stationary of order k** $(k = 1, 2, \ldots)$ if $\forall\ t_1 < \ldots < t_k$ and $\forall h \in T$ the rvs,

$$\underline{X}_{t_1}, \ldots, \underline{X}_{t_k} \quad \text{and} \quad \underline{X}_{t_1+h}, \ldots, \underline{X}_{t_k+h}$$

have the same cdf.

Theorem:- If $\{\underline{X}_t,\ t \in T\}$ is stationary of order (at least) 2 and is a second order process, then

$$\mu_{\underline{X}}(t) = \text{Constant in } t$$

$$C_{\underline{X}}(s,t) = C_{\underline{X}}(s+h, t+h) = C_{\underline{X}}(|t-s|)$$

$h \in T\ \forall s,t \in T$

This motivates a weaker notion of Stationarity. (Covariance Stationarity)

2) Strict Stationarity

Definition:- The $\mathbb{R}^n$- valued stochastic process $\{\underline{X}_t,\ t \in T\}$ has **strict stationarity** if it is stationary of order $k\ \forall\ k = 1, 2, \ldots$

Ex:- • $\{X_t\}_{t=1}^{\infty}$ i.i.d sequence of rvs.

• $\{X_t\}_{t=1}^{\infty}$ $X_t = X\ \forall t$

- Strict stationarity is stringent. Too much to ask for. We need another weaker notions.

3) Covariance Stationarity

Definition :- A second order process $\{X_t, t \in T\}$ is said to be **Covariance (2^{nd} order, weak, widesense)** stationary if the mean function and covariance function are boath translation invariant.

i.e. $\forall\ s,t \in T$ $\quad \mu_X(t) = \text{constant in } t$

$R_X(s,t) = R(|s-t|)$